

Overview

This study explores the complex process of transit and land-use integration in rapidly growing cities in developing countries. It first identifies barriers to and opportunities for effective coordination of transit infrastructure and urban development. It then recommends a set of policies and implementation measures for overcoming these barriers and exploiting these opportunities.

Well-integrated transit and land development create urban forms and spaces that reduce the need for travel by private motorized vehicles. Areas with good access to public transit and well-designed urban spaces that are walkable and bikeable become highly attractive places for people to live, work, learn, play, and interact. Such environments enhance a city's economic competitiveness, reduce local pollution and global greenhouse gas emissions, and promote inclusive development. These goals are at the heart of transit-oriented development (TOD), an urban form that is increasingly important to sustainable urban futures.

This book uses a case study approach. It draws lessons from global best-case examples of transit-oriented metropolises that have direct relevance to cities in developing countries and elsewhere that are currently investing in bus rapid transit (BRT) and other high-capacity transit systems. It also reports the results of two original in-depth case studies of rapidly growing and motorizing cities that introduced extended BRT systems: Ahmedabad, India and Bogota, Colombia. Two shorter case studies—of Guangzhou, China and Ho Chi Minh City, Vietnam—enrich the understanding of factors that are critical to “transforming cities with transit.”

The mechanisms for successfully integrating transit and urban development are complex. The multitude of factors involved includes inherent characteristics of a city (for example, natural and historical conditions); governance structures; institutional settings; public sector initiatives and actions (for example, transit investments, planning regulations, tax policies, and financial incentives); financing instruments; and market responses.

Transit shapes urban development by enhancing accessibility; attributes of land, such as residential and job densities and the degree of land-use mixing, affect travel demand. The case studies presented in this book focus mainly on the impact of transit investment on land use and urban form rather than on the impact of land use on transit demand, as it takes many years for shifts in land use and their eventual impacts on transit demand to unfold and reveal themselves.

Analysis of the interaction between public sector initiatives and market response is especially challenging, because the mechanisms of these interactions are not fully understood. The study probes market response, which reflects consumer and market preferences indirectly, by studying, for example, the influences of BRT improvements on residential land prices and private real estate investment decisions. Cases show that the private market does not always respond to transit investments as would be expected, opting instead to build new housing enclaves away from BRT corridors. Case studies hint at factors that would explain such responses, but much more research is needed on this topic. Accordingly, this study focuses on public sector initiatives and actions (for example, public policy interventions such as zoning reforms and improved feeder bus connections to BRT stations) that work with, react to, and shape the marketplace.

The case analyses reveal key institutional, regulatory, and financial constraints that hamper successful transit and land-use integration. They also highlight opportunities for charting sustainable pathways. The last chapter of the book offers policy recommendations and suggests steps needed to implement them. The recommendations are drawn from best-practice experiences worldwide as well as insights gained in studying the unique challenges faced by Ahmedabad and Bogota as they strive to become more sustainable.

Most of the cases reviewed are from developed countries. Their experiences cannot be directly applied and transplanted without some adaptation and adjustments to local realities. The aim is less to identify specific practices and more to impart key principles and core lessons that can guide the planning and practice of cities currently planning or investing in large-scale transit systems. Recommendations for creating more sustainable cities of the future range from macro-level strategies that influence land development and governance at the metropolitan scale to micro-level initiatives, such as TOD, which can radically transform development patterns at the neighborhood level.

The book will be of interest to a wide and diverse audience, including mayors, city council members, national and local policy makers, urban and transport planners, transit agency officials, developers, staffs of development financial institutions, and others involved with TOD projects in rapidly growing and motorizing cities in the developing world.

Transit and Land-Use Integration

Cities in developing countries are growing at an unprecedented rate. Their population is projected to increase from 2.0 billion in 2000 to 5.5 billion by 2050, accounting for 95 percent of the increase in the world's urban population. With rising incomes, urban residents will expand outward, following the trajectory of automobile-dependent sprawl evident in developed countries.

This trend must be reversed if significant headway is to be made in increasing cities' economic competitiveness through energy- and time-efficiency improvements. Negative externalities such as air pollution and greenhouse gas emissions cannot be reduced and social equity cannot be promoted without action to prevent urban populations in developing countries from following this path of automobile-dependent sprawl.

Transit and land-use integration is one of the most promising means of reversing the trend of automobile-dependent sprawl and placing cities in developing countries on a sustainable pathway. BRT systems deserve special attention, as the majority of future urban growth is expected to take place in medium-size cities (cities with no more than 500,000 people) (UN Habitat 2011) with limited fiscal capacities. BRT systems, which are less costly than other forms of mass transit, such as metros, can meet these cities' traffic demand. With more than 130 BRT systems having commenced operation in the past decade, promoting TOD will be particularly important.

Global Best Cases of Transit and Land-Use Integration

All global best-case examples of transit and land-use integration had a cogent land-use vision—a vision that shaped regional transit investments more than vice versa. Transit was one of several important tools, along with supportive zoning and creative financing, used to make urban visions a reality. Transportation demand management (TDM) measures, including congestion charges and streetscape enhancements that promote nonmotorized transport modes, have also been important. Cities like Hong Kong SAR, China and Tokyo reveal that successful transit and land-use integration can generate revenue and capture value through the development of property and air rights.

Cities like Copenhagen and Singapore have benefited from cogent regional visions that ensure that high-capacity transit investments produce desired urban-form outcomes ([box 0.1](#)). Experiences from these and other cities suggest that station area planning needs to be carried out selectively and judiciously. In many instances, planning efforts should be devoted to developing or redeveloping no more than a handful of rail and BRT stations, in order to allow resources to be concentrated effectively. Doing so increases the odds of a “win-win” arrangement in which both public and private interests can co-participate in the benefits new transit investments yield. Demonstrating that positive

land-use changes are possible in conjunction with transit investment is important for producing models that the larger development community can emulate as well as for convincing lenders that investing in station area projects can be financially remunerative.

Box 0.1 Transforming a vision into a conceptual image of the future metropolis

The first steps in moving transit and land-use integration from theory to reality are formulating a vision and transforming it into a conceptual image of the future metropolis. Prime examples of such a process include Copenhagen's "Finger Plan" and Singapore's "Constellation Plan."

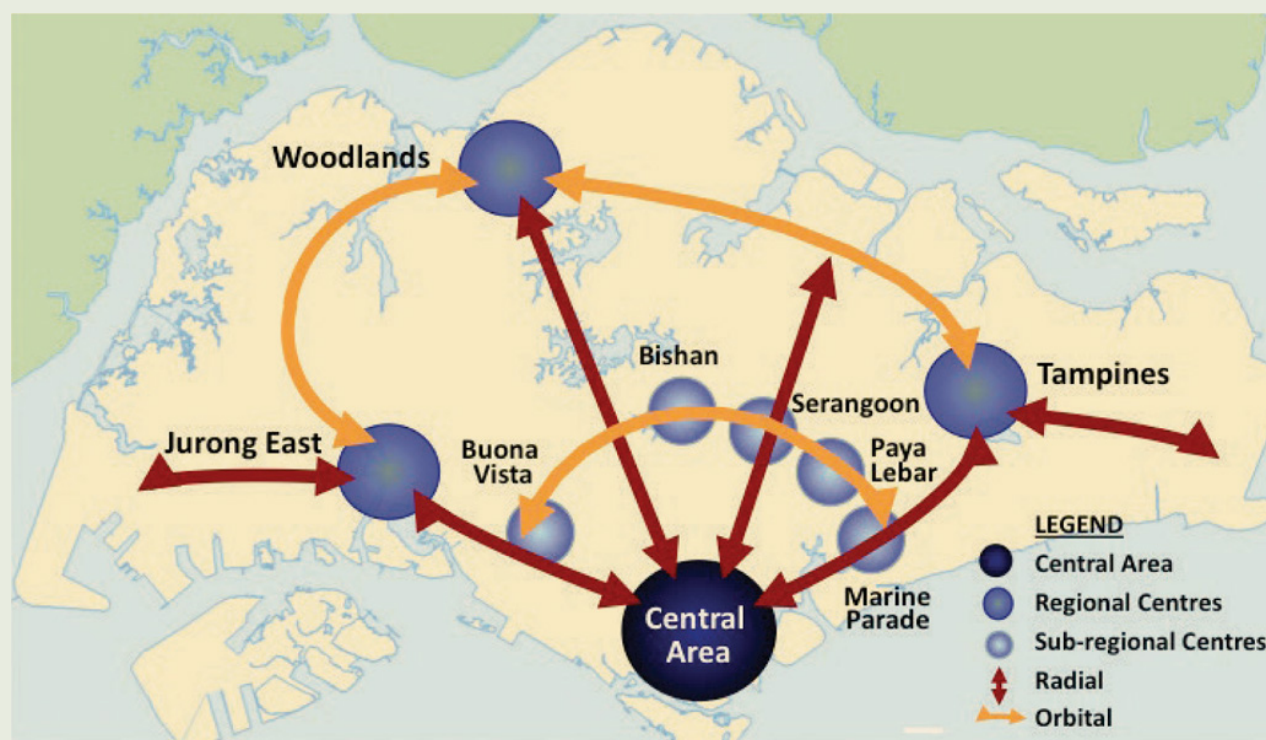
Copenhagen's Finger Plan is a textbook example of a long-range planning vision shaping rail investments, which in turn shaped urban growth ([figure B0.1.1](#)). Local planners identified corridors to channel overspill growth from urban centers early in the planning process. Rail infrastructure was built, often in advance of demand, to steer growth along desired growth axes.

Figure B0.1.1 Copenhagen's "Finger Plan" for urban development



Source: Cervero 1998; reproduced with permission from Island Press, Washington, DC.

Singapore has embraced Scandinavian planning principles that call for radial corridors that interconnect the central core with master-planned new towns (figure B0.1.2). Its spatial plan has the appearance of a constellation of satellite "planets" (new towns), which surround the central core, interspersed by protective greenbelts and interlaced by high-capacity, high-performance rail transit.

Figure B0.1.2 Singapore's "Constellation Plan" for urban development

Source: Singapore Land Transport Authority 2008; reproduced with permission.

Source: Authors.

Planning visions of transit and land-use integration are important, but adequate funding is needed for successful execution. An important lesson from international experience is that transit and land-use integration can yield the income needed to expedite and support the process. Experiences in cities like Hong Kong SAR, China and Tokyo show that “value capture”—an infrastructure financing concept that seeks to capture land value created by new infrastructure, particularly transit—is effective not only for sustainable finance but also for sustainable urbanism. Value capture is particularly well suited for financing transit infrastructure in dense, congested settings, where a high premium is placed on accessibility and the institutional capacity exists to administer the program. Accessibility benefits, which get capitalized into land values, present enormous opportunities for recapturing some of the value created by transit investment, at least as a supplement to fare-box income and other traditional revenue sources.

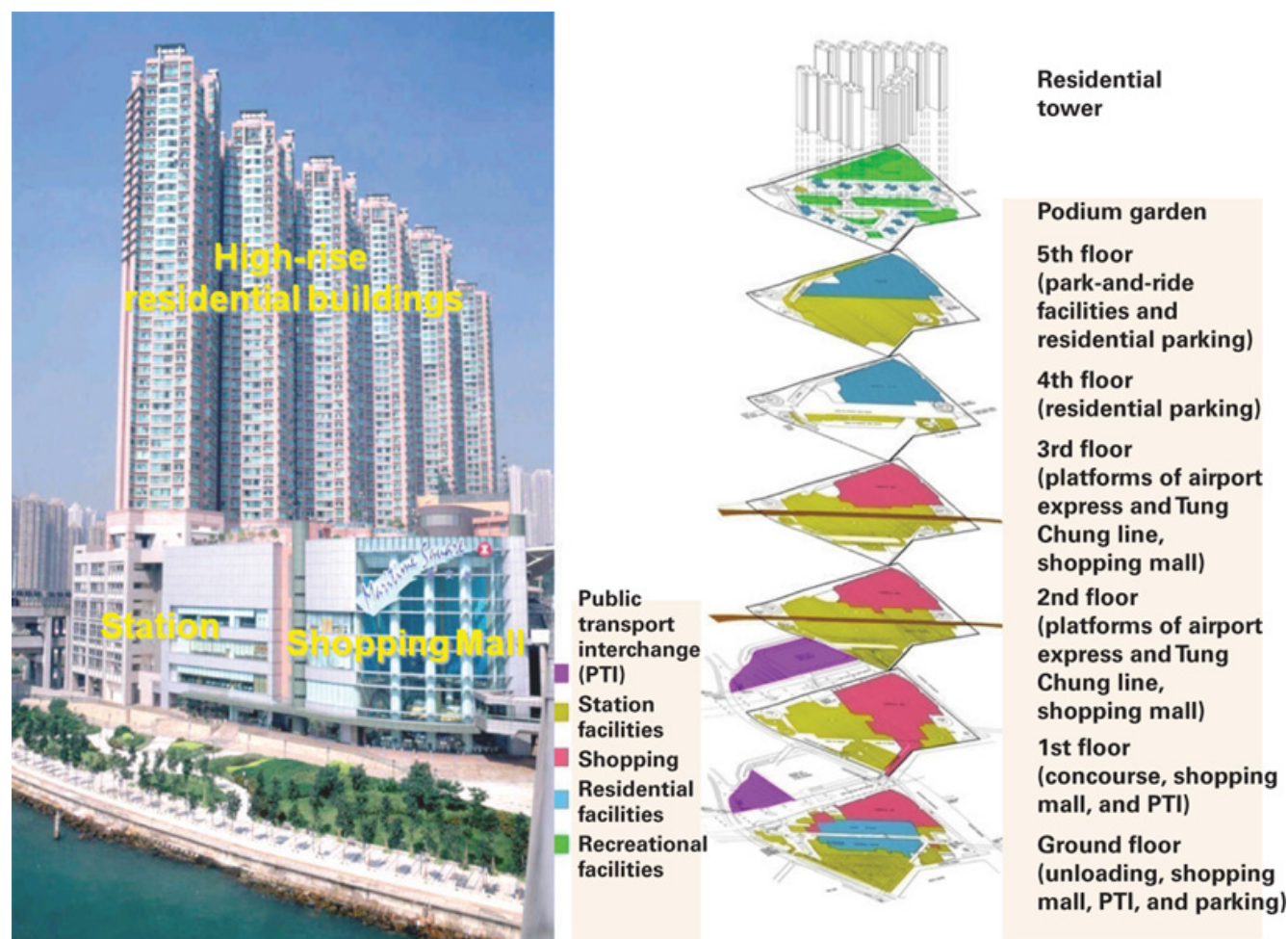
Even in ultra-dense, transit-friendly Hong Kong SAR, China, rail investments are not financially viable on their own: the Hong Kong Mass Transit Railway (MTR) meets investors' demands for return on equity only as a result of its property development. Value capture and joint development not

only generate income to help retire rail-capital investment bonds and finance operations, they also create market demand that ensures high-ridership services. Hong Kong SAR, China's version of a public-private partnership is not about off-loading the cost of building railways to the private sector. Rather, it is about "co-development"—development in which each sector brings its natural advantage to the table (for example, land acquisition powers in the case of the public sector, access to equity capital in the case of the private sector). The resulting "win-win" situation leads to financially viable investments and forges a close connection between rail systems and nearby real estate development that attracts tenants, investors, and transit riders.

Greater Tokyo's private railways have historically practiced transit value capture on an even grander scale, building massive new towns along rail-served corridors and cashing in on the construction, retail, and household service opportunities created by these investments. In both Hong Kong SAR, China and Tokyo, rail and property development has created a virtuous cycle of viable railway operations and a highly transit-oriented built form.

Important to capturing transit value in both cities has been institutional adaptation and change. In Hong Kong SAR, China, MTR executives gained an appreciation over time of the importance of urban design, pedestrian circulation, and public amenities, all of which are particularly important in dense, crowded cities. They also created financially successful Rail + Property (R+P) projects ([figure 0.1](#)). The emergence of the city as an international gateway, combined with its economic transformation from traditional manufacturing to a service-based economy, opened up new possibilities for R+P in both shaping growth and serving new market demands. To its credit, MTR made a conscious decision to build high-quality, mixed-use R+P projects on greenfields en route to the new international airport as well as on brownfields served by central-city railway extensions. These projects have proven to be wise investments: recent-generation R+P projects that functionally and architecturally blend with surrounding communities have outperformed earlier projects in terms of ridership gains and real estate market returns.

Figure 0.1 The Maritime Square residential-retail project developed by the Mass Transit Railway of Hong Kong SAR, China



Source: Courtesy of MTR Corporation Ltd.

Note: Situated above the Tsing Yi Station between the center of the city and the new international airport, Maritime Square features a hierarchy of integrated uses: a shopping mall extends from the ground floor to the third level; the station concourse sits on the first floor, with rail lines and platforms above and ancillary/logistical functions (such as public transit/bus interchange and parking) at or below; above the fourth and fifth floor residential parking lies a podium garden, and above that are luxury residential towers.

Experiences in Tokyo, and to a lesser degree Seoul and Stockholm, also underscore the importance of market adaptation. In Tokyo, the downturn in the real estate market, slower economic growth, and a changing demographic structure prompted private railway companies—both new and old—to seek new market opportunities, most notably infill housing and mixed-use developments around major central-city railway terminals. Such redevelopment complements the earlier towns built by private companies. To appeal to professionals and a more youthful labor force, planners are placing greater

emphasis on creating high-quality urban spaces in and around joint development projects. Resiliency in the design and planning of “new-towns/in-towns” has also been a hallmark of recent urban regeneration successes in Seoul and Stockholm.

Successful value capture schemes depend to a significant degree on a supportive institutional environment. In Washington, DC, a single transit authority—the Washington Metropolitan Area Transit Authority (WMATA)—was formed and given the resources needed to leverage TOD. Particularly important was the formation of a proactive real estate development department within the organization, staffed by individuals with private sector experience to create a more entrepreneurial approach to land development. Creating a town planning and urban design department to ensure that joint development projects are of high quality and architecturally integrated is another key institutional reform, as demonstrated by the case of Hong Kong SAR, China.

Lessons from Case Studies in Developing Countries

Faced with ever-worsening traffic congestion and deteriorating environmental conditions, many cities in developing countries have turned to public transit systems in an effort to reverse course. Transit investments, proponents hope, will also help reverse automobile-dependent patterns of urban growth.

Case studies reveal that transit in developing countries has been guided primarily by the almost singular objective of enhancing mobility (that is, the ease and speed of moving about cities). Motivated by the need to see results while current city leaders are still in office, city administrations and transit agencies often adopt a short-term, narrow focus—for example, on rapidly relieving congestion. In this process, the long-term visions of promoting sustainable patterns of urban growth are often lost. This mindset is reflected in the absence of strategies and regulations to create higher densities along transit corridors and high-quality urban spaces. Both can be vital to increasing transit and nonmotorized transit use, thus reducing private automobile travel.

Two cities—Ahmedabad, India and Bogota, Colombia—serve as case studies for probing the opportunities and challenges posed by integrated transit and urban development. Careful analysis of their BRT projects and urban development trends identified eight major barriers to transit and land-use integration:

- lack of regional coordination at the metropolitan level
- sector silo behavior and practices at the city level
- inadequate policies and regulations for strategically creating “articulated densities” (densities that are strategically distributed across parts of a metropolitan area)

- restrictive national regulations and administrative constraints
- inconsistencies in the planning instruments and deficiencies in their implementation
- inadequate policies, regulations, and supporting mechanisms for redeveloping built-up areas, particularly brownfields or distressed and blighted districts
- neglected urban design at the neighborhood and street level
- financial constraints.

Although their BRT systems have improved overall mobility, these two cities have yet to fully explore the potential benefits of transit and land-use integration. Quantifiable evidence on land-use impacts is limited in both cases, as the initial phases of the BRT systems were constructed primarily to meet existing traffic demand in built-up areas. Since then, market responses to new development opportunities have emerged, particularly along corridors close to economically strategic locations, such as large-scale riverfront development sites or areas closest to international airports.

Lack of regional coordination at the metropolitan level

At the metropolitan scale, governments need to closely coordinate land-use plans, infrastructure investments, and urban services. The management of a metropolitan region is an exceedingly complex task, because it involves diversified governmental entities at multiple governing levels. Through decentralization, the national government devolves some of its decision-making powers and fiscal functions to lower-level governments. However, devolution can sometimes isolate municipalities, making regional collaboration more difficult unless proper regulatory and institutional coordination mechanisms are incorporated.

Political and economic competition among municipalities often impedes the coordination of planning, investment, and service provision across administrative boundaries, a challenge Bogota and its neighboring municipalities currently face. For example, the city's BRT (the TransMilenio) and other regional bus services are not well connected, making commuting for many low-income people who work in Bogota but live in neighboring municipalities difficult. This deficiency arises from the fact that TransMilenio's service jurisdiction does not align with the regional context under which Bogota-Cundinamarca (Bogota's regional department) planning and policy making occurs. Recognizing these institutional shortcomings, in 2011, the national government enacted the Territorial Ordinance Organic Law (Ley Organica do Ordemamiento Territorial [LOOT]) to promote regional coordination. It is too early to gauge its effectiveness.

Sector silo behavior and practices at the city level

At the city level, departments and agencies have varying missions, objectives, budgets, management styles, governance structures, and staff profiles. These differences often hinder the types of cross-sector and interagency coordination needed for transit and land-use integration.

Transportation department staffs often have little knowledge of urban planning and design, and urban planners have little knowledge of transportation, making seamless integration and implementation of the two difficult. Transportation department staffs have limited capacity to take proactive measures toward integrating transit and land use (and often limited interest as well). Staff members from the two departments work under different management and budgetary constraints, and they have little incentive for uncertain cross-sector coordination.

The mobility secretariat and TransMilenio S.A. developed TransMilenio with little coordination with the city's other relevant secretariats, particularly the urban planning secretariat. New urban development near the BRT stations or corridors is limited, for example, except in the airport corridor and some commercial complex developments close to BRT terminal stations. The paucity of development near many TransMilenio stops may be related to the fact that the floor area ratios (FAR) of these areas (which had been kept low, at less than 2.0, except in the central business district and a few other areas) have not been adjusted, even though land values near the BRT stations and corridors have increased. In addition, the BRT station designs are not well integrated with the urban fabric of nearby neighborhoods and streets.

Inadequate policies and regulations for strategically creating articulated densities

Most large cities in Asia and in most developing countries have higher average population densities than land-rich Australia, Canada, and the United States. Ahmedabad (with 18,430 people per square kilometer, according to the 2001 census) and Bogota (with 21,276 people per square kilometer in 2010, according to the Secretary of Planning for the City of Bogota) are among the densest cities in the world. For various reasons, politicians, planners, and residents of most high-density cities in developing countries have a desire to “de-concentrate” and spread development to new areas rather than increase density in existing built-up areas.

It is widely believed that cities in developing countries cannot accommodate additional growth or densification in built-up areas. Indeed, city authorities tend to consider population growth and the limited “carrying capacity” of the city as prime causes of chronic gaps between the demand for and

supply of urban services or their low quality. They tend to prefer development of peripheral greenfields rather than urban redevelopment in the city core, because greenfield development is faster and costs less up front (Burchell and others 2002).

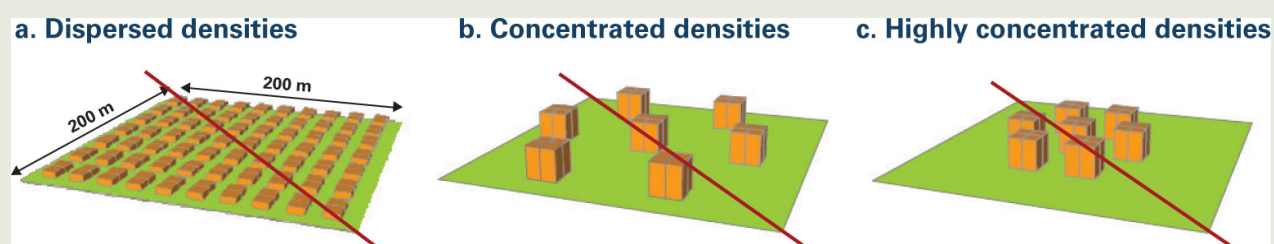
Although expansion of the urban area of rapidly growing cities in developing countries is unavoidable to some extent, these arguments suggest that the only way to accommodate urban growth is to expand outward. This belief ignores the high life-cycle costs related to expanding the infrastructure network (for example, water pipelines and sewerage networks); operating and maintaining that network; and forgoing the high opportunity costs of expanded urban land use.

Although it logically follows that higher densities require an increased level of infrastructure investments, the perception that density leads to the deterioration of urban service provision is not necessarily true. Seoul (with 16,589 people per square kilometer in 2009, according to the UN Department of Economic and Social Affairs [UNDESA]) and Singapore (with 7,025 people per square kilometer [according to UNDESA]), for example, provide efficient and high-quality urban services while maintaining good environmental conditions.

Because of the prevalence of this negative perception of high average urban population density, policy makers and city planners in developing countries are often reluctant to ramp up urban density by allowing a higher FAR ([box 0.2](#)). They also tend to apply either a uniform or too narrow a band of ratios across all city areas, without capitalizing on location premiums, such as proximity to transit stations or corridors. Controlling density without considering the land's economic value prevents cities from effectively managing their land use. In Ahmedabad, for example, the FAR of urban areas is kept low (1.80–2.25) for the entire urban area. In Bogota, except for the central business district and several peripheral areas, where no FAR restriction exists, the FAR is either 0–1.0 or 1.1–2.0, including near BRT stations and corridors.

Box 0.2 Articulated versus average density

What matters most for transit and land-use integration is not average population densities but “articulated densities”—densities that are strategically distributed across parts of a metropolitan area. The layout depicted in panel c of [figure B0.2.1](#) is better suited for mass transit than the one in panel a, even though the two forms have the same average population density.

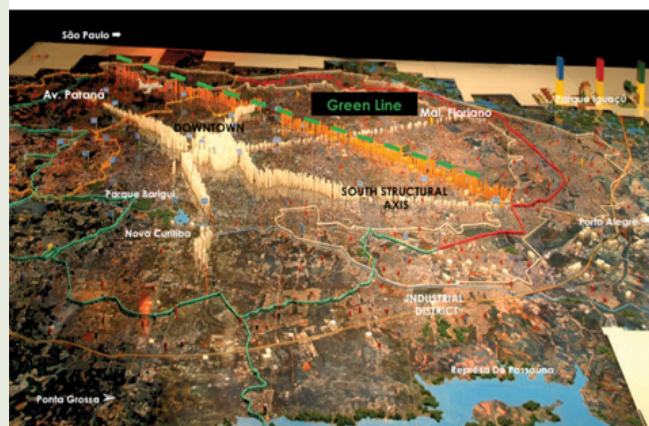
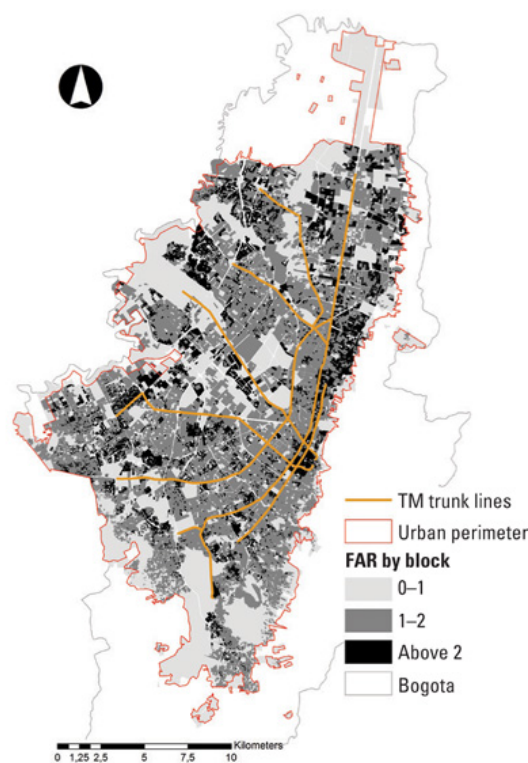
Figure B0.2.1 Importance of articulated density for mass transit

Source: Adapted from OECD 2012.

Note: Red line represents a mass transit line.

Curitiba, Brazil has created articulated densities along its BRT corridors ([figure B0.2.2a](#)).

Bogota, Colombia has not done so along its TransMilenio corridors ([figure B0.2.2b](#)). It maintains a low FAR (0–2), except within the central business district and a few other selected spots.

Figure B0.2.2 Transit-oriented spatial development in Curitiba and Bogota**a. Curitiba, Brazil****b. Bogota, Colombia**

Source: IPPUC 2009.

Note: Buildings in Curitiba, represented in bright yellow and orange, are strategically built along the BRT corridors with good urban planning. The current urban form was developed under a long-term vision; a TOD concept was first outlined in its 1965 Master Plan. Today, the city has lower greenhouse gas emission levels, less traffic congestion, and more livable urban spaces compared with other similar Brazilian cities. The red, green, and orange lines indicate the city boundaries.

Source: World Bank.

The transit-oriented cities analyzed in [chapter 2](#) of the book adopt much wider variations. Within the central business district area alone, these ratios range from 12 to 25 in Singapore, 8 to 10 in Seoul, 1 to 20 in Tokyo, and 1 to 12 in Hong Kong SAR, China. Each of these cities has developed different density ranges in different areas, taking social and economic features into consideration. In general, all allow for high-density construction around transit lines and stations.

Restrictive national regulations and administrative constraints

In addition to the FAR, certain national and local government regulations and administrative deficiencies adversely affect the smooth functioning of land markets. The result is the under- or

oversupply of land, noncontiguous spatial development, and changes in land-use patterns that respond slowly to the values created by transit infrastructure. These regulations are major barriers to transit-oriented spatial development.

In India, many national regulations restricting the market's rational response significantly limit the ability of city planning instruments to redirect real estate development in a way that is consistent with a long-term spatial vision. For example, the Urban Land Ceiling Act (ULCA) of 1976 prevented developers or individuals from owning large parcels, thus hindering the redevelopment of vacant land of the closed textile mills until the law's abolishment in 1999. A high stamp duty on land sales, control over minimum plot sizes, and rent control are other examples of regulations that impede market-based property transactions and development.

Colombia has fewer national or local restrictive regulations hindering the smooth functioning of the real estate market. Problems there relate to the deficiency of the planning process and its implementation rather than regulatory constraints. In Bogota, because of the inconsistency between the Territorial Ordinance Plan (POT) and development plans, it usually takes four to five years to obtain a construction permit or approval to consolidate land (Samad, Lozano-Gracia, and Panman 2012). This problem is one reason why it is difficult for developers to undertake urban redevelopment projects, particularly large-scale, mixed-land, urban regeneration projects that require the consolidation of several pieces of land. These deficiencies in planning and administrative process make it difficult for the private sector to respond to the new development opportunities provided by the increase in the economic value of land near BRT stations and its corridors.

Inadequate policies, regulations, and supporting mechanisms for redeveloping built-up areas

Most transit investments in cities in developing countries take place first in already urbanized areas (as shown in the four case cities in developing countries), because the priority is to address existing unmet traffic demand and reduce congestion. Retrofitting these areas in response to new development opportunities created by transit is more complex and difficult than implementing TOD in greenfield development.

Retrofitting built-up areas is difficult for two reasons. First, private businesses or households own the majority of TOD precinct property; the government has little control over this land. In contrast, the government can control the construction of transit systems, once it secures the right of way and the funds for construction. Second, as its name indicates, the redevelopment of "built-up" areas—whether

they be publicly owned or private property—requires the demolition of physical assets, such as infrastructure and housing stock, and their reconstruction. Such requirements have both economic and social implications, as redevelopment requires substantial costs and resettlement arrangements.

Neglected urban design at the neighborhood and street level

Transit shapes urban development, and land-use patterns influence travel demand. Land-use variables reflect not only trip interchange attributes (for example, travel time and the cost of travel from origin to destination) but also the attributes of the origins and destinations themselves (for example, density and degree of land-use mixture).

Density is not the only important element of land use and built environments. Other elements—including carefully articulated land-use mixtures; safe and smooth accessibility to transit stations (enabled by foot paths, cycle paths, and street lights, for example); and amenities such as benches, parks, landscaping, and libraries—contribute to the development of a good built environment. Some BRT stations in Bogota are located in the medians of six- to eight-lane highways, requiring passengers to walk across long steel access flyovers. The city has not been able to improve the quality of public places along BRT corridors, because the neighborhood redevelopment plan has not yet been implemented. A long line of unadorned walls occupies the space between the BRT corridors. The lack of urban design elements creates a disconnect between the transit system and the surrounding neighborhoods. This lack of integration does not promote the reconfiguration of the city layout along the BRT lines in a way that fosters vibrant urban life and economic activities.

Financial constraints

The huge upfront capital investment required to develop a transit system is one of the greatest obstacles to integrating transit and land use. This obstacle is particularly difficult to surmount because unprecedented urban growth has put severe pressure on the ability of local governments to finance infrastructure investments and urban services.

At the operation stage, cities sometimes need to provide subsidies, either to make up operational deficits or to provide vouchers for low-income riders. Scarce financial resources also sometimes oblige transit companies to choose routes based on the cost of right-of-way acquisition rather than on the long-term development potential of the areas to be served. In such cases, system design is driven primarily by the desire to minimize construction costs—by, for example, siting stations in the median of thoroughfares, reducing costly land takings and disruptions. Opportunities to maximize development are forgone.

Both Ahmedabad and Bogota recognize the benefits of integrating BRT or planned metros and urban development. Building on their experiences, both cities have huge potential to generate sizable revenues through value capture. Both cities recognize that improving the quality of pedestrian access and providing high-quality urban spaces next to transit stations will be pivotal in promoting TOD in future phases. True to the World Bank's Ecological Cities as Economic Cities (Eco²) concept promoting sustainable urban development through cross-sector integration, this planning approach toward integrated urban space will facilitate the creation of cities that are not only cleaner, more livable, and more inclusive but cities that are more prosperous and economically competitive as well.

Toward Transit and Land-Use Integration

Case studies of cities that have aggressively invested in mass transit systems highlight the unique contexts and challenges of integration. Collectively, the insights gleaned from these case experiences point to policy recommendations and implementation measures that deserve careful consideration at different levels of strategic decisions and planning exercises.

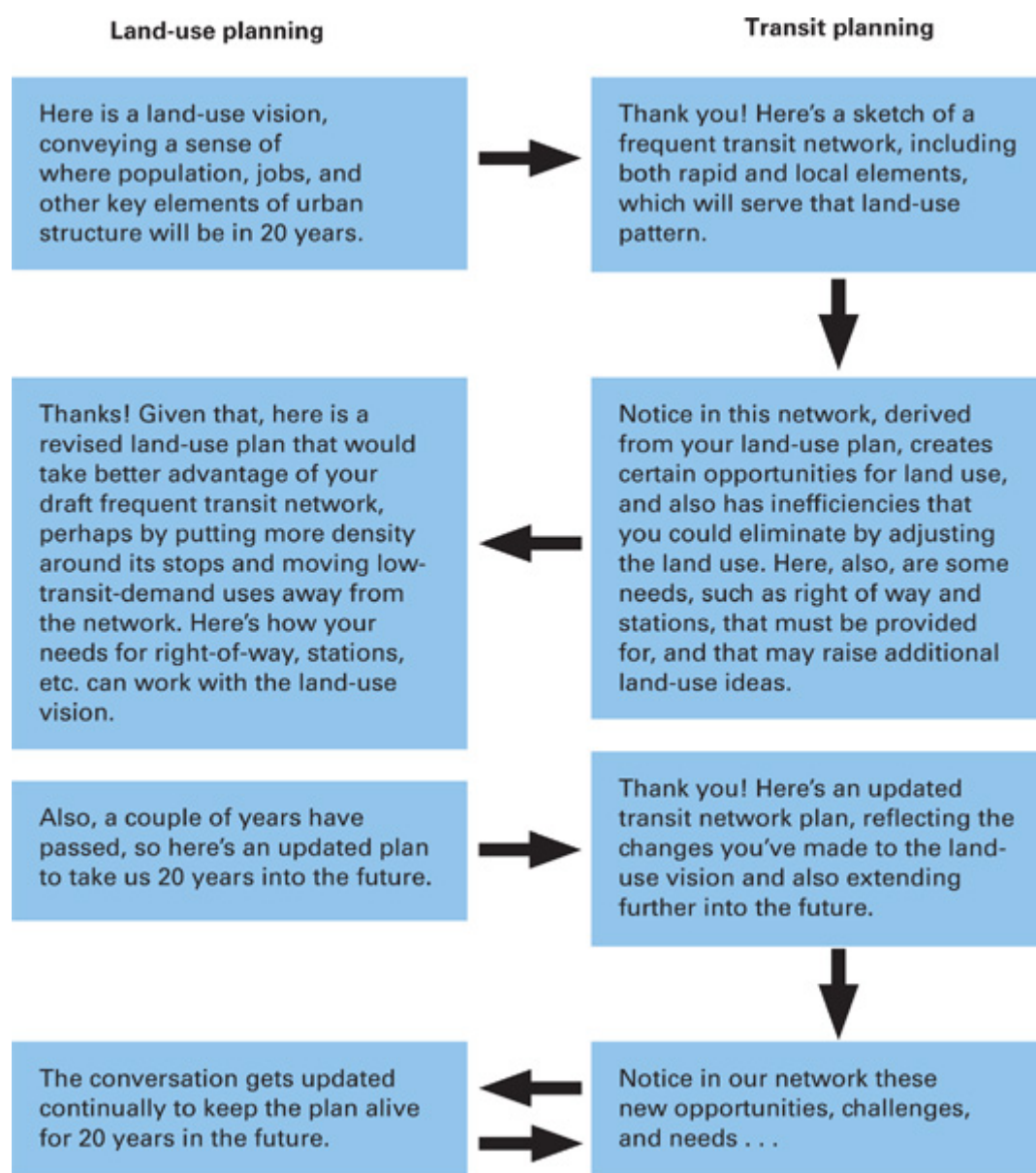
Strategic vision and enabling institutional and regulatory framework

1. Develop strategic plans. A strategic vision is indispensable for successfully integrating transit and urban development. Transit is a means to help create desirable patterns of urban growth, not the other way around. It is, literally and figuratively, a vehicle for connecting people to places, thus contributing to creating the kinds of cities and neighborhoods in which people want to live, work, play, learn, and interact. Cities need to translate their visions of the future into the land-use and infrastructure elements of a statutory master plan, which must be market sensitive, socially inclusive, and rooted in fiscal realities.

2. Create a supportive institutional and government environment. Effective integration requires a robust regional governance structure that gives rise to intermunicipal cooperation; fosters accountability, in order to reduce negative spillovers across jurisdictions; and provides a territorial context for coordinating growth within a region's commuter and labor sheds. Regional institutions need to be endowed with the power of regulatory oversight and the funding capacities to finance investments required for integration. At the city level, the administration must have the capacity and political backing to overcome the inertia formed by parochial, unisectoral entities that view the world in silo terms. Creation of a super-organization that absorbs various agencies to redress this problem is rarely effective. Instead, the administration should focus on establishing consensus on critical, urgent

issues that require institutional coordination and involve key stakeholders in the process. The administration also needs to align budgetary and human resource policies with the objectives of cross-sector cooperation. Walker (2012) states:

The land planners do a long-range sketch of urban structure and this goes up on the wall in the transit planner's office, so that it guides daily thinking as well as long-range planning. The transit planner does a similar sketch of a long-range transit network, and this goes up on the wall in the land-use planner's office. That way, when developments are being approved, the short-term land-use planner can check whether the location is a good or bad one for transit and can judge developments accordingly. Meanwhile, as the long-term land-use planners stare at the transit map, they have new ideas for how to build communities around the proposed line and stations ([figure 0.2](#)).

Figure 0.2 A healthy conversation between transit and land-use planning officials

Source: Walker 2012; adapted with permission from Island Press, Washington, DC.

Note: Similar conversations happen between land use and road planning, or between transit and road planning, etc., for other infrastructure or government objectives.

3. Remove restrictive regulations and set appropriate prices on land. Some national and local government regulations adversely affect the smooth functioning of land markets by distorting prices. The result is an under- or oversupply of land and noncontiguous spatial development. Regulations such as India's ULCA also inhibit the ability of transit infrastructure to change land-use patterns. These regulative barriers to transit and land integration need to be removed in order to ensure the

rational functioning of the land market. It is also important that the national government help manage the balance between private automobile and public transit use by removing various incentives for car use, such as subsidized fuel prices, low parking prices, and low toll charges.

4. Aim for short-term mobility objectives and long-term sustainable urban-form goals, in parallel. Cities are encouraged to try to realize both short-term mobility enhancement objectives and long-term sustainable development. Global best practices indicate that investing in more roads and transportation infrastructure in order to improve mobility in the short term eventually leads to more and longer-distance trips—what is called “induced traffic.” By itself, supply-side expansion often provides only ephemeral congestion relief. Successful integrated spatial development can provide both near-term congestion relief and longer-term urban-form benefits. Effective integration is not automatic, however; it requires strong leadership and institutional efforts and support.

City-level planning approach

5. Create articulated densities. Simply stated, mass transit needs mass. Cities need to promote higher densities along transit-served corridors to ensure a critical mass of trip origins and destinations that will drive up ridership and thus increase cost effectiveness. How densities are organized in relationship to high-capacity transit services, not average densities, matters.

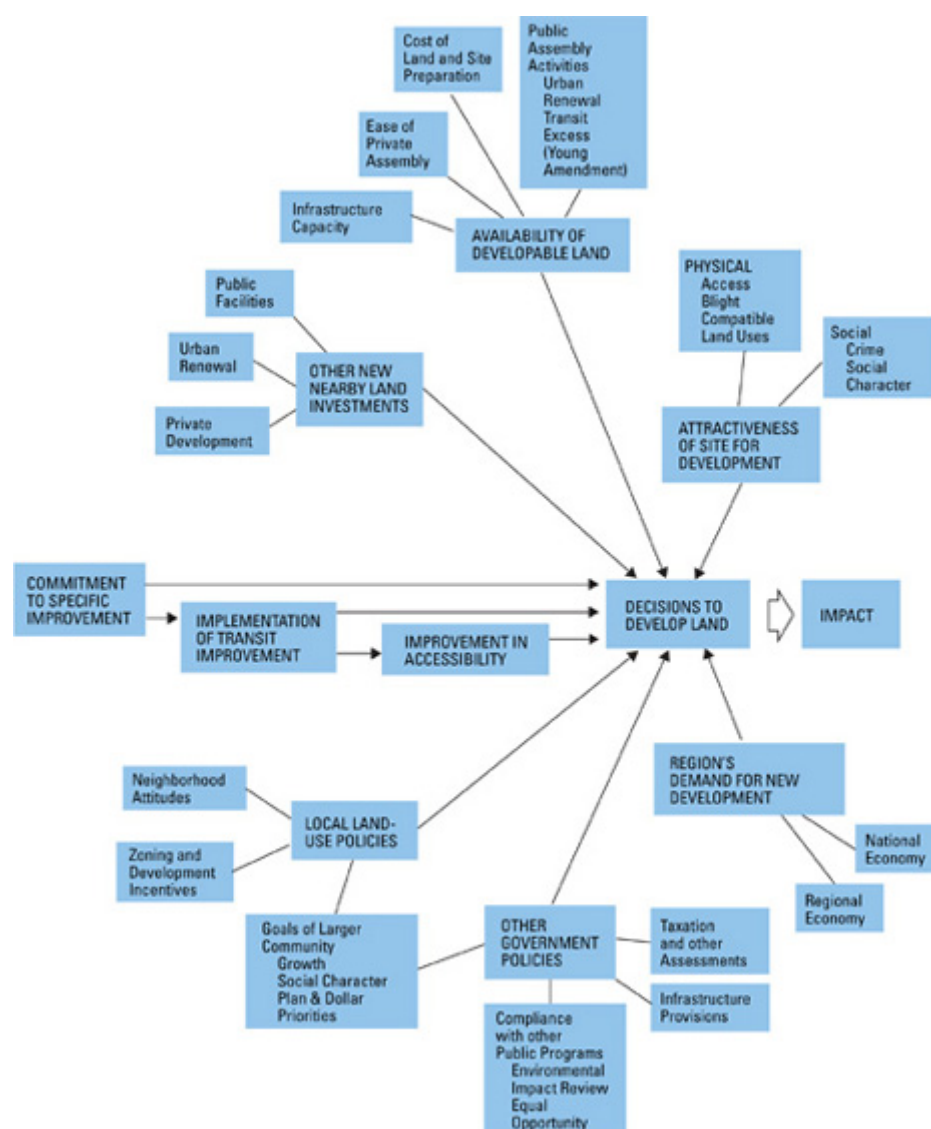
6. Combine higher densities with diverse land uses and pedestrian-friendly design. By themselves, higher densities are insufficient for promoting sustainable urban development. Experience shows that to promote sustainable travel and city form, higher densities must be complemented by diverse land uses, high-quality public places, and pedestrian-friendly street designs. Such physical interventions can shorten trips, encourage nonmotorized travel, keep many trips within the TOD neighborhood, and increase transit use.

7. Create a supportive environment to leverage TOD. Enhancing accessibility through urban transit investments is rarely sufficient to trigger significant land-use changes. A number of factors must be in place if significant land-use shifts are to occur as a result of newly opened transit stations ([figure 0.3](#)):

- the presence of other new nearby land developments, spurred perhaps by public sector urban renewal projects or private developers reacting to market opportunities
- the availability of developable land
- the existence of strong regional demand for new growth

- supportive government policies and measures, including tax incentives, infrastructure investments, and community involvement through participatory planning.

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Figure 0.3 Preconditions for successful integration of transit and land use

Source: Knight and Trygg 1977.

8. Exploit implementation tools. A host of regulatory and incentive-based tools for promoting transit and land-use integration is available, including the following:

- establishment of a redevelopment authority to convert brownfields and distressed urban districts, with the power to underwrite the cost of land development through such mechanisms as tax-increment financing and assistance with land assemblage/acquisition through eminent domain powers
- land consolidation and readjustment schemes to assemble land and finance local infrastructure improvements around major transit stations

- process-related tools, such as streamlining development reviews and fast-tracking the permitting of projects
- TOD and site and street guidelines that illustrate how TODs can contribute to an efficient arrangement of land uses and increase ridership.

Such implementation tools often require the passage of enabling legislation by higher levels of government before introduction by local authorities.

9. Rationalize mainline transit investments and feeder systems. High-quality transit services must be provided if TOD is to yield significant accessibility benefits. Such services are not only time competitive with private automobiles, they are also free of extreme overcrowding, they are reasonably on time, and they provide exceptionally high-quality feeder access. Global experiences point to the importance of providing well-integrated, seamless pedestrian connections between surrounding neighborhoods and transit stops. Particularly important are “green connectors,” which provide perpendicular bikeway and pedway linkages to transit stations and surrounding areas.

Promoting and implementing transit-oriented development

10. Create TOD typologies. Cities in developing countries need to build a typology of transit station environments and TODs, backed by realistic market assessments and illustrative design templates. A diverse portfolio of plans and layouts is more likely to reflect market realities and regulatory factors—such as responding to emerging consumer preferences and ensuring the availability of usable land—than a standardized, one-size-fits-all approach.

11. Create TOD prototypes. For cities with little history of TOD, or even transit and land-use integration, it is important that prototypes be introduced as a way to test the waters in both market and political terms. Each city presents a unique context for creating such prototypes.

12. Combine TOD with transportation demand management (TDM). TOD hardware—the physical design that promotes transit ridership and nonmotorized access—needs supportive software. When introduced with TOD, TDM measures such as congestion pricing and parking controls can yield synergies, such as increases in transit ridership from combining incentives for using public transit with disincentives for using automobiles.

13. Mainstream social development in TOD (make cities inclusive). TOD should not be just about creating economically efficient and environmentally friendly urban spaces. It should also help address

the most serious problem facing many cities in developing countries: crippling urban poverty and deprivation. Land markets typically respond by increasing land prices in and around transit stations, which often displaces low-income households. Where possible, city authorities should aggressively pursue affordable housing policies and developer incentives to ensure that affordable housing is built close to transit stops.

Financing scheme

14. Pursue sustainable finance through value capture. Vertical growth, rather than horizontal expansion, is vital to accommodating urbanization pressures and combating the negative impacts of automobile-dependent sprawl. Strategically targeted densities suitable for transit use can create a more compact city. However, higher densities require expanded or upgraded urban infrastructure, including transit services. Ahmedabad plans to construct a new metro connecting its downtown and the state capital, Gandhinagar. Bogota plans to construct a new metro system to increase passenger capacity in the extended central business district. As rail is more expensive than BRT, the biggest challenge for cities planning major transit upgrades is to secure adequate financial resources in light of limited budgets.

Value capture is the most promising fiscal tool available for generating new revenue. It can directly occur through land ownership or joint development, as the cases of Hong Kong SAR, China and Tokyo reveal. It can also occur indirectly, by extracting surplus from other property owners through the imposition of betterment taxes, the creation of benefit assessment districts, and the reaping of higher regular property tax proceeds, for example.

Successful value capture schemes depend, to a significant degree, on a supportive institutional environment. Particularly important is the formation of a proactive real estate development department within a transit agency, staffed by seasoned professionals with private sector experience who bring an entrepreneurial approach to land development. Transit company management should foster an organizational culture that embraces property development as a legitimate mission of the organization, as successful transit operations are not just about running trains and buses on time but also about shaping markets and ensuring high levels of future ridership through carefully placed development. Equally important to increasing urban densities and creating needed institutional capacity is ensuring a supportive legislative environment that empowers transit authorities to pursue joint development projects.

The most challenging issue in applying value capture in developing countries is the lack of transparency in property rights registration systems and the complex procedures associated with land

acquisition and transaction. Without straightforward, transparent, and efficient regulations and procedures for land registration and transactions, the successful application of value capture financing schemes is difficult, if not impossible.

Core Lessons on Transit Investments and Urban Growth

Seven lessons about using transit investments to reshape urban forms can be drawn from global experiences:

1. *The impacts of transit-related land use are greatest before an upswing in regional growth.*
Investment timing strongly influences whether significant land-use shifts will occur. For many rapidly developing cities, investing in transit during growth spurts can translate into appreciable land-use impacts, assuming the institutional capacity and wherewithal to leverage land development opportunities are present.
2. *Transit systems generally reinforce and often accelerate decentralization.* By improving accessibility to different parts of a region, extensive railway and BRT networks, like their highway counterparts, generally encourage suburbanization to some degree. However, transit-oriented growth can follow a much more sustainable pattern than the one fostered by highways.
3. *Proactive planning is necessary if decentralized growth is to take the form of subcenters.* An aggressive effort to leverage the benefits of mass transit services can lead to more concentrated forms of decentralized growth.
4. *Radial high-capacity transit systems such as BRT and metros help keep downtowns economically viable.* These systems lead to increased employment growth in urban centers, because such centers receive the largest incremental gains in regional accessibility. However, the regional shares of jobs and retailing that are downtown often still fall in the wake of new transit investments, because of the decentralization trends mentioned above. They would have fallen even more, however, were it not for central business district–focused railway services.
5. *Under the right conditions, railways and high-capacity busways can spur central city redevelopment.* When government agencies are willing to absorb some of the risks inherent in redeveloping depressed and economically stagnant neighborhoods, railways and high-capacity busways can help attract private capital and breathe new life into struggling areas. However, an unwavering public commitment is necessary to underwrite redevelopment costs and provide needed financial investments.

6. *Other pro-development measures must accompany railway and high-capacity busway investments*

. In addition to financial incentives, software policies are needed to make transit hardware attractive to land developers. These policies include various financial and tax incentives and process basis tools.

7. *Network effects matter.* For fixed-guideway railway and high-capacity busway systems to induce large-scale land-use changes, they must provide the geographic coverage and regional accessibility of their chief competitors, limited-access freeways and highways. The addition of exclusive-guideway services can create spillovers and synergies, benefiting not only the newly served corridors but existing ones as well.

Roles of Development Financial Institutions

Given that a denser, transit-oriented, built form results in more cost-effective transit investments, a sensible policy position would tie outside financial assistance to bona fide local efforts to improve coordination and integration. In addition to providing financial assistance for physical investments, financial institutions must help national governments address regulatory constraints and help cities develop their institutional capacities to design and implement prototype TODs and value capture schemes. This approach requires a hybrid lending instrument that combines policy and regulatory reforms, capacity development, and physical investment components.

Conclusion

This report seeks to build on the Eco² concept (which promotes sustainable urban development through cross-sector integration) by focusing on integrated transit and land-use development as a promising strategy for advancing environmental sustainability, economic development, and inclusive development. If done well, integrated spatial development—particularly the physical linkage of transit investments and urban development—can create positive and meaningful Eco² outcomes. Achieving such effective integration requires a cogent, forward-looking strategic vision; an enabling institutional framework; and sustainable financial models.

A point emphasized throughout this report is the importance of developing visions of urban futures. Experiences from Scandinavian cities like Copenhagen and Stockholm reveal that a clear and compelling regional vision is critical if high-capacity transit investments are to produce desired urban-form outcomes.

It is also imperative that there be a financial model to sustain TODs. Such a model can occur at two levels and in two directions. The macro, top-down perspective of sustainable financing is critical for cities competing regionally or globally. Pursuing a strategy of compact, mixed-use, high-quality development—whether TOD or smart growth in general—at the citywide or even metropolitan level can become an effective regional economic development tool, making city regions more economically competitive in the global marketplace. A high-quality, functional regional transit system that links high-quality urban centers creates the kind of built environment that is of increasing importance to amenity-conscious firms, employers, and employees. Blending good urban design with compact, mixed-use TOD becomes an effective tool for attracting outside investments, knowledge-based companies, and businesses, which increasingly place a premium on livability and quality of place. Cities that manage to create great environments around transit stations can enjoy a competitive edge in recruiting and retaining highly sought firms, which are instrumental in fostering regional economies. Experiences with orienting growth in transit-served corridors matched by high-quality urban design—in cities such as Hong Kong SAR, China; Seoul; Singapore; and Stockholm—reveal that good urbanism is wholly compatible with economic prosperity and competitiveness. Given that cities globally contribute 75 percent of gross domestic product (GDP), this macro perspective should also be of prime interest to national governments.

At the micro, bottom-up level, financial sustainability can take the form of value capture as a tool for generating revenue—funds that go to pay for not only the stations, tracks, and vehicles but also for all the upgrades and enhancements that are needed within a half-kilometer, five-minute walkshed of the station to create a high-quality environment for living, working, playing, learning, and interacting. The revenue could also finance social housing and associated facilities for low-income people. As experiences in Hong Kong SAR, China and Tokyo reveal, using some of the profit from selling land or development rights to the private sector to embellish public spaces and pedestrian environments around stations can lead to better projects that generate higher profits as well as higher property tax revenues. The resulting TOD is not only environmentally but also financially sustainable. A virtuous cycle is set in motion in which denser, high-quality TODs generate higher profits and fiscal revenue, some of which can go into creating additional high-quality TODs.

As new urban growth shifts to cities in the developing world, unprecedented opportunities will arise for linking land development and transit infrastructure. While motorization rates are subsiding in the developed world, they are growing exponentially elsewhere. The potential for payoffs from successful integration of transit and land use in the Nairobis, Jakartas, and Mexico Cities of the world is therefore enormous.

Given that the vast majority of future urban growth that is projected for cities will occur in cities with fewer than half a million people (UN Habitat 2011), a bus-based form of smaller-scale TOD interlaced with high-quality infrastructure for pedestrians and cyclists may be appropriate in many settings. Many cities in the developing world have the kinds of conditions needed if railway and BRT investments are to trigger meaningful land-use changes, including rapid growth, rising real incomes, and increased motorization and congestion levels.

Effective BRT requires supportive planning policies and zoning regulations, public sector leveraging and risk-sharing opportunities, and the capacity to manage the land-use shifts that are put in motion by transit infrastructure investments. Cities in developing countries are well positioned to learn from international experiences with transit-induced changes in land use. Few, however, have adopted the practices of cities like Hong Kong SAR, China; Tokyo; and Washington, DC in engaging private investors to co-finance capital investments through ancillary land development. Such value capture initiatives could go a long way toward putting cities in developing countries on more sustainable pathways—in terms of both facility financing and future patterns of urban development. Among all the tools available, TOD combined with value capture offers the most promise for putting the Eco² concept of sustainable growth and economic prosperity into practice.

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CHAPTER 1

Introduction: Critical Challenges Facing Cities and Urban Transport

Worldwide, the urban transportation arena finds itself at a critical crossroads, as the growing presence of private automobiles influences not only the way people move about cities but the shape and form of cities themselves. This chapter examines the challenges posed by urban sprawl in the wake of rapidly increasing automobile ownership and usage as well as associated problems, such as the economic productivity losses from worsening traffic congestion, environmental deterioration, and social inequity. It highlights the need to integrate transit¹ and land use as a critical step toward achieving sustainable urban development, building on the World Bank's Ecological Cities as Economic Cities (Eco²) model, which promotes sustainable urban development through cross-sector integration.

The role of transit in sustainable urban development is increasingly being recognized and promoted as a way to moderate climate change and increase the mobility of the poor. At the 2012 Rio+20 Conference, for example, international development banks, including the World Bank, announced a “game changer” commitment to sustainable transport and pledged substantial financial support over the next decade for this purpose (WRI 2012).²

This study explores the complex process of transit and land-use integration in rapidly growing cities in developing countries. It first identifies barriers to and opportunities for effective coordination of the transport infrastructure and urban development. It then recommends a set of policies and implementation measures for overcoming these barriers and exploiting these opportunities.

The study uses the comparative case study method to identify and learn from global best-case practices in promoting transit-oriented development (TOD)—compact, mixed-use, pedestrian-friendly development organized around a transit station—on a metropolitan scale. It analyzes the market-driven and policy-shaped interactions between bus rapid transit (BRT) systems (box 1.1) and urban development in four cities in developing countries. It pays particular attention to the impacts of public interventions—such as transit investments, planning regulations, tax policies, and financial incentives—on the ability to successfully integrate transit and urban development, all within the framework of the institutional and regulatory realities and constraints.

Box 1.1 What is bus rapid transit?

Bus rapid transit (BRT) is the name given to sophisticated bus systems that have their own lanes on city streets. These systems use bus stations instead of bus stops, a design feature that allows passengers to pay before boarding the bus. Bus stations allow for faster, more orderly boarding similar to the procedures used on metro and light rail systems. Stations have elevated boarding platforms level with the bus floors, so that passengers do not need to climb steps to get on the bus. Electronic signage tells passengers when the next bus is arriving. BRT is faster, safer, more efficient, and more user-friendly than traditional bus systems.

Many cities are now choosing BRT, for two important reasons: cost and convenience. Building a heavy rail system can cost 10 times as much as building a BRT system. Light rail, which is common throughout Europe, is less expensive than heavy rail, but it can still cost more than four times as much as BRT. What's more, cities that opt for BRT can see results very quickly, as installing a system can take just two years. By contrast, building an underground metro can drag on for a decade.

Figure B1.1.1 Bus rapid transit in Curitiba, Brazil



Source: Photo of Curitiba, Brazil from EMBARQ website (<http://www.embarq.org>).

Expanding Cities, Shrinking Green Space

Cities in developing countries are growing at an unprecedented rate. Over the next several decades, most of the world's urban population growth is projected to take place in developing countries, whose populations are expected to increase from 2.0 billion in 2000 to 5.5 billion by 2050. The number of urban inhabitants in developed countries, by contrast, is expected to stabilize at about 1 billion. Angel and others (2011) note that if developing countries continue to experience annual population growth rates of 2.5 percent and built-up densities continue to decline by 1.5 percent a year, their land coverage—often referred to as “sprawl”—will expand by 4.0 percent a year. At this rate, the world's cumulative area of built-up, impervious surfaces will double in 17 years and triple in 27 years. The long-term ecological consequences of converting land from natural habitats and open space to urban functions—diminished water supplies, the release of more pollutants into the air, heat-island effects, and lost agricultural land, for example—could be devastating.

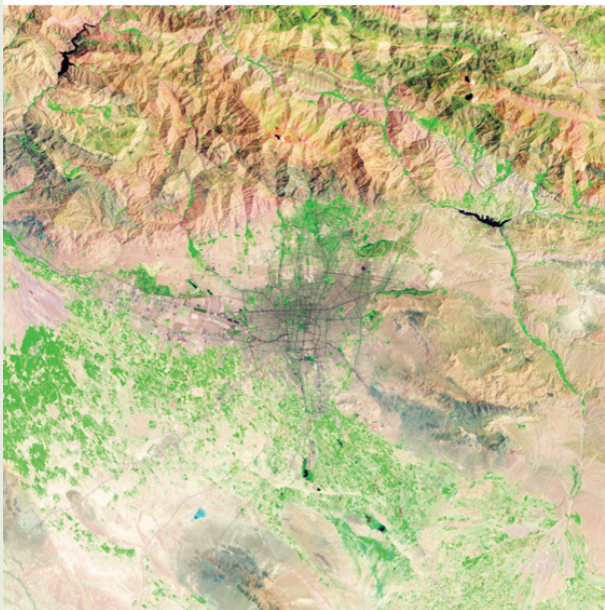
Urban population growth and spatial expansion are expected to accelerate in the coming years, as migration from the countryside to cities continues (boxes 1.2 and 1.3). Cities have always attracted people from agrarian settings seeking better living, educational, and employment opportunities. At the same time, fast-growing cities create jobs that further accelerate their economic growth. Urbanization then further intensifies, as people and growth spill out from city boundaries, sometimes in informal settlements that offer the only means of shelter to unskilled, poorly educated rural migrants.

Box 1.2 Urban growth in Tehran

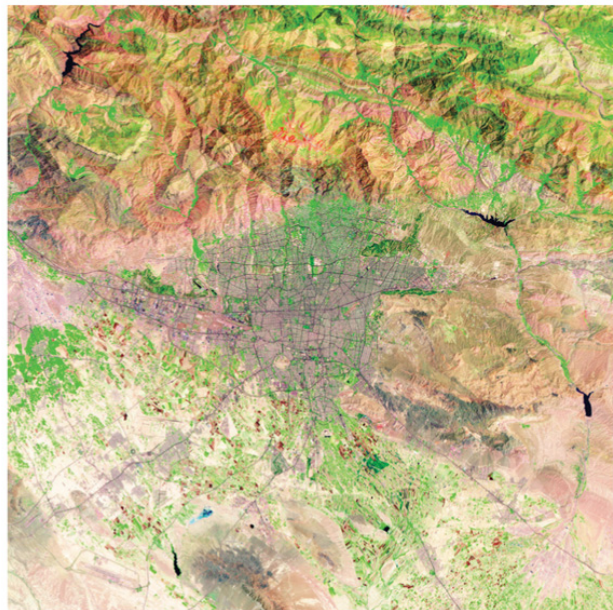
Tehran is one of the world's fastest-growing cities, with a population that increased from 6 million in 1985 to more than 10 million in 2009 (NASA Earth Observatory 2009). Color images of Tehran's expansion reveal that built-up areas have extended along roads along the city's outskirts (figure B1.2.1). To manage population growth and development, Tehran is currently upgrading and building new transport systems, including urban railways and BRT lines.

Figure B1.2.1 Aerial views of metropolitan Tehran, 1986 and 2009

a. 1986



b. 2009

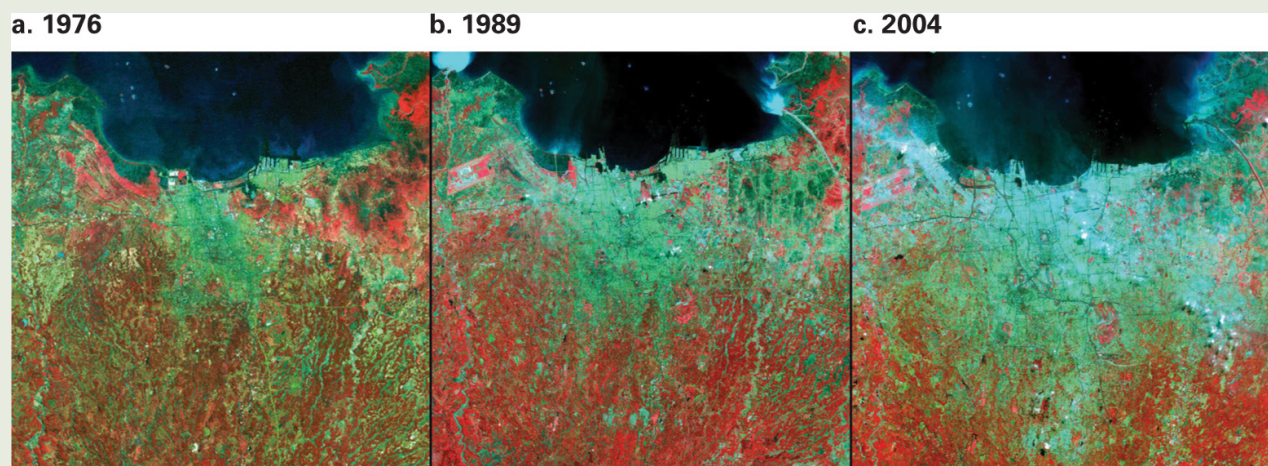


Source: NASA Earth Observatory 2009.

Box 1.3 Urban expansion in the Jakarta metropolitan area

The population of Indonesia's capital city more than doubled in 30 years, rising from 6 million in the mid-1970s to 13 million in the early 2000s. In 1976, developed areas were mainly along Jakarta's northern coastal line (shown in light green in [figure B1.3.1](#)). By 1989, the city's urbanized area had expanded toward the eastern side of the metropolis. By 2004, Jakarta's land coverage extended southward, overtaking large amounts of land with vegetation and natural habitats (shown in red).

Figure B1.3.1 Aerial views of metropolitan Jakarta, 1976–2004



Source: NASA Earth Observatory 2005.

The images also reveal the transformation of Jakarta's built environment, marked not only by more land coverage but also by higher densities in general ([figure B1.3.2](#)). Many of the city's high-rise structures have been and continue to be built in an ad hoc, piecemeal manner, without a guiding long-range vision of physical form or institutional coordination.

Figure B1.3.2 Density of metropolitan Jakarta, 1988 and 2011

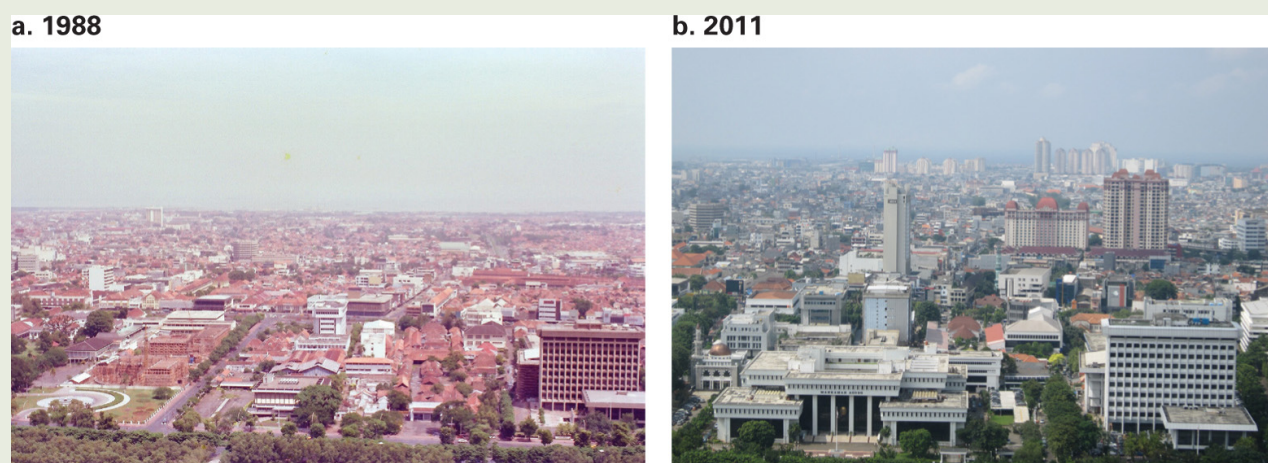


Photo by Sadao Orishimo.

The current speed of urbanization is extremely rapid. “Globally, towns and cities with populations equivalent to Malmö, Sweden—about 200,000 inhabitants—are built every day, 365 days a year” (Reuterswärd 2009).

Cities in the developing world are poised to play an increasingly central role in the global economy. They often provide relatively low-cost labor, less-stringent regulatory constraints, and a newly emerging consumer base. As they actively compete head-to-head regionally and globally, cities will continue to strive for agglomeration economies—that is, the economic benefits conferred by ease of access and face-to-face contact among specialized businesses and labor. Such market forces spawn economic clusters and concentrations within cities.

With rising incomes and population increases, cities will also expand spatially—more specifically, outward. The correlation between economic wealth and urban spatial coverage is strong and positive: big cities generally average higher gross domestic product (GDP) per capita than smaller ones.

Of concern is the fact that many rapidly growing cities around the world are poorly planned, managed, and governed. The absence of a long-term vision and spatial plans for where and how to develop, as well as a history of informal settlements or limited coordination among key institutions and stakeholders, undermine the ability to effectively and sustainably manage urban growth. When urban growth is poorly managed, cities and their inhabitants suffer the inescapable consequences: degraded urban environments, lost economic productivity, and widening income disparities and inequities.

To make matters worse, urbanization is outpacing the growth in municipal revenues, squeezing cities financially. As a consequence, cities in developing countries struggle to provide basic services like clean water, sanitation, and primary education, much less modern infrastructure like thoroughfares or high-capacity transit.

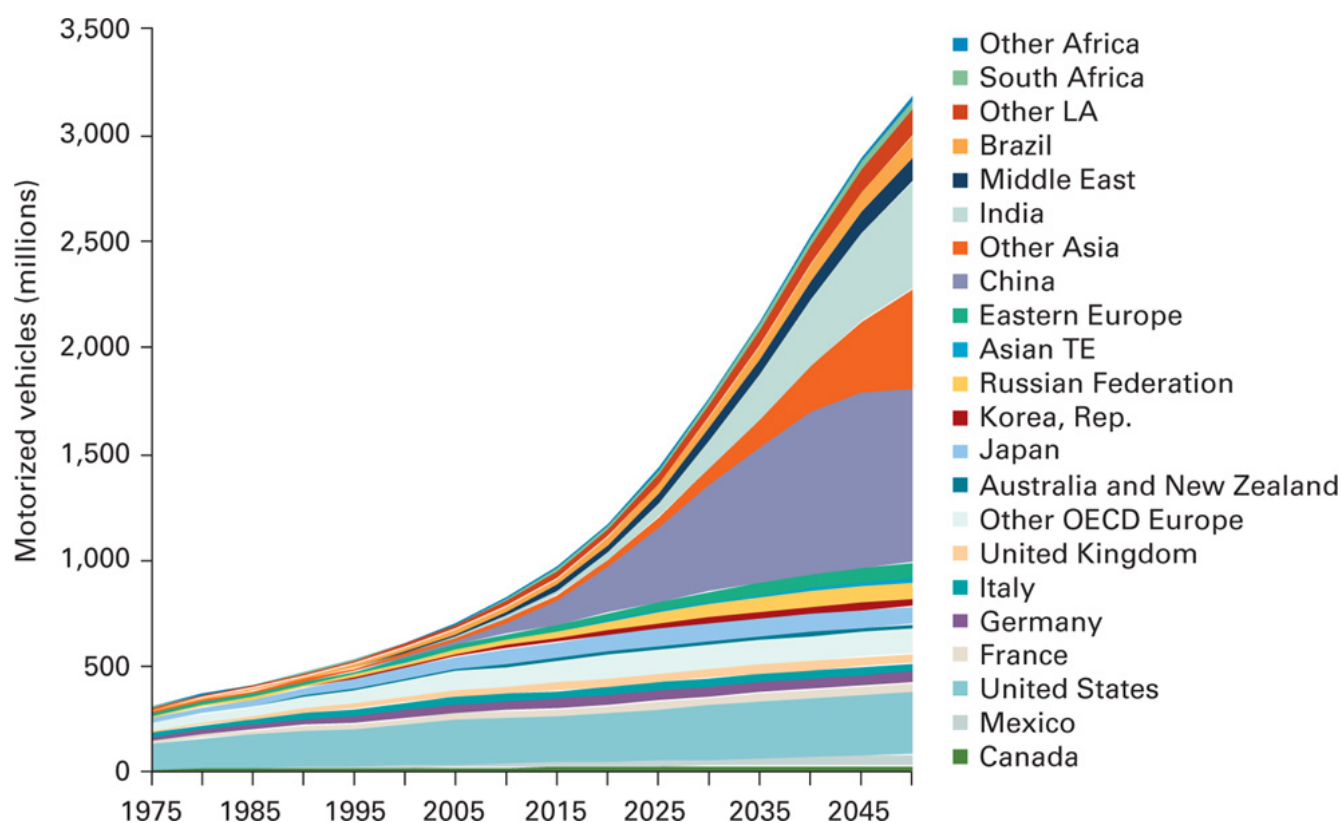
Cities for Cars, Not for People

One of the most critical challenges facing cities worldwide is automobile-dependent urban development.³ With economic growth, many cities in developing countries have begun to follow the trajectory of motorization that developed countries once followed—but at a much faster rate. Worldwide, rising incomes are fueling automobile ownership, spurred by increasingly affordable vehicles. Many urban inhabitants—often the middle-income population—are shifting from public or nonmotorized transportation to private automobiles. In addition, higher incomes allow people to afford more spacious houses on large suburban lots, using their own cars to reach jobs and educational

opportunities in the city. An automobile-oriented built form—marked by spread-out development, noncontiguous land uses, large city blocks that are unfriendly for pedestrians, and strip development—is an inevitable consequence of an automobile-dependent lifestyle.

Of particular concern is the fact that motorization rates are increasingly faster in the developing world. With rising incomes and modernization come more cars, trucks, and motorcycles. Motorcycles are viewed as stepping stones to full-blown car ownership; they also serve as an indicator of changes in income levels. The number of motorcycles in Bangkok increased at an average annual rate of 34 percent in the 1990s (World Bank 1996). In many parts of China and India, motorization growth rates in cities today are believed to exceed 20 percent a year. These increases are, in part, supply driven. In China, for example, mainstreaming of the automotive industry and government investments in superhighways have contributed to the exponential increase in the numbers of motorized vehicles. By 2050, China is projected to have 900 million motor vehicles—more than the total number of cars, trucks, and motorcycles in the world today (figure 1.1). At the current rate of motorization, 2.3 billion automobiles will be added worldwide between 2005 and 2050, and more than 80 percent of them will end up on the roads and highways of developing countries (Chamon, Mauro, and Okawa 2008).

Figure 1.1 Actual and projected number of motorized vehicles in the world, 1975–2050



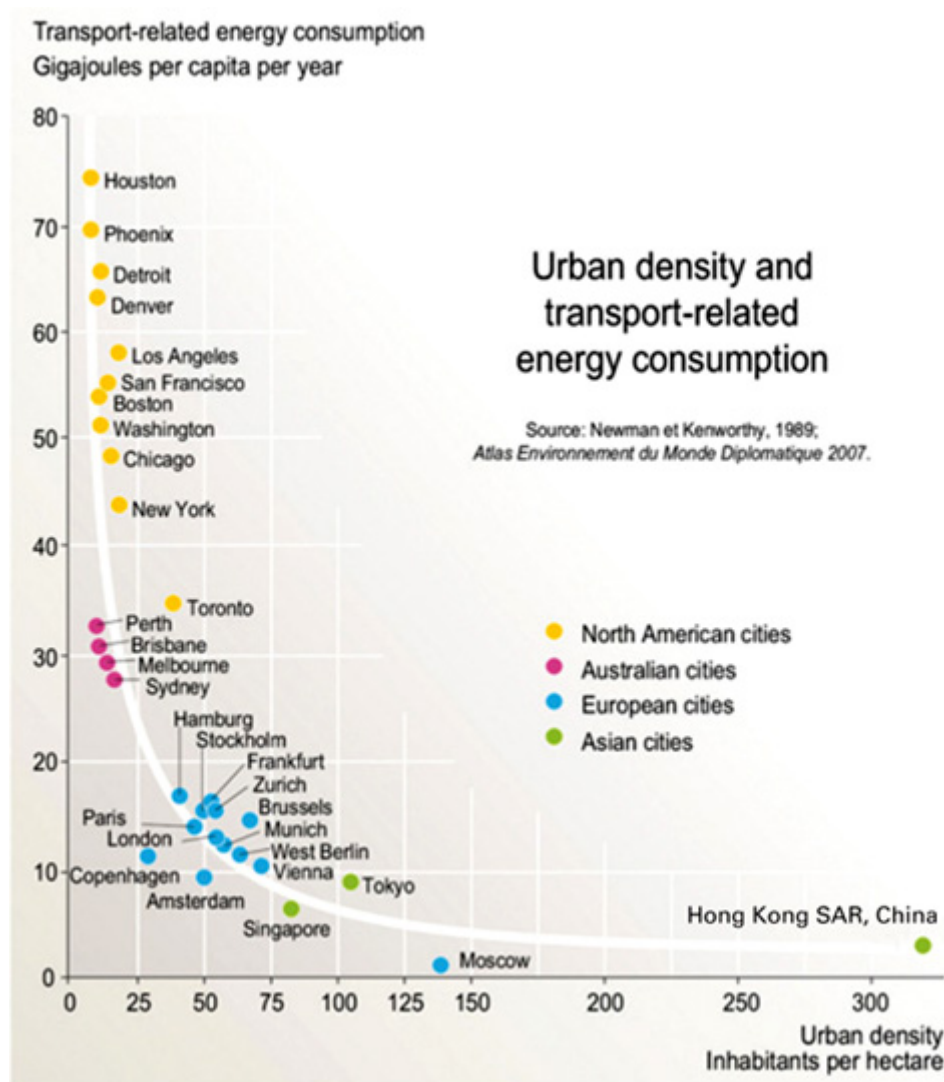
Source: Fulton and Cazzola 2008.

Rapid motorization has contributed to deteriorating traffic conditions worldwide, and congestion levels are worsening day by day. A 2010 study by IBM examined “global commuter pain” by polling 8,192 motorists in 20 global cities across 6 continents. According to the study, traffic congestion levels rose markedly between 2007 and 2010, despite the worldwide recession. With a 24 percent annual growth rate in registered vehicles, traffic conditions deteriorated the most in Beijing over this period.

In a follow-up survey in 2011, almost half of respondents reported that traffic had continued to worsen. Motorists in Moscow and Nairobi reported the worst commutes, with 45 percent of respondents in Moscow and 35 percent in Nairobi claiming they spent three or more hours a day stuck in traffic (IBM 2011). The survey found numerous side effects of traffic, including stress, road rage, respiratory ailments from dirty air, and bodily injuries and deaths from traffic accidents. Traffic congestion also has real and enduring economic consequences: over the course of a month, traffic caused about half of survey respondents to cancel a planned trip for work, shopping, recreation, entertainment, or dining out. Delays in the movement of people, goods, freight, and raw materials in and out of cities become a drag on economic productivity, reducing a city’s competitiveness. One study estimates the economic loss from congestion-induced delays in Bangkok during the 1990s at more than \$4 million a day (World Bank 1996).

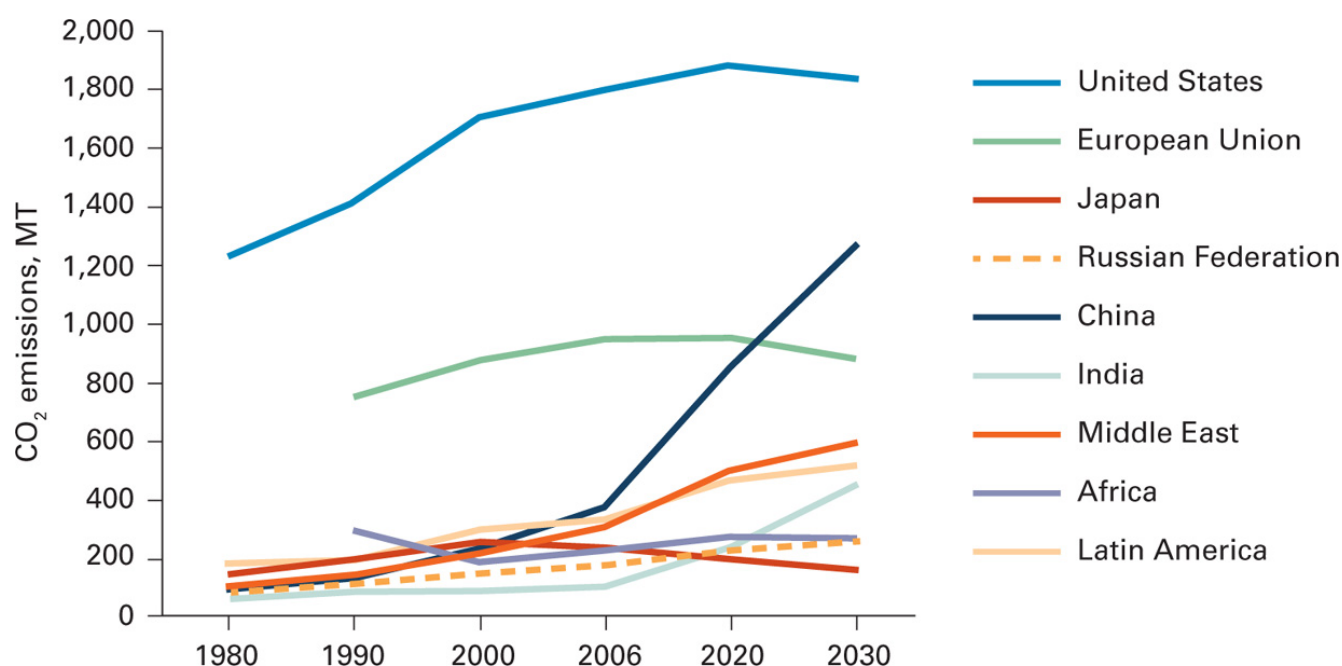
Other negative consequences of automobile-dependent growth include excessive energy consumption and pollution. There is a strong association between urban density and energy consumption. One study finds that low-density cities made possible by high levels of car ownership average considerably higher energy consumption per capita than high-density cities (Newman and Kenworthy 1989) ([figure 1.2](#)). The difference partly reflects the fact that residents depend largely on private cars for travel in cities where fewer public transit services are available. In addition, air pollution worsens and the number of traffic accidents rises when car ownership rates increase. Pollutants emitted through the tailpipes of cars, trucks, and buses—including lead, ozone, and suspended particulates—can trigger debilitating upper-respiratory illnesses; over the long run, they can damage body organs from prolonged exposure. Carbon dioxide (CO₂) emissions from cars and trucks are increasingly worrisome, especially at the global scale, because their environmental effects threaten to destabilize climates, leading to rising sea levels, flooding, extreme heat waves, and droughts.

Figure 1.2 Urban density and transport-related energy consumption in selected cities



Source: Bournay 2008, based on Newman and Kenworthy 1989, and *Atlas Environnement du Monde Diplomatique 2007*; used with permission.

The transport sector is a major contributor to global climate change. It is a significant source of CO₂ emissions, generating about one-fourth of total emissions, with 18 percent coming from road transport (UNEP 2010). CO₂ emissions from the road sector continue to increase, with their share expected to reach one-third of the total by 2050. Over the next two decades, emissions from emerging markets, particularly China and India, are expected to grow the fastest (figure 1.3).

Figure 1.3 Actual and projected global CO₂ emissions from the transport sector, 1980–2030

Source: Fulton and Cazzola 2008.

Note: CO₂ = carbon dioxide; MT = metric ton.

The frequency of traffic accidents, which has decreased in developed countries, has increased in developing countries: the number of road fatalities per 10,000 inhabitants is often four times higher in developing countries than in developed ones (World Bank 1996). In Africa, the number of road fatalities increased 26 percent between 1987 and 1995 (Jacobs, Aeron-Thomas, and Astrop 2000).

Social aspects are also important considerations. Low-income populations in cities are the most vulnerable in automobile-dependent cities, for several reasons. First, automobile-dependent urban forms provide limited choices on where to live, often forcing people who cannot afford a car to live in informal settlements on the urban fringe. As a consequence, they often have limited access to basic urban services, jobs, and educational opportunities.

Second, even where transit services are available, the cost is often substantial. One study found that poor families devote as much as 20 percent of total household income to transportation, compared with just 5 percent for wealthier households (Gomez-Ibanez and Meyer 1993). Money spent on transit is money that cannot go to meeting other pressing needs, such as food, shelter, health care, and education.

Third, in automobile-dependent cities, the majority of public investment goes to automobile-related facilities, such as roads and parking, taking money away from the upgrading of public transit or nonmotorized transport facilities, such as footpaths, bike lanes, and bike parking. Low-income

residents benefit disproportionately from investments in public transit and bikeways, because they are much more reliant on these relatively low-cost modes than other income groups. Directing scarce national, regional, and municipal revenues to roads and highways does little to improve the livelihoods of people who do not own cars.

Fourth, “time pollution”—the amount of time wasted in trying to complete essential tasks, because of congestion or the need to travel long distances—often affects low-income people the most. Time spent traversing long distances in the city means less time available at the intended destination—working, studying, shopping, or socializing. Studies show that the poor spend much more time commuting than other income groups, often traveling from informal settlements on the urban periphery to city centers, where formal and informal employment opportunities exist. A study of Mexico City in the 1990s, for example, finds that 20 percent of the underprivileged respondents surveyed spent more than three hours a day commuting to work, and 10 percent spent more than five hours (Schwela and Zali 1999). When commuting long distances, they needed to make multiple transfers on privately operated transportation and other informal carriers (Cervero 2000).

Fifth, low-income residents are highly vulnerable to the health effects of airborne pollutants and traffic accidents. Walking, cycling, or using other micro paratransit vehicles to get around, they face exposure to polluted air and aggressive motor vehicles every day. In Bangladesh and India, more than three-quarters of deaths on roads are pedestrians and people transported by pedal power (World Bank 1996). In Manila, almost half of medication used by residents is for illnesses related to air pollution (Romualdez 2012).

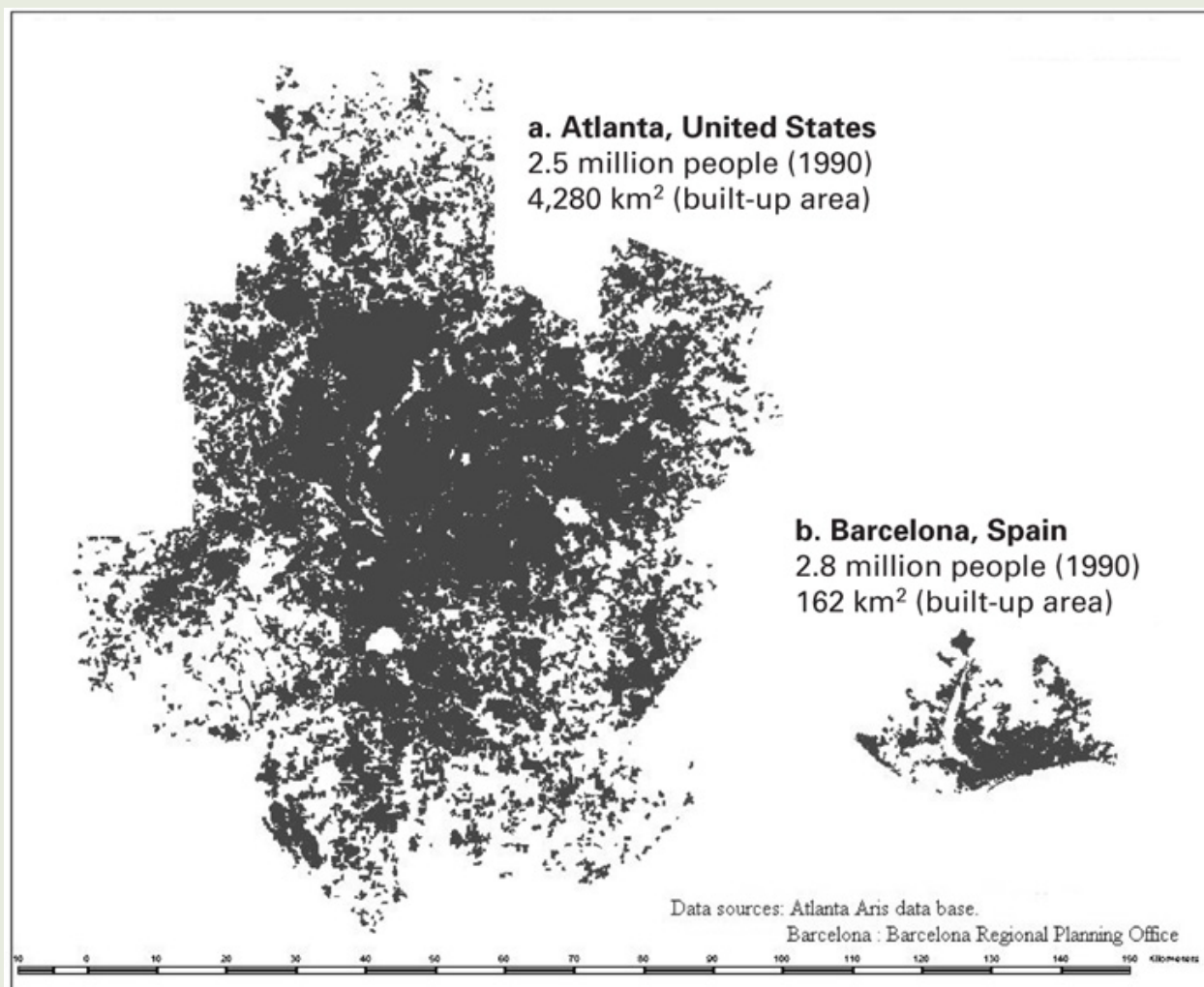
Integrating Transit and Land Use toward Sustainable Urban Development

Reversing automobile dependency is not easy. Yet history shows that transit investments, if well integrated with urban development, promote sustainable urban forms with promising outcomes. Atlanta, a sprawled low-density city, and Barcelona, a compact high-density city, show dramatic differences in their private car use, energy consumption, and environmental footprints ([box 1.4](#)).

Box 1.4 A tale of two cities: Atlanta and Barcelona

Guiding development to reduce low-density expansion is a key feature of sustainable urbanization. [Figure B1.4.1](#) reveals huge differences in the environmental footprints of two global cities, Atlanta and Barcelona. In 1990, the two cities had similar size populations of about 2.5 million people. Atlanta, however, had more than 26 times the built-up land area of Barcelona.

Figure B1.4.1 Built-up area of Atlanta and Barcelona, 1990



Source: Bertaud 2003.

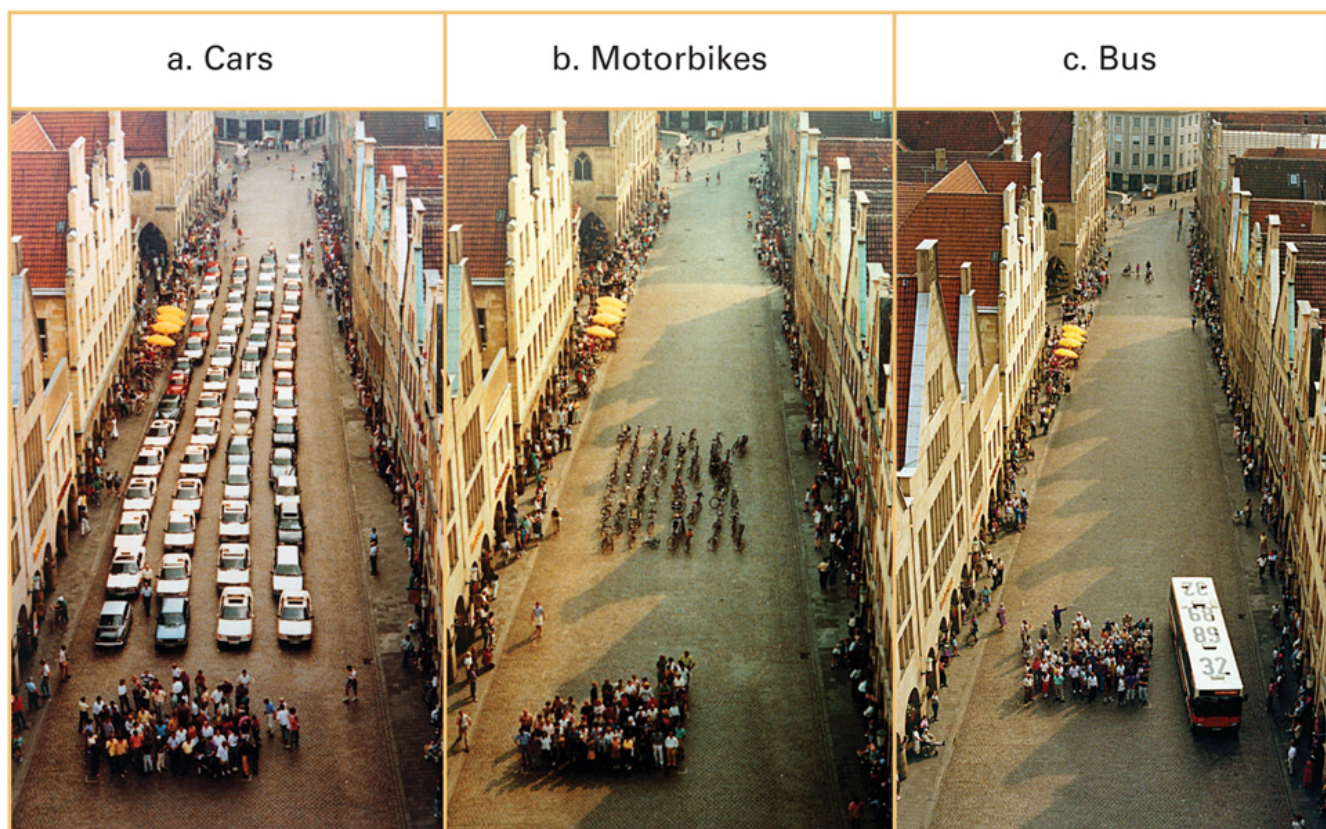
Having less paved, built-up areas has allowed Barcelona to preserve green spaces. Less sprawling development reduces the cost of providing and extending urban infrastructure investments; it also makes transportation operation and maintenance, water and sanitation, and energy distribution systems less expensive.

Barcelona's energy use is considerably lower than Atlanta's. Because vehicle kilometers traveled per capita are much lower in Barcelona than Atlanta, far less fuel is consumed in the transport sector and CO₂ emissions are considerably lower. Additional benefits of compact urban form may include economic productivity gains from urban agglomeration and increased opportunities to instill a sense of community through increased social interactions.

Reversing the Culture of Automobile-Oriented Development

Two key initiatives for reversing automobile dependency include (a) shifting travel from private motorized vehicles to nonmotorized and public transport, and (b) reducing sprawl by promoting TODs that are served by high-quality public transit services. Public transit is inherently the most resourceful form of mobility, measured in consumption of energy or land, as illustrated by [figure 1.4](#), which compares the amount of road space required for about 80 travelers by different modes of transportation (GTZ 2004). Reducing the amount of impervious surfaces given over to roads and parking not only shortens trip distances, it also reduces heat-island effects and water pollution (from oil-stained run-off of rain into streams).

Figure 1.4 Urban space taken up by cars, motorbikes, and buses



Source: GTZ 2004.

Cities designed to reduce travel distances encourage walking, cycling, and use of the public transit system. “Cities of short distances” are also cities with lower levels of air pollution, energy consumption, and CO₂ emissions. In the long term, efficient urban form makes cities more economically competitive and environmentally sustainable. It also helps build social capital by allowing people from all walks of life to come into regular, day-to-day contact with one another. Denser cities are also more socially just. By providing high levels of access to everyone, regardless of income, they enable the less-privileged segments of society to achieve upward mobility.

One practical approach to reducing traffic congestion, greenhouse gas emissions, and other pressing environmental problems is to promote TOD (Calthorpe 1993; Cervero and others 2005; Curtis, Renne, and Bertolini 2009) ([box 1.5](#)). Experiences show that well-designed TOD not only increases ridership of public transit by drawing more travelers out of cars and into trains and buses, it can also serve as a hub for organizing community development and revitalizing long-distressed urban districts (Bernick and Cervero 1997; Cervero 1998).

Box 1.5 What is transit-oriented development?

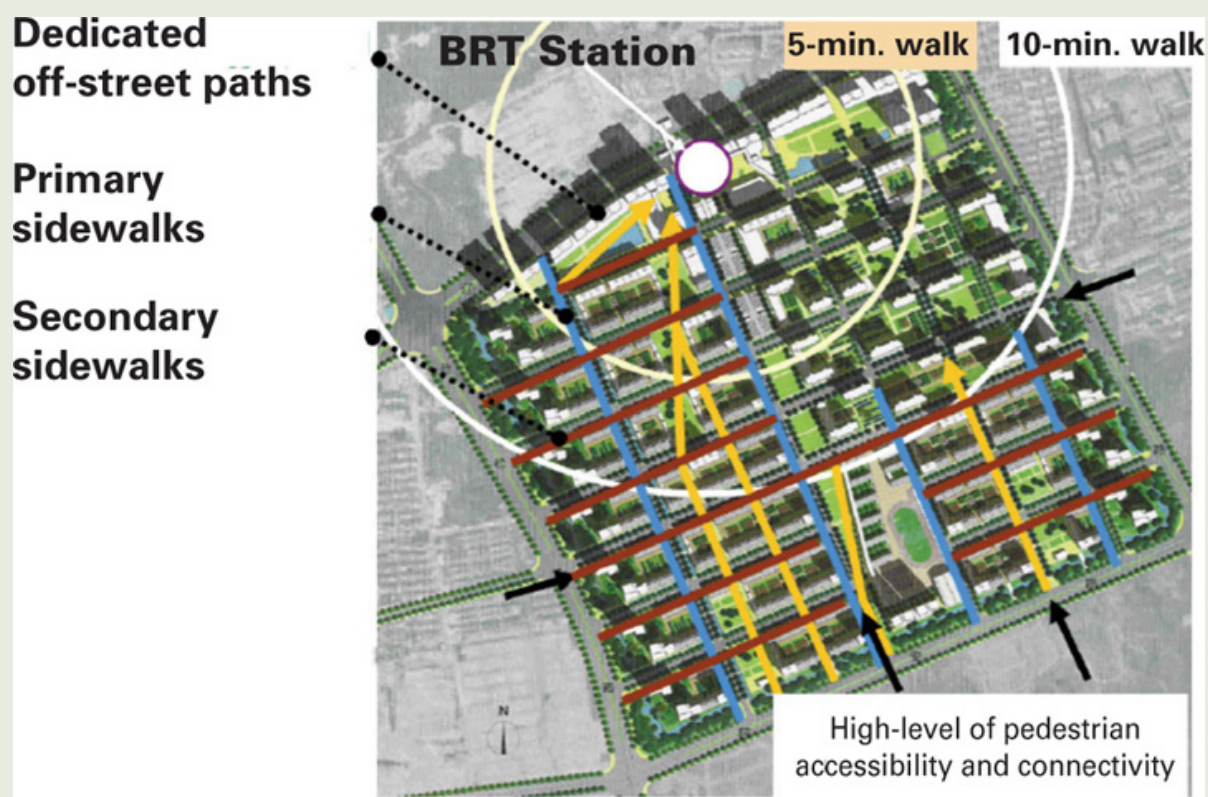
Transit-oriented development (TOD) is characterized by two main features:

- proximity to and a functional relationship with transit stations and terminals and service provision by high-quality public transit (BRT systems, underground trains, and so forth)
- compact, mixed-use buildings and neighborhoods that, because of their design, encourage walking, cycling, and use of public transit by residents, employees, shoppers, and visitors.

The ingredients of successful TOD include strategic (macro) and design (micro) elements such as a strong development climate and master plans for multiuse, high-intensity developments supported by implementation plans. They also include investments that promote the following:

- easy and direct pedestrian, bicycle, and public transit access ([figure B1.5.1](#))
- good signage and a pleasant environment to attract substantial pedestrian flows
- significant regional accessibility to major job and activity centers
- short, direct connections between transportation modes and transit facilities
- bicycle lanes and parking facilities that feed stations
- attractive facilities that are well integrated with the surroundings (public spaces, street furniture, and so forth)
- safe and secure designs, including adequate lighting
- effective parking management around stations
- environmentally friendly technology options, such as shared fleets of alternative (electric) vehicles located in neighborhoods.

Figure B1.5.1 Key features of the eco-block concept



Source: Fraker 2009.

Note: The eco-block concept maximizes pedestrian access to transit stations. It is illustrated here using a location in China.

Research shows that the impacts of TOD are realized over the long term and depend on the quality of related microdesigns and the rate of an area's demographic and economic growth. Lund, Cervero, and Willson (2004) studied residential and commercial sites in major cities in California. They found that factors related to TOD, particularly proximity to urban and commuter rail stations, increased ridership on rail and buses by as much as a factor of three to four relative to control sites. The ridership benefits of TOD are also found in rapidly urbanizing settings. Cervero and Day (2008a, 2008b) surveyed households relocating to suburban TOD and non-TOD sites in Shanghai to assess the impacts on travel behavior. They found that TOD sites experienced increased public transit ridership, improved access to regional jobs (as measured by employment locations within a radius equivalent to one hour in travel time), and reduced commuter times per household worker.

Figure B1.5.2 Transit-oriented development in metropolitan Washington, DC

a. Connection between metrorail and feeder bus system



b. Road and metrorail connection at Clarendon, Virginia



Source: Zimmerman 2008.

Note: The photo in panel a shows a high-quality public transit corridor between Arlington, Virginia, and Washington, DC, with an underground metropolitan train and a feeder bus system. The corridor exhibits many elements of good macro-level planning and TOD, including higher densities around high-quality public transit (the metro) in an otherwise car-oriented environment. After 20 years of mixed-use development around stations (such as Clarendon, Virginia, pictured in panel b), the corridor has become transit supportive in its form and design.

Mixed land use is a particularly important element of TOD. Intermixing housing, offices, retail shops, and other urban amenities in close proximity to public transit stations integrates long-distance travel by transit and short-distance, within-neighborhood travel by foot.

TOD concepts should be carefully adapted to reflect the local urban context, social and cultural considerations, and market realities. A one-size-fits-all approach to TOD planning and design is not recommended, particularly for rapidly growing cities in developing countries. Instead, cities should develop typologies of transit station environments and TODs that correspond to local conditions.

When coupled with transportation demand management (TDM) measures, TOD can become even more effective. TDM aims to alter the demand for travel in order to reduce traffic congestion and improve environmental conditions. It includes measures that modify the physical designs of cities and

streetscapes as well as various incentive mechanisms that change the cost of transportation to users. Physical design examples include pedestrian- and bicycle-friendly street designs, traffic calming, and intermixing of land uses to shorten trip distances. Examples of TDM include congestion pricing, parking controls, and car sharing. Experience shows that combining TOD and TDM measures can induce people to use public transit and nonmotorized transportation and to reduce levels of car ownership and usage (Guo and others 2011).

In general, cities take on a polycentric form as they grow. Connecting high-density centers—be they central business districts, residential areas, or mixed-use subcenters—by high-quality transit is critically important to sustainable urban development. Interlinking mixed-use urban centers with high-capacity transit can create two- or multiple-way travel flows, allowing travel demand to be more evenly spread out throughout the day.

Another way to materially enhance transit services is to design a superior system of feeder services that allows smooth and seamless transfers and connections among different transport modes. Trunk-line services, such as metros, provide economies of scale; smaller vehicle feeder services, such as minibuses, provide economies of scope. The two modes can be highly complementary. A well-integrated metropolitan-wide public transport system that links regional railways, metros, BRT systems, and local buses is essential if rising automobile usage and ownership is to be reversed. Physically linking nonmotorized transportation with public transit—through dedicated bike lanes that funnel into public transit stations, for example—is another essential part of a highly interconnected and viable transportation network that offers mobility options to the private car.

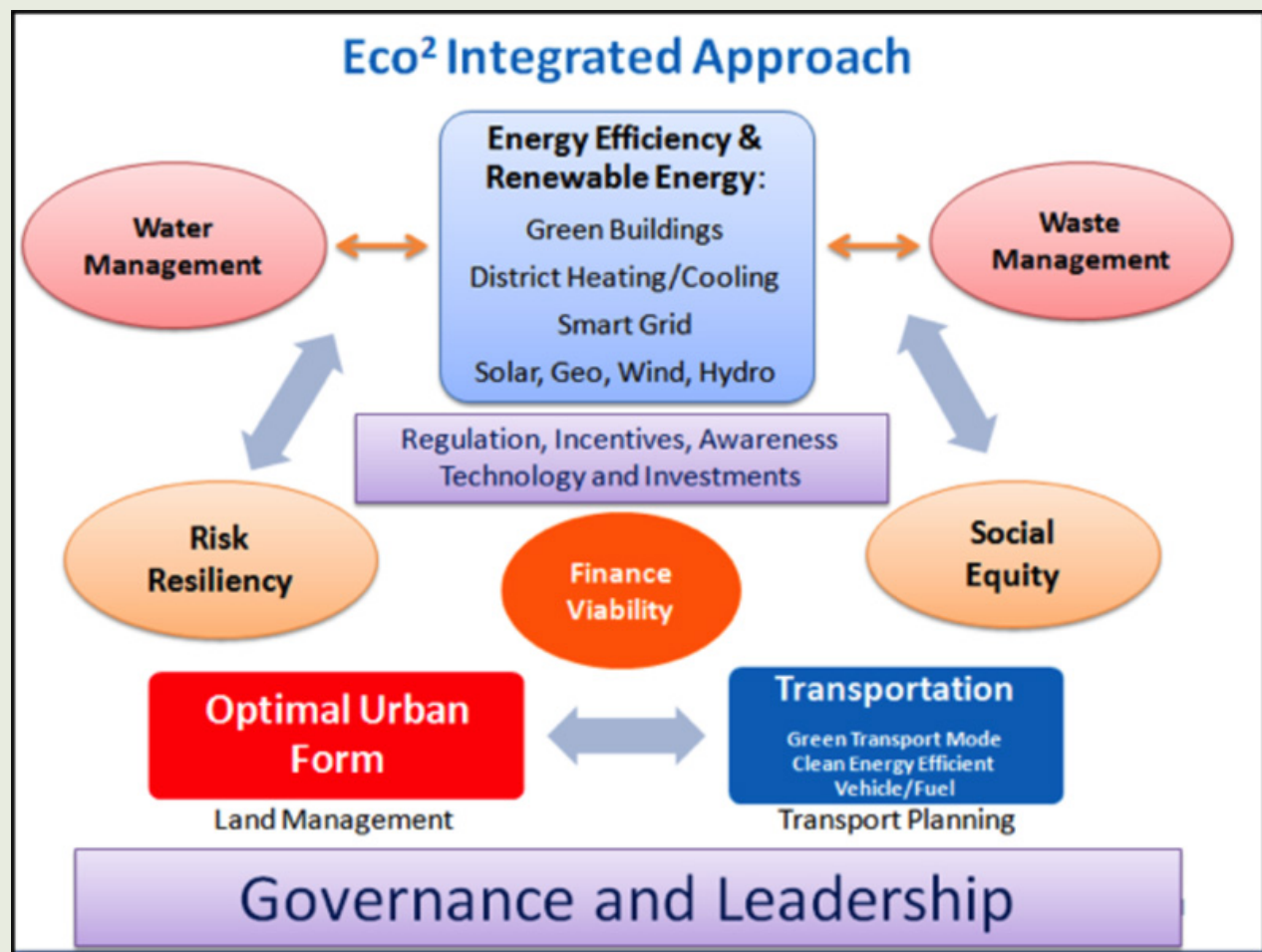
Integration: Perceived Consensus and Neglected Reality

Interest in the role of transport investments and policies in fostering efficient urban growth has received considerable attention in recent years. In 2010, the World Bank launched an initiative called “Eco² Cities: Ecological Cities as Economic Cities” (Suzuki and others 2010). The initiative aims to help cities in developing countries achieve economic and environmental sustainability and enhance social equity. The “one-system approach,” which promotes cross-sector integration in spatial development, is one of the four principles of Eco² (box 1.6). Integrating transit and land use is one of the most promising strategies for achieving urban sustainability.

Box 1.6 Economic cities as ecological cities: The World Bank’s Eco² initiative

Eco² Cities is a World Bank initiative that helps cities in developing countries achieve greater ecological and economic sustainability, while increasing human and social well-being. It provides a framework to plan, manage, and invest in sustainable urban systems that are integrated and beneficial in the long term. Four principles underlie the Eco² framework:

- Principle 1, a city-based approach, focuses on the need to enable and strengthen the leadership, capacity, and decision-making abilities of cities and their regional planning institutions. It also emphasizes the need to enhance the unique historic, cultural, and ecological resources of each city.
- Principle 2, a platform for collaborative design and decision making, focuses on compounding the benefits of urbanization by leveraging and combining the unique capacities and resources of all stakeholders. It supports an inclusive and fair process of urban development and decision making that involves and empowers all stakeholders.
- Principle 3, a one-system approach, strives to create a “resource regenerative and multifunctional” city. Sectors, policies, and budgets—as well as natural and man-made systems—need to work together across spatial scales and administrative jurisdictions so that the city functions effectively as one system.
- Principle 4, an investment framework that values sustainability and resilience, focuses on broadening the scope and extending the timeframe within which policies, plans, and investment options are assessed for costs, benefits, and risks. It supports decision-making approaches that value natural, cultural, and social capital.

Figure B1.6.1 The Eco² integrated approach to development

Source: Suzuki and others 2010.

Note: Available online at www.worldbank.org/eco2.

City leaders and policy makers increasingly recognize that cities that enjoy high levels of mobility and accessibility are also economically more productive, vibrant places to live and attractive places to run a business. They also have a smaller environmental footprint and score high on the quality-of-life index. Thoughtful leaders realize that careful integration of transit with urban development is one of the best ways to place cities on a sustainable pathway.

History shows that transit investments powerfully shape the growth of cities by changing accessibility (the ease of traveling from one place to another) (Cervero and Seskin 1995; Giuliano 2004). In the absence of regulatory restrictions in a growing environment, investment in high-capacity, high-quality transit services unleashes land-use shifts and urban form adjustments by yielding accessibility benefits. Because there is a limited supply of highly accessible locations, enhanced accessibility prompts developers to acquire land, intensify development, and introduce land-

use mixes to maximize profits. Land-use and location shifts in turn affect daily activity patterns and thus travel demand. Shifts in travel, as reflected by flows within the network, affect the performance of the transportation system.

Despite these benefits, urban development that integrates transit and land use rarely takes place—in developing countries or developed ones. Major reasons for this may lie in the distinctive nature of the planning and implementation of transportation and land use. Transportation planning often embraces a linear, technological perspective; land-use planning tends to be more flexible, ad hoc, and passive, thus making collaboration difficult (Gakenheimer 2011; [table 1.1](#)). Weak institutional and regulatory frameworks for regional and local planning, an orientation toward short-term project investments, and constrained technical and financial capacity also make urban development and management difficult. And a myriad of urban actors—including multiple layers of governments and their agencies, private developers, and informal sectors, including squatters—make coordination of transportation and urban growth exceedingly complex.

Table 1.1 Constraints on coordinating transit and land-use planning

Feature	Transit planning	Land-use planning
Objectives	Objectives focus on economic and affordable accessibility.	Objectives are diverse, extending to housing, social equity, and economic development.
Planning approach	Planning involves analysis-based quantitative approach, focusing mainly on traffic demand.	Planning approaches are mixed, depending on area of focus.
Scale for planning and implementation	Planning focuses on metropolitan-wide integrated linear systems; infrastructure is indivisibly constructed.	Planning concerns are often locally focused on communities and districts; implementation is tied to parcels.
Implementation power	Governments have power to implement projects.	Individuals and private actors have more decision-making power than governments.
Scale of projects affecting administrative strength	Projects have large, publicly funded capital budgets, strengthening administrative power.	Projects involve smaller, publicly funded capital budgets, weakening the power of administration with respect to individuals and the private sector.
Length of planning vision	Sector requires long-term vision.	Ad hoc and shorter plans can exist in parallel with long-term visions.

Source: Authors, based on Gakenheimer 2011.

The difficulty of integrating transit and land development is evident in projects managed by the World Bank. “Cities on the Move” (2002), a strategy for urban transport developed by the Bank, underscored the need to link urban development and the transport sector in order to promote livable cities, economic competitiveness, financial sustainability, and good urban governance and management. But saying and doing are two different things. A review of the Bank’s urban transport project portfolio from 1999 to 2009 indicates that only 2 of 55 projects addressed the linkage between transport and urban development (Mitric 2011). Over this period, uncoordinated growth continued in many cities in developing countries.

Coordinating transit and land use is not easy. As highlighted by the cases reviewed in this study, many barriers stand in the way, including deeply entrenched institutional and political ones. The articulation of a long-range vision and a spatial development strategy is critical to jump-starting the process. The political will and commitment to execute the vision is equally important.

Two cities in Brazil—Curitiba (population 3 million) and São Paulo (16 million)—have followed significantly different approaches to managing urban growth, with very different outcomes. Curitiba was planned as a city with well-defined linear densities along BRT-served corridors. In contrast, São Paulo has mostly unplanned densities, which are poorly linked to the city’s transit system (Acioly 2000) (figure 1.5). Despite large differences in population sizes, the number of annual public transit trips per capita in 2000 was similar in the two cities (355 for Curitiba, 330 for São Paulo).

Figure 1.5 Urban form of Curitiba and São Paulo, Brazil

a. Linear densities in Curitiba



b. Unplanned densities in São Paulo



Photos by Robert Cervero.

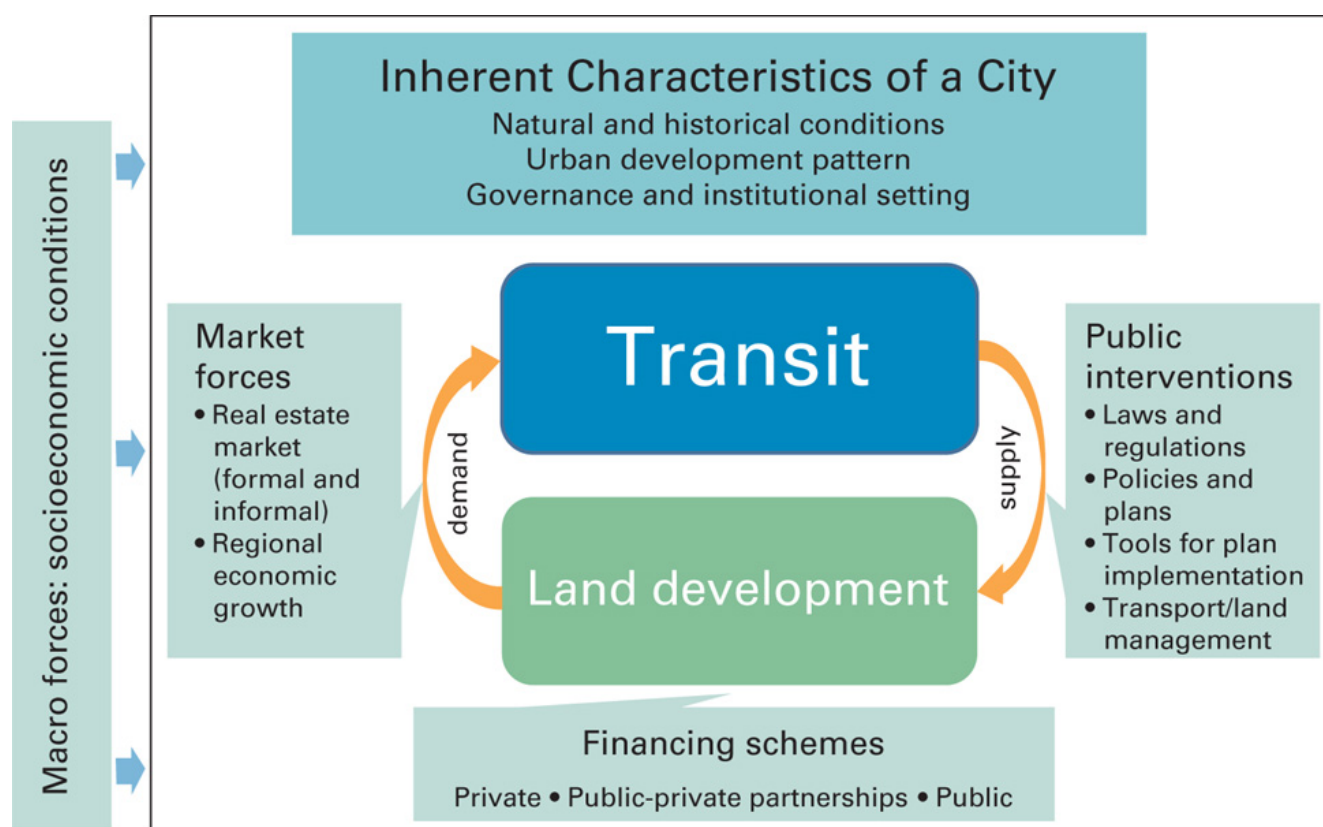
Curitiba’s transit-oriented built form stems from good urban planning. Early in the planning process, Curitiba’s leaders adopted a master plan that sought to channel growth along designated corridors, mix land uses, intensify land development at key BRT stations, and introduce high-quality urban designs that encouraged pedestrian access to the BRT corridor. In contrast, São Paulo’s pattern of

growth has been largely market driven, producing a more chaotic urban form that is difficult to serve by public transit and encourages people who can afford to travel by private car (Cervero 2000).

Study Objectives, Framework, and Methodology

This study identifies ways to promote more sustainable forms of urban growth that result in increasingly sustainable travel behavior in rapidly developing cities of the world. It examines factors that encourage or hinder built forms that are conducive to the use of public transit; highlights lessons on the prerequisites to and policy outcomes of successful transit and land-use integration from around the world; and offers case-based insights on how cities can overcome barriers to sustainable urban growth and become more transit oriented when opportunities present themselves.

Figure 1.6 presents a normalized perspective on transit and urban-form relationships. It shows that transit investments can bring about urban form and land-use adjustments by enhancing accessibility, typically in the form of more compact TODs. Such shifts in the built environment can in turn influence the demand for public transit, as higher densities lead to higher ridership levels. Density, however, is but one dimension of the built environment that influences travel. Other variables that shape travel demand are diversity, design, destination accessibility, and distance to transit (the so-called “D” variables) (see Ewing and Cervero 2010 for detailed discussions). Transit and urban form are co-dependent; the need to carefully coordinate the two lies at the heart of promoting sustainable urban futures. The in-depth case studies presented in this book focus mainly on the right-hand side of figure 1.6: how public interventions—in particular, transit investments, planning strategies, regulation, tax policies, and incentives—influence urban form. Analysis of market responses to transit investments receives less attention, largely because data are constrained and the market response to public interventions requires further detailed analysis. Some insights are provided on the ridership-inducing effects of TOD, which are often a product of public interventions and private market forces. However, because it often takes time for these impacts to unfold and reveal themselves, this part of the relationship receives less empirical attention.

Figure 1.6 Study framework

Source: Authors.

Figure 1.6 reveals other forces that shape the relationship between transit and urban form that are examined in this study. One is a set of macro forces that are outside the sphere of local policy influence, such as shifting socioeconomic trends and gasoline prices. Such forces can powerfully influence the relationship between transit and land use. Higher gasoline prices, for example, promote both mixed-use higher densities and transit ridership.

Also at play are market forces, which affect the choice of business and residential locations. For example, the shift toward knowledge-based industries in a city's economic base can encourage spatial clustering, prompting firms to seek denser, transit-served locations. Deliberate policy interventions, such as zoning regulations and parking codes, can influence private development around transit station areas and thus ridership. Financial instruments play a pivotal role in determining whether there is the fiscal capacity to create a transit-oriented built form.

This two-way dynamic between transit and land development, and the powerful array of forces that shape this relationship, occurs at multiple geographic scales: neighborhoods, corridors, districts, and regions. Correspondingly, it is often important that institutional capacity be in place, as well as the wherewithal to coordinate planning and implementation activities at and between each spatial level.

This study draws lessons from global best-case examples of transit-oriented metropolises that have direct relevance to cities in developing countries. It also reports the results of two in-depth case studies of rapidly growing and motorizing cities that introduced extended BRT systems: Ahmedabad, India and Bogota, Colombia. Two shorter case studies—of Guangzhou, China and Ho Chi Minh City, Vietnam—provide additional insights into factors that are critical to transforming cities with transit. All four cases represent cities that are aggressively investing in high-capacity transit systems at a time of rapid growth, meaning that these investments are in a position to significantly shape their urban futures. Factors that have both contributed to and inhibited sustainable patterns of growth are identified, and possible ways to overcome barriers are suggested.

The qualitative approaches used in conducting the case studies included informant interviews of public officials and other stakeholders, reviews of background reports and secondary materials, and site visits to BRT station areas. In addressing the degree to which BRT investments have shaped urban form and built environments, the case studies pay particular attention to the following concerns:

- the degree to which local authorities defined and articulated a land-use vision of the future (visioning the future)
- the degree to which implementation tools were introduced and procedural mechanisms put into place to translate visions of urban-regional futures into on-the-ground actions (tools and processes)
- the roles and contributions—past, present, and potentially future—of public transit investments and services in shaping and giving rise to urban form (public transit and urban development)
- the core lessons that are potentially transferable to other global city regions (core lessons).

Factors that may have hindered spatial integration are also identified. In several instances, matched pair analyses were conducted to investigate whether there were differences in land development between BRT station areas and otherwise similar control neighborhoods that were beyond the typical walking distance to a station.

The case studies also focus on the important issue of whether BRT investments and their corresponding urban development impacts improved access and living conditions for the underprivileged. Only if disadvantaged and marginalized populations are better off as a result of transit and land-use integration can such interventions be considered successful contributors to development.

In addition, the cases focus on BRT systems because BRT systems are being built at a rapid-fire pace throughout the developing world, due to their lower investment costs in comparison to metros and other rail options and their relatively short construction periods. Currently, more than 130 BRT

systems exist worldwide, and at least as many are in various stages of planning, advanced engineering, and construction. In addition, BRT is probably the most suitable high-quality transit service for cities with populations of less than 500,000, which account for a large share of future global urban population growth.

The findings and recommendations of this study are applicable to other transit modes as well, however, because core relationships between transit and urban development hold regardless of the mode adopted. Although a metrorail investment in a busy urban district might prompt developers to build taller office buildings than a BRT investment, the fundamental relationship is the same: accessibility improvements conferred by transit encourage densification.

Structure of This Book

The rest of this book is organized as follows. [Chapter 2](#) describes successful efforts by cities to integrate public transit and land development. These cities offer core lessons on how investment in transit can shape urban growth and how the built environment enhances demand for public transit.

[Chapter 3](#) presents two in-depth cases in developing countries: Bogota, Colombia and Ahmedabad, India. Both cases show that BRT investments have significantly enhanced urban mobility across their metropolitan areas but that in the absence of proactive planning, market forces have often steered urban growth away from BRT corridors. These cases also reveal that planning for mobility took priority over concerted efforts to reshape the cities and achieve a long-term vision of urban form through BRT investments. In both cities, government agencies at all levels are beginning to realize the importance of integrating public transit with urban development as a means of enhancing their cities' economic competitiveness, improving environmental conditions, and increasing social equity. [Chapter 3](#) also introduces two shorter case studies, Guangzhou, China and Ho Chi Minh City, Vietnam, which show other initiatives of transit and land-use integration.

[Chapter 4](#) identifies key constraints in integrating transit and built environments in rapidly growing cities of the world, as well as recommendations on ways to overcome barriers and promote successful transit and land-use relationships. Topics addressed include short-term needs versus long-term visions, regional and institutional coordination for planning and implementation, and regulatory constraints for strategic development. Based on lessons from global best-case practices introduced in [chapter 2](#), and the four case studies examined in [chapter 3](#), the concluding chapter makes specific recommendations on how to overcome barriers and promote successful transit and land use relationships.

Notes

1. The terms *transit* and *public transport* are used interchangeably in this book. *Transit* is often used in the United States; *public transport* is more common throughout the rest of the world.
2. Other international development banks include the African Development Bank, the Asian Development Bank, the CAF Development Bank of Latin America, the European Bank for Reconstruction and Development, the European Investment Bank, the Inter-American Development Bank, and the Islamic Development Bank.
3. In this context, *automobiles* refer to vehicles with engines, including cars, trucks, and motorcycles.

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CHAPTER 2

Lessons from Sustainable Transit-Oriented Cities

Global experiences show that cities on sustainable pathways are able to successfully link public transit investments and urban development. This was a core message of the book, *The Transit Metropolis: A Global Inquiry* (Cervero 1998). This chapter builds upon that work, updating and extending half the cases from *The Transit Metropolis*. Several case cities were also added to highlight recent good practices in transit and land-use integration and to examine innovative land value capture financing methods.

Cities that have successfully integrated transit and urban development have done so in one of two ways. Adaptive cities altered their urban form—through higher densities and mixed land-use patterns, for example—to support what is inherently the most resourceful form of motorized mobility: high-capacity, high-quality transit services, such as metros and bus rapid transit (BRT). Other cities focused on adaptive transit, modifying transit in less traditional ways to allow it to better serve market-driven, largely low-density patterns of urban and suburban development.

This chapter first reviews the experiences of seven adaptive cities—cities that created a compact, mixed-use, walking-friendly built form that has enabled high-quality, high-capacity public transit services to thrive. The cities include two Scandinavian cities (Copenhagen and Stockholm); four Asian cities (Hong Kong SAR, China; Seoul; Singapore; and Tokyo); and one North American city (Washington, DC).

These seven cases are followed by a review of two cities with successful BRT systems, Curitiba, Brazil and Ottawa, Canada. Although Curitiba is included in this group, it has more attributes of an adaptive city (that is, high-rise, lineal patterns of development) than an automobile-centric city oriented to flexible, adaptive forms of transit. It is the common feature of BRT services that links this group. Because [chapter 3](#) focuses on two BRT cities, Ahmedabad, India and Bogota, Colombia, the experiences of bus-based Curitiba and Ottawa are particularly germane to the themes of this book.

Most of the cases reviewed are from developed countries, whose experiences are not directly applicable or easily transferred to rapidly developing cities like Ahmedabad and Bogota. For this

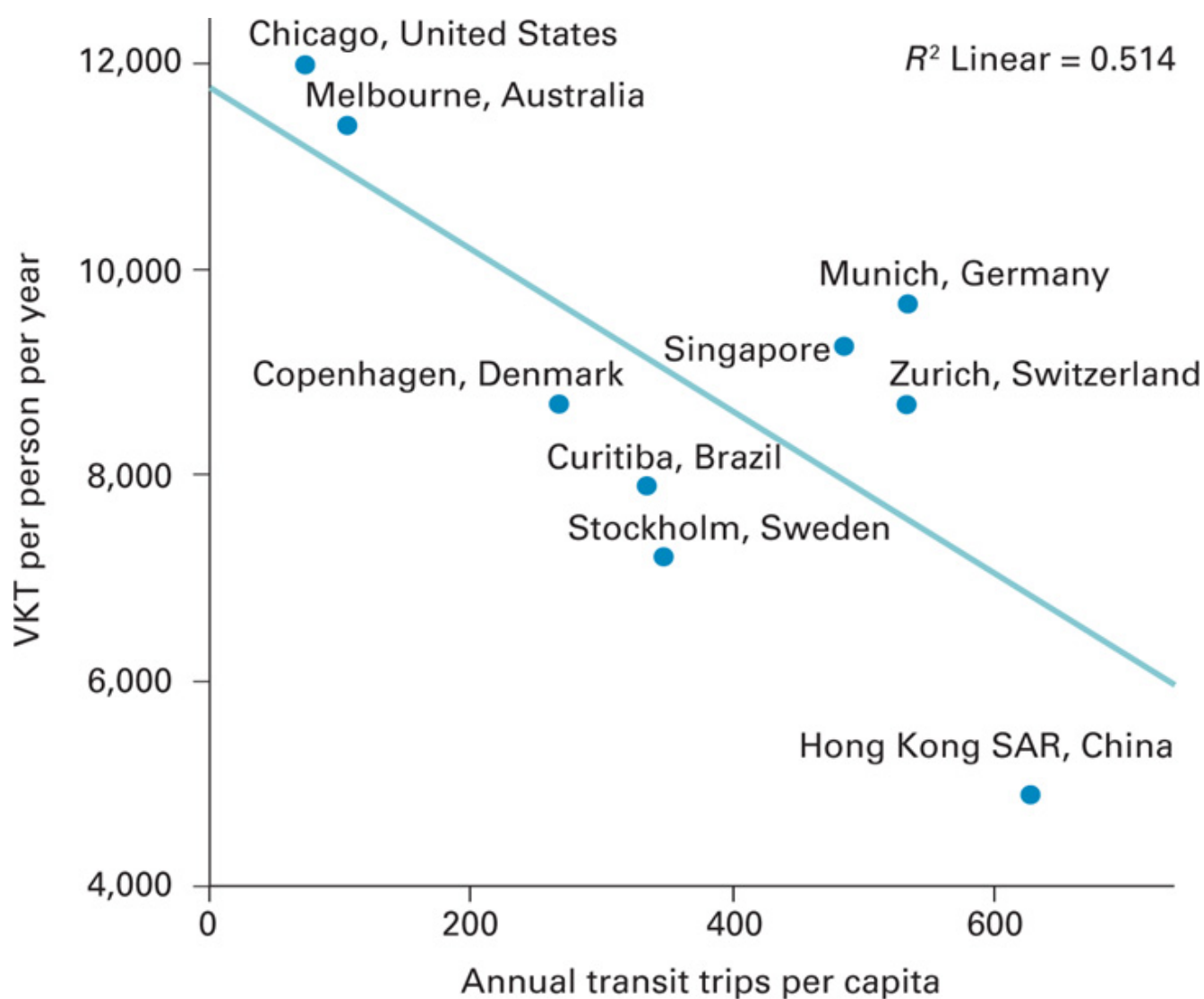
reason, the aim of this chapter is less to shape the precise practices of rapidly growing cities in developing countries, and more to impart key principles and lessons that can guide the planning and practices of cities currently planning for or investing in large-scale transit systems.

The cities reviewed differ immensely in geographic size, population, economic bases, socio-demographic makeup, and intensities of transit usage (see [annex table 2A.1](#) for comparative background information on nine of these cities). They range from Denmark's capital, Copenhagen, with a metropolitan population of about 1.2 million, to the world's two largest megalopolises—metropolitan Tokyo and Seoul, each with more than 24 million inhabitants. The share of daily trips by transit ranges from 21 percent in Ottawa to more than 60 percent in Hong Kong SAR, China; Seoul, Republic of Korea; and Singapore. Such wide variation underscores the difficulties of generalizing and drawing lessons that are applicable to all settings and contexts. Clearly, creating sustainable cities of the future calls for adapting core lessons and insights gained from best-case practices to the political, cultural, and institutional realities of particular urban settings.

A signature feature of successful transit metropolises is that they are environmentally sustainable. Another is that compact urban forms are generally needed for high ridership levels, and thus positive environmental outcomes, to be attained. That is, mass transit needs mass. An increasingly critical argument is that transit-oriented sustainable urban forms are fully consistent with economic prosperity and productivity.

Global statistics, using some of the city-region cases in this report, shed light on these matters. The International Association of Public Transport (UITP) publishes the Mobility in Cities database, which provides background data on a number of international cities, including some of the case cities.

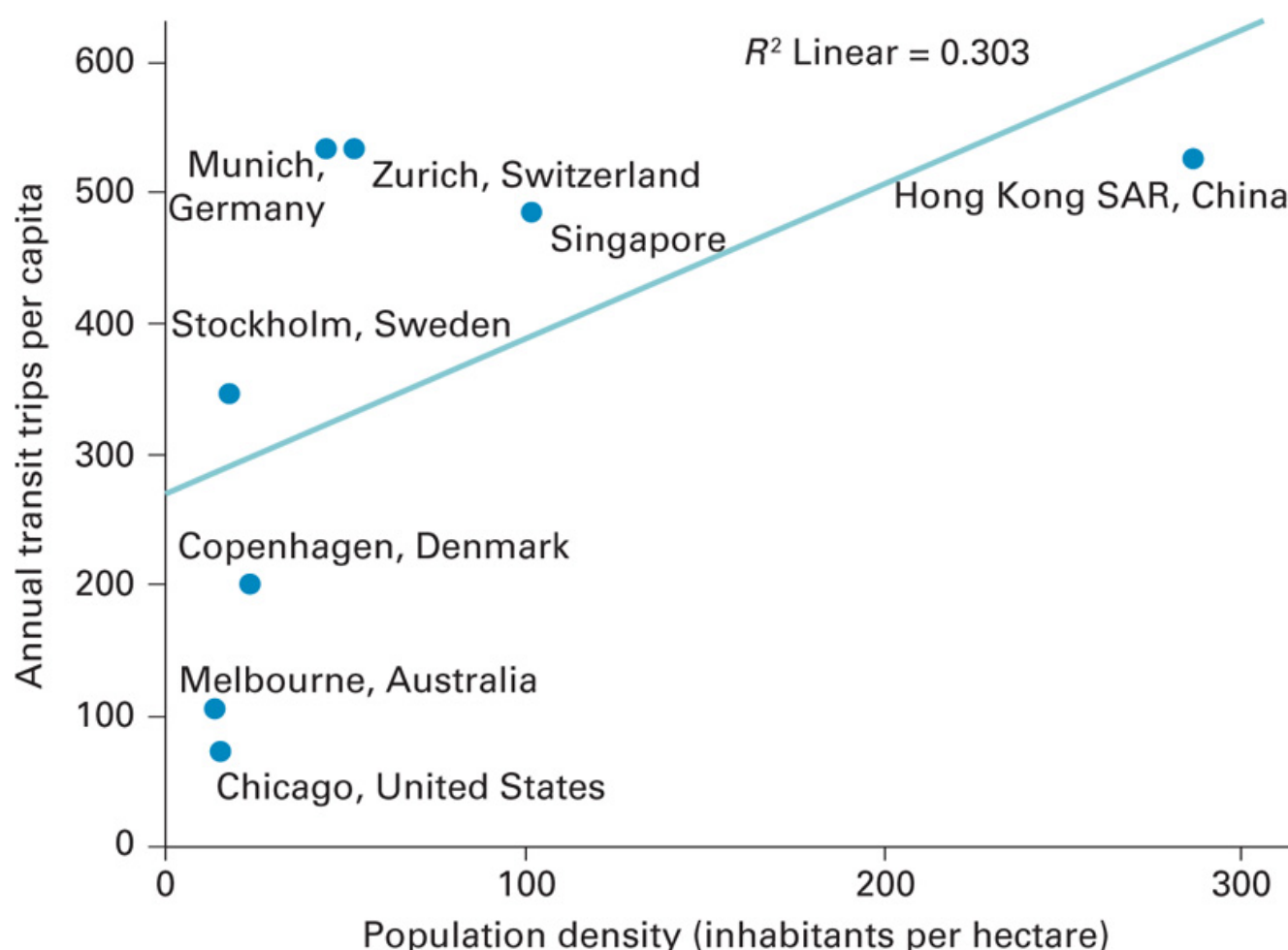
Vehicle kilometers traveled (VKT) per capita is widely viewed as the best aggregate metric for gauging sustainability in the urban transport sector. As VKT per capita increases, so does fossil fuel consumption; tailpipe emissions (for example, carbon dioxide [CO₂] and photochemical smog); and land consumption from roadway expansion. [Figure 2.1](#) reveals a strong negative association between transit usage and VKT per capita. Cities such as Stockholm; Hong Kong SAR, China; and Curitiba, Brazil stand out for their comparatively small environmental footprints.

Figure 2.1 Transit ridership and vehicle kilometers traveled per capita in selected global cities

Source: Authors, based on data from UITP 2006.

Note: VKT = vehicle kilometers traveled.

The positive association between population density and transit ridership is shown in [figure 2.2](#). Although not sufficient, compact patterns of development are usually a necessary feature of a successful transit system, at least when measured on the basis of per capita transit ridership.

Figure 2.2 Population density and transit ridership in selected global cities

Source: Authors, based on data from UITP 2006.

The ways in which relationships between transportation and urban form influence economic productivity have garnered increased policy attention in recent years. Some research finds that compact and highly accessible cities are associated with relatively high levels of labor productivity (Prud'homme and Lee 1999; Cervero 2001, 2009). Well-designed cities and efficient pricing of infrastructure that helps slow VKT growth can also promote economic growth, studies reveal. A report by the Center for Clean Air Policy (2011), for example, finds that states in the United States with lower VKT per capita tend to have higher gross domestic product (GDP) per capita. Although correlations do not prove causality and other researchers have reached opposite conclusions (QuantEcon 2009), most observers agree that the aim should be less about encouraging physical movement and more about designing communities and pricing resources to maximize economic and social interactions.

For the global cities studied in the UITP database, low VKT and high transit ridership are at least not associated with low economic performance (table 2.1). European cities with world-class transit

systems, such as Munich and Zurich, for example, have high average GDP per capita, high transit ridership, and relatively modest VKT per capita. Zurich is one of the wealthiest cities in the world, ranking first in 2011, according to the City Mayor's database. Its high per capita level of transit ridership is matched by commercial real estate values that are among the highest in the world; its quality of life is rated the highest in the world (Mercer 2002); it has one of the lowest vehicle ownership rates in the developed world (40 percent of households have no cars); and its air quality is among the best of any European city (Cervero 1998; Mees 2009).

Table 2.1 Transit ridership, vehicle kilometers traveled, and GDP in selected cities

City	Transit trips/ person/year	Vehicle kilometers traveled/person/year	GDP per capita (\$)
Hong Kong SAR, China	627	4,880	27,600
Munich, Germany	534	9,670	45,800
Zurich, Switzerland	533	8,690	41,600
Singapore	484	9,240	28,900
Stockholm, Sweden	346	7,210	32,700
Curitiba, Brazil	334	7,900	6,800
Copenhagen, Denmark	268	8,700	34,100
Melbourne, Australia	105	11,400	22,800
Chicago, United States	73	12,000	40,000

Source: Authors, based on data from UITP 2006.

Transit and Land-Use Integration in Adaptive Cities

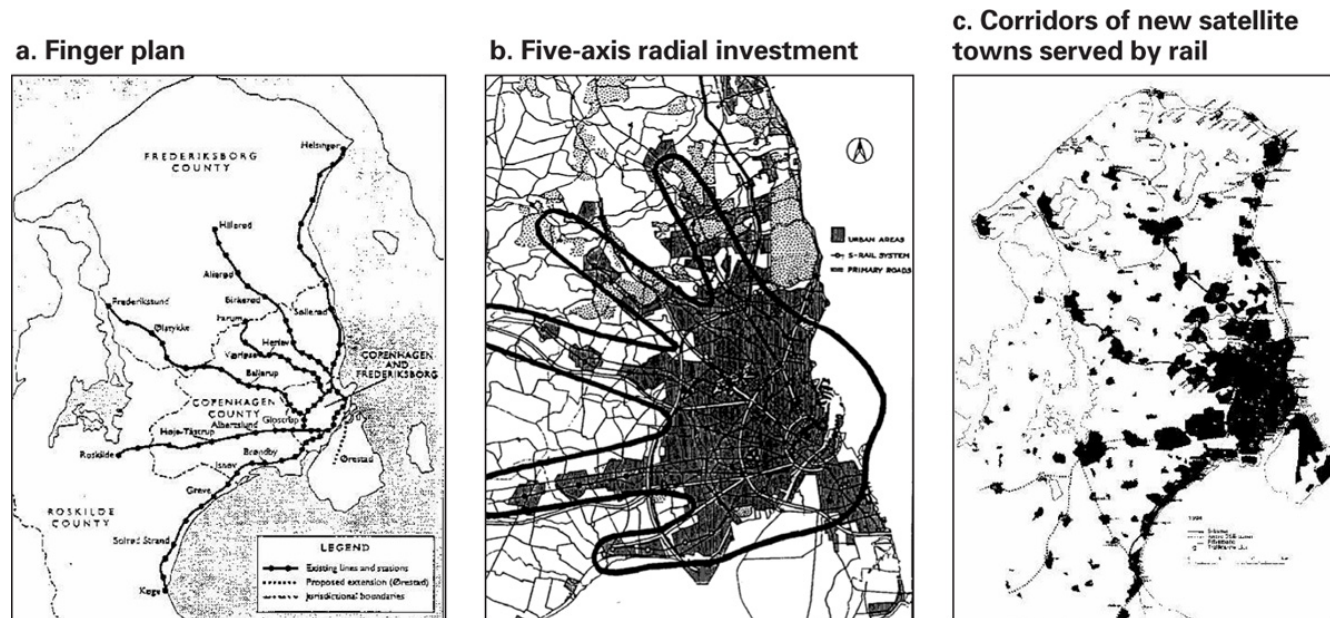
This section reviews experiences in seven adaptive cities: Copenhagen; Stockholm; Hong Kong SAR, China; Seoul; Singapore; Tokyo; and the Washington, DC metropolitan area, where metrorail investments were used to create a more compact, mixed-use, pedestrian-friendly urban form. It highlights the planning, design, institutional, and financial approaches used to leverage transit-oriented station areas and corridors.

Copenhagen: Transit Oriented and Bike Friendly

A textbook example of long-range planning visions shaping rail investments, which in turn shaped urban growth, comes from Copenhagen, with its celebrated “finger plan.” Early in the planning process, planners identified corridors for channeling overspill growth from the urban centers. Rail infrastructure was built, often in advance of demand, to steer growth along desired growth axes.

Greenbelt wedges set aside as agricultural preserves, open space, and natural habitats were designated and major infrastructure directed away from districts with these features. The evolution of Copenhagen—from a finger plan to a directed rail-investment program along defined growth axes to a finger-like urbanization patterns—is shown in figure 2.3.

Figure 2.3 Evolution of Copenhagen’s “transit first” plan



Source: Cervero 1998; reproduced with permission from Island Press, Washington, DC.

On the periphery of Copenhagen are new towns that are bike and pedestrian friendly. Suburban towns of 10,000–30,000 inhabitants, like Ballerup, Brønby, and Høje-Taastrup, are laced by greenways that connect neighborhoods, schools, retail centers, and pocket parks to inviting rail stops. About half of residents in these middle-class, master-planned new towns take a train to work, and four out of five walk, bike, or take a bus to their community's rail station.

Copenhagen planners have long embraced the notion that industrial progress should not encroach on the rights and needs of pedestrians and cyclists. They created one of the first and the longest car-free streets in Europe, Strøget, which, during summer, accommodates some 55,000 pedestrians, often shoulder to shoulder. Street life is not viewed only in terms of foot traffic but also with regard to stationary activities. Jan Gehl, a noted urban designer from Copenhagen, sold city leaders on the idea that great public spaces accommodate not only busy pedestrians but also casual sitting, relaxing, and milling about. Today, some 80,000 square meters of public squares—big and small, grand and modest—dot central Copenhagen.

One of the chief ways of meeting the needs of cyclists has been the expropriation of car lanes and curbside parking for their exclusive use. Between 1980 and 2005, Copenhagen's inventory of bike

lanes increased from 210 to 410 kilometers within an area of about 90 square kilometers. Over the same period, the number of bike trips rose 80 percent. By 2005, 36 percent of journeys to work in Copenhagen were by bicycle, the highest mode split of any capital city in Europe. The city's master plan, *Eco-Metropolis for 2015*, commits Copenhagen to become "the world's best city for bicycling" (City of Copenhagen 2008). Copenhagen has set an ambitious goal of 50 percent of its citizens biking to work or school by 2015. New, separated cycle-tracks are being added in hopes of achieving this goal.

To further boost cycling, Copenhagen introduced a short-term bike lease program, called City Bikes, in 1995. More than 2,000 white bikes have been placed at some 140 bike stands throughout the city. In addition to improving rail access, the program reduced on-vehicle carriage of bikes, freeing up train capacity for passengers. City Bikes is overseen by a local nonprofit organization that hires hundreds of "rehabilitees" (prisoners who are being rehabilitated) to maintain the bikes. The organization reports that 55 percent of the rehabilitees get jobs after participating in the program (Foundation City Bikes 2009). Statistics reveal how well articulated Copenhagen's "access shed" is for transit riding. A 2002 survey of 15 suburban rail stations found that people walked 38–100 percent of access trips up to 1 kilometer; for trips 1–2 kilometers away, cycling accounted for 40 percent of access trips. Beyond 2 kilometers, buses handled two-thirds of access trips. Even 4 kilometers from stations, twice as many access trips were by bicycle as by car. Danish designers have found that acceptable walking and cycling distances can be stretched considerably by creating attractive, visually stimulating, and safe travel corridors. Partly by doing so, Copenhagen succeeded in reducing CO₂ emissions per capita by 25 percent between 1990 and 2008 (City of Copenhagen 2010).

A transit-oriented built form that is pedestrian and bike friendly is not the only factor accounting for Copenhagen's low annual VKT per capita of 8,700 (see [table 2.1](#)). Also critical have been national policies that aim to moderate car ownership and usage. Every four years since World War II, the national government has issued policy guidelines aimed at shaping the land-constrained country's physical development. Over the years, a series of national directives have called for targeting Greater Copenhagen's future growth around rail transit stations. National infrastructure funds are tied to compliance with these directives. Although they do not carry the force of law, Denmark's national directives clearly imply that localities are to make good faith efforts in encouraging transit-oriented development (TOD). The nation's Ministry of Environment has veto power over proposed local development projects. These veto powers have been exercised sparingly over the years, in large part because most localities strongly support sustainable patterns of development.

In addition to national directives that mandate that major trip generators be sited near rail stops, Denmark adds taxes and fees that typically triple the retail price of a new car. At 250 motor vehicles per 1,000 inhabitants, Copenhagen's vehicle ratio is about half that of large German cities like Hamburg and Frankfurt.

Local policies also restrict car travel. Central-city road capacity in Copenhagen has been kept constant since 1970; outside the core city, additional road capacity must be matched by at least as many square meters of additional bike lanes and bus lanes. Parking has also been restricted, particularly near rail stops. The outsourcing of parking to peripheral areas has led to a 2–3 percent annual reduction in core-area parking. Central-city bus services have been enhanced by a system of reserved lanes and signal prioritization.

Also strengthening Copenhagen's standing as a transit metropolis has been the expansion of rail services. The city has built new rail "fingers," notably, an automated, fully grade-separated line to the new-town of Orestad, south of the center city ([figure 2.4](#)).

Figure 2.4 Copenhagen’s “finger plan” for urban development



Source: Cervero 1998; reproduced with permission from Island Press, Washington, DC.

Rail services preceded development in Orestad, a clear case of building transit first to guide development. From the start, Orestad was a true mixed-use community, designed as a place to live, work, shop, learn, and play. Particular attention was paid to the livability of the new community. Housing was built close to parks and canals and connected by plazas and pathways. Cafes and squares were sited to attract customers who arrive by foot. Neighborhood parks were designed to meet the diverse recreational needs of residents. In contrast to the drab, standardized appearance of Copenhagen’s early-generation TODs, Orestad also features a variety of architectural styles carefully planned to interact and blend with one another. Many of Orestad’s signature buildings were designed by world-class architects, and several have won prestigious design awards. Orestad’s diversity is underscored by one of the largest car-free housing developments anywhere, called Urbanplanen.

Another rail line being built in the region follows a circular route, providing “cross-finger” connections. This Cityringen metro line will serve a number of districts outside the city proper not served by the S-Train commuter rail system. The sale of land whose value has appreciated in anticipation of new rail services has helped finance new investments like Cityringen, the Orestad line, and Copenhagen’s first light rail line, Letbanen, being built parallel to the region’s third ring road.

Copenhagen planners have also taken advantage of new technologies to enhance the transit riding experience. In 2009, the city introduced a new state-of-the-art mobile ticket for transit trips. Customers are now able to buy and display tickets on their mobile phones for transit rides on the metro, suburban commuter rail lines, and city buses.

Stockholm: First-Generation Transit Necklace, Second-Generation Urban Regeneration

The last half century of strategic regional planning has given rise to regional settlement and commuting patterns in Greater Stockholm that have substantially reduced dependency on automobiles in middle-income suburbs. Stockholm’s investment in radial rail lines has given rise to a “string of pearls” urban form and a balanced use of land for work and housing. By consciously establishing a balance between jobs and housing along rail-served axial corridors, Stockholm planners produced directional flow balances in commuting periods. During peak hours, 55 percent of commuters are typically traveling in one direction on trains and 45 percent are heading in the other direction.

Stockholm’s transit mode share is nearly twice that found in larger rail-served European cities, such as Berlin; it is even higher than inner London’s market share. Perhaps most impressive, Stockholm is one of the few places where automobile use appears to be receding. Between 1980 and 1990, Stockholm was the only city among a sample of 37 cities that registered a per capita decline in automobile use—a drop of 229 annual kilometers of travel per person (Kenworthy and Laube 1999). Its VKT per capita remains among the lowest in the world (see [table 2.1](#)). An independent analysis by Siemens AG/McKinsey (2008) finds Stockholm’s per capita CO₂ emissions from transportation lower than four other rail-served and considerably larger global cities (London, New York, Rome, and Tokyo).

These statistics do not mean that Stockholm is “anti car.” In fact, Stockholm has a relatively high level of car ownership (555 cars per 1,000 inhabitants) (European Commission 2012). In a well-designed transit metropolis like Stockholm, residents simply drive less; they are more judicious and discriminate in their use of the cars more than car owners in other cities. Most Stockholmers use public transport to get to work, selectively using cars where they have natural advantages, such as for grocery shopping or weekend excursions.

Stockholm is credited with spearheading TOD in the age of the motorway, in the form of master plans for new towns like Vällingby, where the rail stop sits squarely in the town center. Upon exiting the station, passengers step into a car-free public square surrounded by shops, restaurants, schools, and community facilities. The civic square, adorned with benches, water fountains, and greenery, is the community's central gathering spot—a place to relax, socialize, and hold special events, such as public celebrations, parades, and demonstrations. Sometimes the square doubles as a place for farmers to sell their produce or street artists to perform, changing chameleon-like from an open-air market one day to a concert venue the next. The assortment of flower stalls, sidewalk cafes, newsstands, and outdoor vendors dotting the square, combined with the musings and conversations of residents sitting in the square, retirees playing chess, and everyday encounters among friends, adds color and breathes life into the community. A community's rail station and its surroundings are thus more than a jumping off point to catch a train—they should also be the kinds of places people are naturally drawn to. If done well, TODs are “places to be,” not “places to pass through” (Bertolini and Spit 1998).

Development of Hammarby Sjöstad. The first generation of TOD in metropolitan Stockholm was on former greenfields, such as Vällingby and Kista. More recently, a push has been made to redevelop brownfields. The most notable example is Hammarby Sjöstad, an eco-community that has taken form along a recently built inner-ring tramway.

The development of Hammarby Sjöstad marked an abrupt shift in Stockholm's urban planning practice. For decades, new towns had been built on peripheral greenfield sites. Hammarby Sjöstad is one of several “new-towns/in-towns” created based on Stockholm's 1999 city plan, which set forth a vision of “building the city inward.”

Consisting of some 160 hectares of brownfield redevelopment, Hammarby Sjöstad stands as Stockholm's largest urban regeneration project to date. Because the project focuses on a new inner-city transit line and is designed for energy self-sufficiency and minimal waste, it has been called a “green TOD” (Cervero and Sullivan 2011). Just as the greenfield town of Vällingby pioneered TOD, Hammarby Sjöstad is a paragon of TOD with green urbanism and green architecture ([table 2.2](#)).

Table 2.2 Green transit-oriented development attributes of Hammarby Sjöstad, Sweden

Built environment	Green transportation		Green urbanism	
	Infrastructure	Programs and policies	Energy	Open space, water, and stormwater
Brownfield Infill Former army barracks: • High density along light rail boulevard (8 stories) • TOD: Mixed use with ground-floor retail and a wide range of goods and services	“Tvärbanan” light rail line: 3 stops in district: • 5 minutes to major station • 10–30 minutes to all parts of Center City • 7-min peak headway 2 bus lines Ferry Bike lanes and pedestrian bridges Ample bike parking at every building Car-sharing for 3 companies, 37 vehicles Near congestion toll boundary Pedestrian-friendly design/complete streets	Transit-Boulevard is focus of activity/commerce Grid streets increase connectivity/calm traffic Convenient bike parking/storage at every building	Waste converted to energy: • Food waste and wastewater sludge converted to biogas and used for heating • Combustible waste burned for energy and heat • Paper recycled Heat recaptured for reuse Combined heat and power plant Low-energy construction and energy saving measures: • Efficient appliances • Maximum insulation and triple-glazed windows	Stormwater treatment: • Rainwater collection • Maximum permeable surfaces • Purify runoff through soil filtration Ample open space: • Inner courtyards • Parks • Playgrounds • Green median • Borders large nature reserve with ski slopes Preservation of existing trees and open space Reduced water flow faucets and low-flush toilets

Source: Cervero and Sullivan 2011.

The community’s signature transit element is a new tramway, Tvärbanan, which runs through the heart of the community along a 3-kilometer boulevard (Hammarby Allé and Lugnets Allé). In TOD fashion, taller buildings (mostly six to eight stories) cluster along the transit spine, and building heights taper with distance from the rail-served corridor. Trams run every seven minutes in peak hours and provide five-minute connections to Stockholm’s metro underground network and commuter trains. Rail stations are well designed and fully weather protected, and they provide real-time arrival information. The city’s buses run on biogas produced by local wastewater processing.

Parks, walkways, and green spaces are also prominent throughout Hammarby Sjöstad. Where possible, the natural landscape has been preserved. Bike lanes run along major boulevards, ample bike parking can be found at every building, and bike and pedestrian bridges cross waterways. Design features that are integral to TOD, like buildings that go up to the sidewalk line (that is, no set-backs),

offer comfortable and secure walking corridors with clear sight lines. They also bring destinations together and, by creating side friction (parallel movements that prompt vehicles to slow down and encourage pedestrian activities), slow traffic.

The presence of three car-sharing companies, which together provide access to 37 low-emission vehicles, has further reduced the need to own a car in Hammarby Sjöstad. The area was designed with just 0.25 parking spaces per dwelling unit (though this rate has inched up some in recent years). All commercial parking is for a fee, and rates discourage long-term parking. The neighborhood sits just outside Stockholm's congestion toll boundary, adding another incentive to use public transport, walk, or bike when heading to the central city.

Hammarby Sjöstad's green urbanism is found in energy production, waste and water management, and building designs. The annual energy use of buildings in Hammarby Sjöstad is set at 60 kilowatt hours, a third less than for the city as a whole. All windows are triple-glazed and walls thoroughly insulated. Other conservation measures include extra heat insulation, energy-efficient windows, on-demand ventilation, individual metering of heating and hot water in apartments, electrically efficient installations, lighting control, solar panels, fuel cells, reduced water flow, and low-flush toilets.

The ecological feature of Hammarby Sjöstad that has garnered the most attention is the fully integrated closed-loop eco-cycle model. This clever system recycles waste and maximizes the reuse of waste energy and materials for heating, transportation, cooking, and electricity.

Also impressive is the community's approach to water management. All storm water, rainwater, and snowmelt is collected; purified locally through sand filter, stormwater basins, and green roofs; and released in purified form into a lake. A preserved oak forest, ample green surfaces, and planted trees help collect rainwater to ensure cleaner air and provide a counterbalance to the dense urban landscape.

Hammarby Sjöstad is well on its way to becoming a low-carbon eco-community. Relative to conventional development, the project reduced air, soil, and water emissions and pollution by 40–46 percent; nonrenewable energy use by 30–47 percent; and water consumption by 41–46 percent (Cervero and Sullivan 2011). Similar to the rest of Stockholm, 95 percent of all waste produced by Hammarby Sjöstad's household is reclaimed.

Environmental benefits have accrued from Hammarby Sjöstad's relatively high share of nonmotorized (walking and bicycling) trips. In 2002, the project's mode splits were public transit 52 percent, walking/cycling 27 percent, and private cars 21 percent ([table 2.3](#))—a much more ecological split than in suburban neighborhoods of Stockholm with similar incomes (Grontmij 2008). Noncar travel shares are thought to be considerably higher today. Car ownership has also fallen, from 66 percent of households in 2005 to 62 percent in 2007, in line with averages for the denser, core part of

Stockholm (Grontmij 2008). Residents' carbon footprint from transportation in 2002 was considerably lower than comparison communities (438 versus 913 kilograms of CO₂ equivalent/apartment/year) (Grontmij 2008). Stockholm's goal is to become fossil fuel free by 2050.

Table 2.3 Mode splits for journeys in various parts of Stockholm County

(percent)

Mode of transportation	Inner city	Southern suburbs	Western suburbs	Hammarby Sjöstad
Car	17	39	43	21
Public transit	36	28	23	52
Bike/walk	47	32	34	27

Source: Authors, based on data from Grontmij 2008.

Another barometer of Hammarby Sjöstad's environmental benefits is the community's relatively healthy local economy. Median household income was higher and the unemployment rate lower than the city of Stockholm as a whole in 2006, and land prices and rents have risen more rapidly over the past decade than most other parts of the Stockholm region (Grontmij 2008).

Transport demand management and pedestrian- and transit-oriented projects. Congestion pricing is an important component of creating a functional and livable core city. Stockholm's electronic road pricing (ERP) scheme, introduced in 2004 on a trial basis, charges motorists for entering the central city on weekdays, using a graduated price scheme. Buses, taxis, eco-fuel cars, and drivers coming and going from the isolated island of Lidingö are exempted.

As traffic conditions and the quality of public transit services improved, citizens' support grew steadily (Eliasson and others 2009). In a referendum in 2007, 53 percent of Stockholm residents voted to make the road pricing trial permanent. During the first two years of the permanent scheme, peak-period traffic volumes within the pricing zone fell 25 percent (removing 1 million vehicles from the road a day), CO₂ emissions fell 14 percent, and daily toll revenues reached about \$300,000 (Eliasson and others 2009).

Some revenues from congestion tolls have gone to enhance transit services; most have been used to upgrade road facilities, including the South City tunnel project (Södra Lanken), which removes through-traffic from central-city surface streets, and a western bypass project. Revenues are also going toward a major inner-city land reclamation project in the Slussen area. Guided by a master plan by noted architect Norman Foster, Slussen—which has been called a tangled mass of highway

overpasses—is to become a pedestrian- and transit-oriented central-city infill project. Slussen will feature attractive public spaces and water terraces that link pedestrians on a historic route into the old city.

Also in the works are makeovers of Stockholm's first-generation transit villages. Vällingby's upgrading is being called one of Sweden's largest suburban renewal projects. Skarholmen, Spanga, and Kista Science City are also being renovated as second-generation TODs.

Hong Kong SAR, China: Profitable Transit

Any visitor to Hong Kong SAR, China, instantly recognizes that public transit is the lifeblood of the city. The city boasts a wide array of transit services, including a high-capacity railway network, surface-street trams, ferries, and an assortment of buses and minibuses. In late 2007, the city's main passenger rail operator, Mass Transit Railway Corporation (MTRC), merged with the former Kowloon-Canton Railway Corporation, forming a 168-kilometer network of high-capacity, grade-separated services on the island of Hong Kong; the Kowloon peninsula; the Northern Territories (to the Chinese border); and, through a recent extension, to the new international airport. In 2000, more than 90 percent of all motorized trips were by public transit, the highest market share in the world (Lam and Bell 2003; Cervero and Murakami 2009).

The combination of high urban densities and high-quality public transit services has not only produced one of the highest levels of transit usage in the world (570 annual public transport trips per capita), it has also substantially driven down the cost of motorized travel. In 2002, more than half of all motorized trips made by city residents were half an hour or less (ARUP 2003). Motorized travel consumes, on average, about 5 percent of the city's GDP. This figure contrasts sharply with more automobile-oriented global cities, such as Houston, Texas, and Melbourne, Australia, where more than one-seventh of GDP goes to transportation (IAPT 2002). Hong Kong SAR, China residents enjoy substantial travel cost savings even in comparison with much larger global cities with extensive railway networks, like London and Paris.

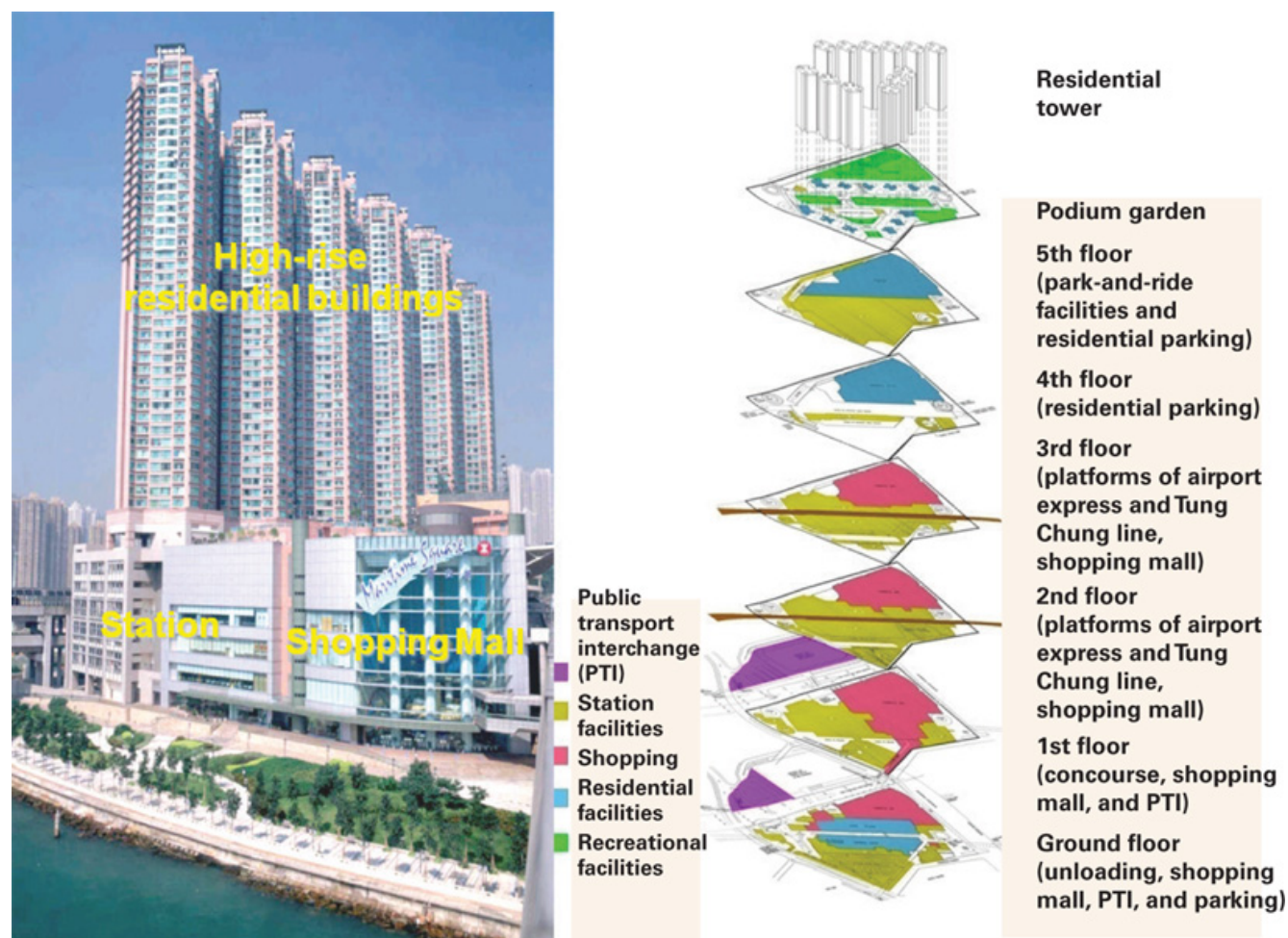
Hong Kong SAR, China is one of the few places in the world where public transit makes a profit. It does so because the city's rail operator, MTRC, has adopted the "Rail+Property" (R+P) program (Cervero and Murakami 2009). R+P is one of the best examples anywhere of transit value capture in action. Given the high premium placed on access to fast, efficient, and reliable public transit services in a dense, congested city, the price of land near railway stations is generally higher than elsewhere, sometimes by several orders of magnitude. MTRC has used its ability to purchase the development rights for the land around stations to recoup the cost of investing in rail transit and turn a profit. The

railway has also played a vital city-shaping role. In 2002, about 2.8 million people, 41 percent of the population, lived within 500 meters of a railway station (Tang and others 2004).

The profit motive accounts for MTRC's active involvement in land development. As a private corporation that sells equity shares on the Hong Kong stock market, MTRC operates on commercial principles, financing and operating railway services that are self-supporting and yield a net return on investment. The full costs of public transit investments, operations, and maintenance are covered by supplementing fare and other revenues with income from ancillary real estate development, such as the sale of development rights, joint ventures with private real estate developers, and the operation of retail outlets in and around subway stations. The local government is MTRC's majority stockholder, ensuring that the company weighs the broader public interest in its day-to-day decisions. At the same time, the ownership of 23 percent of MTRC's shares by private investors exerts market discipline, prompting the company to be entrepreneurial.

Between 2001 and 2005, property development produced 52 percent of MTRC's revenues. By contrast, railway income, made up mostly of fare-box receipts, generated 28 percent of total income. MTRC's involvement in all property-related activities—development, investment, and management—produced 62 percent of total income, more than twice as much as fares. An example of an R+P project that has yielded both high rates of financial returns and high ridership (and thus fare-box income) is Maritime Square at the Tsing Yi Station ([figure 2.5](#)).

Figure 2.5 The Maritime Square residential-retail project developed by the Mass Transit Railway of Hong Kong SAR, China



Source: Courtesy of MTR Corporation Ltd.

Note: Situated above the Tsing Yi Station between the center of the city and the new international airport, Maritime Square features a hierarchy of integrated uses: a shopping mall extends from the ground floor to the third level; the station concourse sits on the first floor, with rail lines and platforms above and ancillary/logistical functions (such as public transit/bus interchange and parking) at or below; above the fourth and fifth floor residential parking lies a podium garden, and above that are luxury residential towers.

Timing is crucial in MTRC's recapturing of the value added by rail service. MTRC purchases development rights from the local government at a "before rail" price and sells these rights to a selected developer (among a list of qualified bidders) at an "after rail" price. The difference between land values with and without rail services is substantial, easily covering the cost of railway investments. When bargaining with developers, MTRC also negotiates a share of future property development profits or a co-ownership position from the highest bidder. Thus, MTRC receives a "front end" payment for land and a "back end" share of revenues and assets in kind.

MTRC has hardly been the sole financial beneficiary of R+P. Society at large has also reaped substantial rewards. Between 1980 and 2005, it is estimated that Hong Kong SAR, China received nearly \$140 billion (in current dollars) in net financial returns. This estimate is based on the difference between earned income (\$171.8 billion from land premiums, market capitalization, shareholder cash dividends, and initial public offer proceeds) and the value of injected equity capital (\$32.2 billion from land grants). Thus, the government has enjoyed huge financial returns and seeded the construction of a world-class railway network without having to advance any cash to MTRC. Moreover, the \$140 billion figure is only the direct financial benefit. The indirect benefits—for example, reduced sprawl, air pollution, and energy consumption and higher ridership through increased densities—have increased net societal returns well beyond \$140 billion.

Hong Kong SAR, China has long had tall towers perched above railway stations. But density alone does not make a good TOD. Often missing was a high-quality pedestrian environment and a sense of place. Most first-generation R+P projects featured indistinguishable apartment towers that funneled pedestrians onto busy streets and left them to their own devices to find a subway entrance. Growing discontent over sterile station-area environments and older buildings' sagging real estate market performance prompted MTRC to pay more attention to principles of good town planning and urban design.

In 2000, MTRC created a town planning division within the corporation to pursue land development strategies that met corporate financial objectives while also enhancing station-area environments. Before 2000, R+P projects followed the existing development pattern rather than anticipated development; yet, with an in-house town planning department, MTRC became more proactive. The company is now ahead of market demand, building high-quality, pedestrian-friendly TODs to steer growth. Research shows that the design of high-quality walking environments has increased financial returns per square meter for R+P projects (Cervero and Murakami 2009). Pedestrian-friendly R+P projects have contributed to sustainable urbanism as well as sustainable finance, as these benefits have been capitalized into land prices.

Seoul: Bus Rapid Transit and Urban Land Reclamation

At the turn of the 21st century, Seoul embarked on a bold experiment in urban regeneration, principally involving the reclamation of urban space given to the automobile in the post-Korean War era. Through the leadership of Myung-Bak Lee, the former mayor of Seoul and the current president of the Republic of Korea, the city has sought to strike a balance between transit infrastructure as a provider of mobility and public space as an urban amenity. In good part, this effort has been prompted

by the desire to make Seoul globally competitive with the likes of Hong Kong SAR, China, Shanghai, and other East Asian heavyweights of by emphasizing quality of life every bit as much as mobility and large-scale infrastructure development.

Like many modern metropolises in East Asia, over the past several decades Seoul has followed a pattern of American-style sprawl, fueled by steady economic growth and the concomitant meteoric rise in private automobile ownership. However, population densities in Seoul (10.4 million inhabitants spread over a land area of 605 square kilometers) have historically been and remain high by global standards. The city of Seoul itself, along with the port city of Incheon and surrounding Kyunggi Province, constitute the Seoul Metropolitan Area (also called the Seoul National Capital Area), with more than 23 million inhabitants—the world's second-largest urban agglomeration. In 2006, Seoul and Incheon combined had the sixth-highest population density in the world (16,700 people per square kilometers). Because of such high densities, the Seoul metropolitan government has over the years aggressively sought to decentralize growth, mainly in the form of building master-planned new towns sited on the region's periphery.

A severe housing shortage and rising housing rents (fueled largely by land speculation) during the 1980s prompted the Korean government to build new towns quickly, as a mass-produced commodity. Twenty-six new towns have been built in the greater Seoul Metropolitan Area over the past three decades. Most were built in a modernist Le Corbusier style, as mid- to high-rise “towers in the park.”

The shortage of central-city land combined with the presence of a protective greenbelt that surrounds the city of Seoul resulted in most new towns being built in the region's far-flung fringes (Jun and Hur 2001).

New-town development achieved its goal of stabilizing housing prices, but it did so at a price: congestion on radial links to the urban center worsened, tailpipe emissions and fuel consumption rose, and demand for expensive highway infrastructure mushroomed. Between 1990 and 1996, the average commuting distance of new-town residents increased 70 percent (Jun 2000). Longer trips combined with increased car ownership inevitably translated into steadily worsening traffic congestion: average speeds during evening hours in the Seoul Metropolitan Area fell from 24 kilometers an hour in 1998 to 17 kilometers an hour in 2003. Daily commuting expenses rose sharply, as well: one study estimated the out-of-pocket cost (ignoring the value of time and externalities) incurred by new-town residents living outside of Seoul's greenbelt at \$12 a day (Jun and Bae 2000).

By the late 1990s, rumblings could be heard within public policy circles that the region's new towns were a failed experiment, exacerbating traffic congestion and reducing environmental quality. Some feared that such factors, along with the productivity losses from long commutes, were becoming a

drag on economic growth, prompting companies and their workers to relocate elsewhere in the country and possibly even abroad. The idea of reurbanizing Seoul's central areas through "new-towns /in-towns" began to surface.

Myung-Bak Lee led the charge of reinvesting in the central city and regenerating Seoul. In 2001, Lee ran for mayor of Seoul, largely on a platform of reinvigorating the central city as means of creating a more sustainable yet productive city. Lee campaigned on the premise that Seoul could achieve a better balance between function and the environment by reordering public priorities so as to emphasize quality of place. Before becoming mayor, Lee found and led the Hyundai Group for three decades, Korea's largest builder of public works and infrastructure projects, including highways.

Lee won a decisive victory. Upon assuming office, in early 2002, he moved quickly on his campaign promises. His vision called not only for expanding public transit services but also for reducing the ecological footprint of private cars by reclaiming urban space consumed by roads and highways, especially space used to funnel new-town inhabitants in and out of the central city. Why scar the interior of the city, he reasoned, to funnel suburbanites to office jobs in the core? A major culprit was the network of elevated freeways that converged on central Seoul—facilities that severed longstanding neighborhoods, formed barriers and created visual blight, cast shadows, and sprayed noise, fumes, and vibrations on surrounding areas. Although freeways provided important mobility benefits, Lee recognized that those benefits had to be weighed against their nuisance effects, particularly in today's amenity-conscious workplace.

The removal of a six-kilometer elevated freeway in the heart of Seoul, Cheong Gye Cheon (CGC), and the restoration of an urban stream and pedestrian-friendly greenway was a natural choice to launch Lee's vision of a more sustainable urban landscape for the city. Change was swift. By February 2003, a plan for the freeway removal was completed; five months later, the freeway had been completely dismantled. Some two years later, in September 2005, the restored CGC stream and linear greenway was opened to the public, following a major public celebration and ribbon cutting by Mayor Lee ([figure 2.6](#)). The entire cost of the freeway demolition and stream restoration was \$313 million.

Figure 2.6 Transformation of Seoul's Cheong Gye Cheon from an elevated freeway to an urban greenway

a. Elevated freeway



b. Urban greenway



Source: Seoul Metropolitan Government 2003.

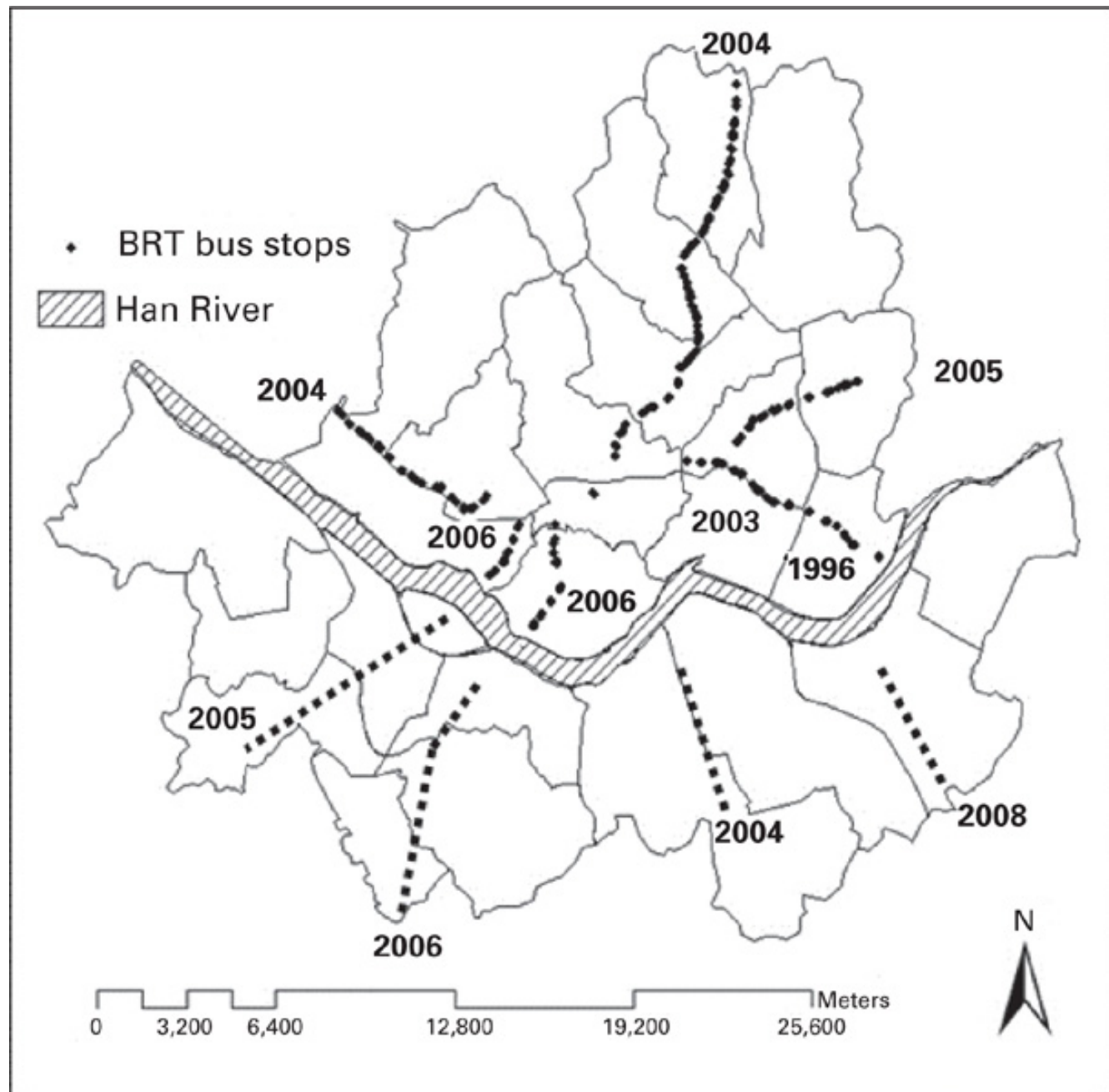
Equally important in symbolic terms was the mayor's decision to convert a massive 1.3 hectare surface-street intersection to an oval-shaped grass park in front of Seoul's City Hall, the nerve center of the city. The huge swath of real estate devoted to car maneuvers in front of City Hall, an architectural icon and one of the busiest locations in the city, created an extremely pedestrian-hostile environment. Today, the green oval is one of the city's most popular gathering places.

Mayor Lee made his policy intentions clear when publicly stating that the transformation of space for cars to space for people represented “a new paradigm for urban management in the new century” (Seoul Metropolitan Government 2003, p. 1). His views were partly shaped by what was happening in several Latin American cities at the time, especially Curitiba, Brazil, which he visited at the invitation of Jaime Lerner, former mayor of Curitiba. Mayor Lee defended the roadway removal projects on the grounds that “we want to make a city where people come first, not cars.” The diminution of roadway capacity represents, in many ways, a recasting of public priorities. In Seoul's case, it marked a shift from building infrastructure that enhances automobility to infrastructure that enhances public amenities and the quality of urban living. A longer-term objective was to encourage more households to settle in the central city and in redevelopment districts, thus reversing the centrifugal flow of residents to Seoul's outskirts and beyond—in effect, creating a new form of urban development.

In and of itself, the withdrawal of road capacity in an increasingly automobile-dependent society will do little to enhance the quality of urban living. Mayor Lee understood that public transit had to be substantially expanded and upgraded to absorb the traffic (169,000 cars a day on the CGC freeway) displaced by large-scale reductions in roadway capacity. The city did so partly by extending subway

lines. More important was the 2004 opening of seven new lines of exclusive median-lane buses (stretching 84 kilometers, later expanded to 162 kilometers) and 294 kilometers of dedicated curbside bus lanes. [Figure 2.7](#) shows the staged expansion of BRT services in Seoul.

Figure 2.7 Bus rapid transit corridors in central Seoul



Source: Cervero and Kang 2011.

Seoul's BRT investments have paid off handsomely. Bus operating speeds have increased from an average of 11 kilometers to more than 21 kilometers per hour, and speeds have increased in some passenger-car lanes ([table 2.4](#)). BRT buses, moreover, carry more than six times as many passengers per hour as buses operating on regular mixed-traffic lanes. And because they are less subject to the

vagaries of ambient traffic flows, buses operating in dedicated lanes have become more reliable: the travel-time variation of Seoul's BRT buses is, on average, five times less than that of buses operating on nonexclusive lanes (Seoul Development Institute 2005). Protected lanes have also reduced the number of accidents, which declined 27 percent one year after BRT services were introduced. Because of these service enhancements and safety improvements, ridership on BRT buses increased 60 percent faster than on non-BRT buses between 2004 and 2005.

Table 2.4 Operating speeds of cars and buses in Seoul before and after opening of exclusive median-lane bus lanes

(kilometers per hour)

Road	Before (June 2004)	After (August 2004)	Percentage change
<i>A</i>			
Bus (exclusive lane)	11.0	20.3	85.0
Car (other lane)	18.5	19.9	7.6
<i>B</i>			
Bus (exclusive lane)	13.1	22.5	72.0
Car (other lane)	20.3	21.0	3.4
<i>C</i>			
Bus (exclusive lane)	13.0	17.2	32.0
Car (other lane)	18.0	19.1	6.1

Source: Seoul Development Institute 2005.

Note: Seven new lines of exclusive median-lane buses were opened in 2004.

BRT was just one of several transit transformations introduced in Seoul. In the early 2000s, skyrocketing operating deficits prompted the metropolitan government to create a semipublic transit organization that set and enforced rules and standards on bus routes, schedules, and private operating practices. Many routes were reorganized into a timed-transfer and pulse-scheduling arrangement. All bus services were classified into one of four types of services: red (long-distance and intercity services), blue (trunk services), green (feeder services), and yellow (circular services). Red long-distance intercity lines link satellite cities with one another and downtown Seoul. Blue trunk lines connect the subcore and central-city Seoul. Green feeder buses mainly funnel passengers to subway stations and express bus stops. Yellow circular lines orbit the urban core.

Also introduced at the time of land reclamation was a sophisticated smart fare card that has allowed for efficient distance-based pricing and integrated bus-rail fares. A real-time traffic information system with message boards and in-vehicle navigation aids was installed to guide traffic flows and

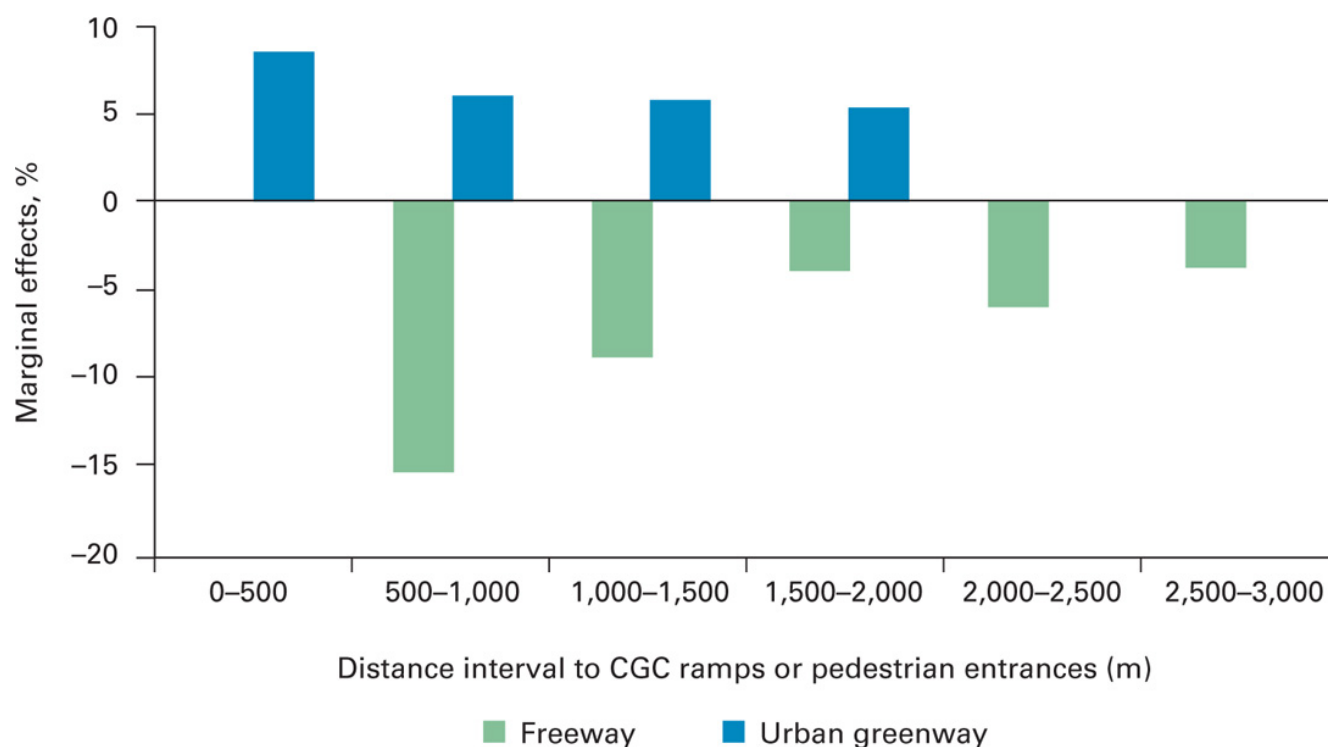
alert motorists to downstream hot spots. Curbside parking was substantially curtailed to help expedite traffic flows. More draconian was the introduction of a scheme that requires motorists to leave their cars at home once every 10 days (based on the last number of their license plate).

Land market responses to the BRT investments as well as projects like the CGC freeway-to-greenway conversion suggest that the net impacts have been positive (Kang and Cervero 2009; Cervero and Kang 2011). In a crowded, congested, and land-constrained city like Seoul, increased accessibility prompted property owners and developers to intensify land uses along BRT corridors, mainly by converting single-family residences to multifamily units, apartments, and mixed-use projects.

Land markets capitalized these accessibility gains, particularly in parcels used for condominiums and higher-density residential uses. Land price premiums of about 5–10 percent have been recorded for residences within 300 meters of BRT stops (Cervero and Kang 2011). For retail shops and other nonresidential uses, premiums were more varied, ranging from 3 percent to 26 percent over a smaller impact zone of 150 meters from the nearest BRT stop.

Greenways supported by expanded transit services further boosted land prices and development activities along high-amenity corridors. Housing prices within 3 kilometers of the elevated freeway fell, reflecting a disamenity; once the corridor was transformed to a greenway, prices of homes within 2 kilometers rose as much as 8 percent (Kang and Cervero 2009) ([figure 2.8](#)).

Figure 2.8 Residential property values before and after transformation of Seoul's Cheong Gye Cheon freeway into an urban greenway



Source: Kang and Cervero 2009.

Commercial parcels also increased following the freeway-to-greenway conversion, as did the concentration of high-value-added industries that hire white-collar professionals and “creative class” workers (Kang and Cervero 2009). Given these changes in prices, one could argue that residents of Seoul valued the quality of the urban space more highly than a freeway. Quality of place won out over automobility as a desirable urban attribute.

In addition to the attractions of greenery and open space, part of the land-value gains conferred by the freeway-to-greenway conversion likely also reflected indirect environmental benefits. Air pollution levels along the CGC corridor have fallen since the stream restoration and greenway conversion. Concentrations of fine-grained particulate matter (PM_{10}) along the corridor were 13 percent higher than Seoul’s regional average before the conversion; after the conversion, they were 4 percent below the region’s average (Seoul Development Institute 2005). Concentrations of nitrogen dioxide (NO_2), a precursor to the formation of photochemical smog, fell from 2 percent above the regional average when the freeway was in operation to 17 percent below the regional average after the greenway was in place.

Many urban centers suffer a heat-island effect, with temperatures higher than surrounding suburban and rural areas because of greater surface area coverage. Seoul is no exception. The spillover

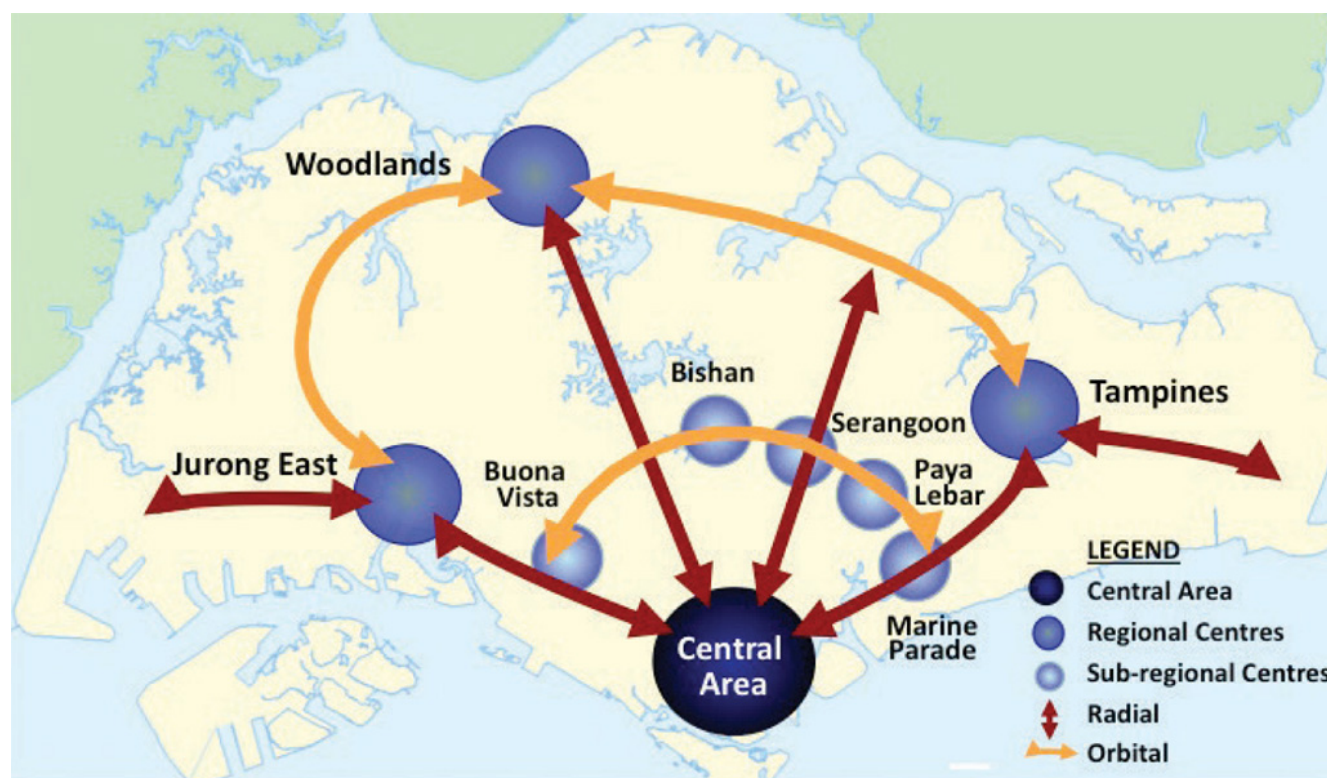
“cooling” benefits from the freeway-to-greenway conversion are revealed by a heat-island study that finds that ambient temperatures along the central-city stream were 3.3°C lower and temperatures along the greenway 5.9°C lower than along a parallel surface arterial five blocks away (Hwang 2006).

The longer-term consequences of Seoul’s bold experiment with urban land reclamation are yet to be seen. The hope and expectation of many urban planners and green politicians is that reclamation will slow the pace of new-town development, reduce the spatial mismatch between development north (primarily employment) and south (primarily housing) of the Han River, and spur redevelopment of former industrial land. More balanced infill development, planners hope, will make Seoul an attractive global city that appeals to international finance and businesses as well as tourists, professionals, and foreign investors.

Singapore: Transit-Oriented Development Empowered by Transportation Demand Management

The city-state of Singapore is internationally renowned for its successful integration of transit and regional development, placing the urbanized island of 5.1 million inhabitants on a sustainable pathway both economically and environmentally. Its transformation over the post–World War II period from a backwater port awash in third-world poverty to a dynamic, modern, industrialized city-state has been remarkable.

As part of a national economic development strategy, Singapore has embraced Scandinavian planning principles that call for radial corridors that interconnect the central core with master-planned new towns. Its structure plan, called the Constellation Plan, looks like a constellation of satellite “planets,” or new towns, that surround the central core, interspersed by protective greenbelts and interlaced by high-capacity, high-performance rail transit. Radial rail links interconnect Singapore’s high-rise urban core with the hierarchy of subcenters, and a looping mix of heavy and light rail lines connects the subcenters ([figure 2.9](#)).

Figure 2.9 Singapore's Constellation Plan

Source: Singapore Land Transport Authority 2008; reproduced with permission.

Like Stockholm and Copenhagen, this rail-served settlement pattern has produced tremendous transportation benefits. VKT per capita is among the lowest of any urbanized region in the world with per capita GDP of more than \$25,000, and the annual number of transit trips per capita is high (484 in 2006) (UITP 2006).

Singapore adopted the approach of building new towns that are not independent, self-contained units but rather nodes with specialized functions that interact with and depend on one another. Some satellite centers are primarily industrial estates, others are predominantly dormitory communities; most are mixed-use enclaves. About three-quarters of residents of master-planned new towns work outside their town. Most, however, commute within the radial corridor that connects their town to Singapore's central business district. Travel is thus predominantly within, not between, rail-served corridors. The dispersal of mixed land uses along corridors has created two-way travel flows and spread travel demand more evenly throughout the day.

Singapore's progressive "transit first" policies complement its transit-oriented Constellation Plan. The city has introduced a three-tier fiscal program that comes as close to "getting the prices right" within the urban transport sector as any city in the world. The first tier of charges is subscription fees for owning a car. Made up of high registration fees, import duties for automobile purchases, and a

licensing surcharge based on a quota system (a certificate of entitlement that is indexed to congestion levels), these charges principally cover the fixed costs associated with providing basic levels of road infrastructure and parking facilities. The second tier of charges is use related, in the form of fuel taxes and parking fees. These charges cover the incremental costs for scaling road capacity to traffic volumes and maintaining roadway infrastructure. The third set of charges—in the form of real-time ERP—forces motorists to internalize the externalities they impose when using their cars during peak hours. Fees fluctuate according to congestion levels, forcing motorists to bear some of the costs they impose on others in the form of time delays and air pollution. Within a month of initiating ERP in 1998, traffic along a main thoroughfare fell 15 percent and average rush hour speeds rose from 36 to 58 kilometers per hour. Vehicle quotas, congestion prices, and an assortment of fees and surtaxes (which add as much as 150 percent to a car's open market value) have reduced the annual growth of Singapore's vehicle population from 6 percent in 1997 to less than 3 percent in 2010—a remarkable achievement for a city in which per capita incomes have risen faster over the past two decades than virtually anywhere in the world.

Charging motorists more to own and use cars is but one form of transportation demand management (TDM) found in Singapore. As in Europe, car sharing has gained a foothold in Singapore, the only Asian city where this is the case. Singapore also has an off-peak vehicle licensing scheme that allows vehicles holding such licenses to be used only during the morning and evening off-peak periods Mondays through Saturday and any time Sunday.

Although higher prices and TDM have boosted transit usage, their influences are being eclipsed by rising income, which continues to push up Singapore's rates of car ownership and motorization. Singapore has among the most affordable housing (thanks to government provisions of mass-produced units), freeing up personal income for the second most costly durable good purchased by households, the private car. In the early 1990s, the ratio of the average housing price to average annual household income (2.3) was far lower than the ratio of the average price of a new car to income (3.7). Over the 1974–95 period, the price elasticity for car ownership was -0.45 , compared with an income elasticity of 1.00, according to one study (Chu, Koh, and Tse 2004). Even if automobile prices increase at twice the rate of household incomes, such elasticities suggest that motorization rates will continue to rise in Singapore.

Rising congestion is reflected by statistics on the density of cars on land-constrained Singapore's fairly fixed supply of road supply. In 1995 (the year the vehicle quota system was introduced), the

number of vehicles per kilometer of road was 180; by 2010, the figure had risen to 250. Car ownership increased by only 11 percent between 2000 and 2005; between 2005 and 2010, it increased 39 percent (Singapore Government n.d.).

In the face of this rapid rise in car ownership, Singapore is turning to higher congestion tolls as a way to temper motorization. The logic of raising congestion tolls is expressed in the long-term master plan: “Although congestion charges such as ERP encourage motorists to consider whether and when to drive, ownership costs are sunk costs and may in fact result in motorists driving more rather than less. Hence, as we expand the electronic road pricing (ERP) system, we will continue to shift the focus of our demand management strategies from ownership taxes to usage charges” (Singapore Land Transport Authority 2008, p. 57).

Singapore’s centralized form of governance has allowed land development and transit services, which are overseen by different authorities (the Urban Redevelopment Authority and the Land Transport Authority), to be closely coordinated, both institutionally and financially. Revenues generated from high vehicle ownership and usage charges, for instance, go to the general treasury, which channels them into vastly enhanced and expanded transit services as well as the construction of the armature of rail TODs (for example, sidewalk networks, civic squares, bus staging areas). Because of the island-state’s world-class transit service offerings and TOD built form, congestion tolls are politically possible, as for a significant share of trips, travel times are lower using public transit than private cars. Of 8.9 million daily motorized trips made in Singapore in 2010, 4.5 million were by rail or bus transit (Singapore MRT 2011). Long-range planning goals call for raising this share to two-thirds.

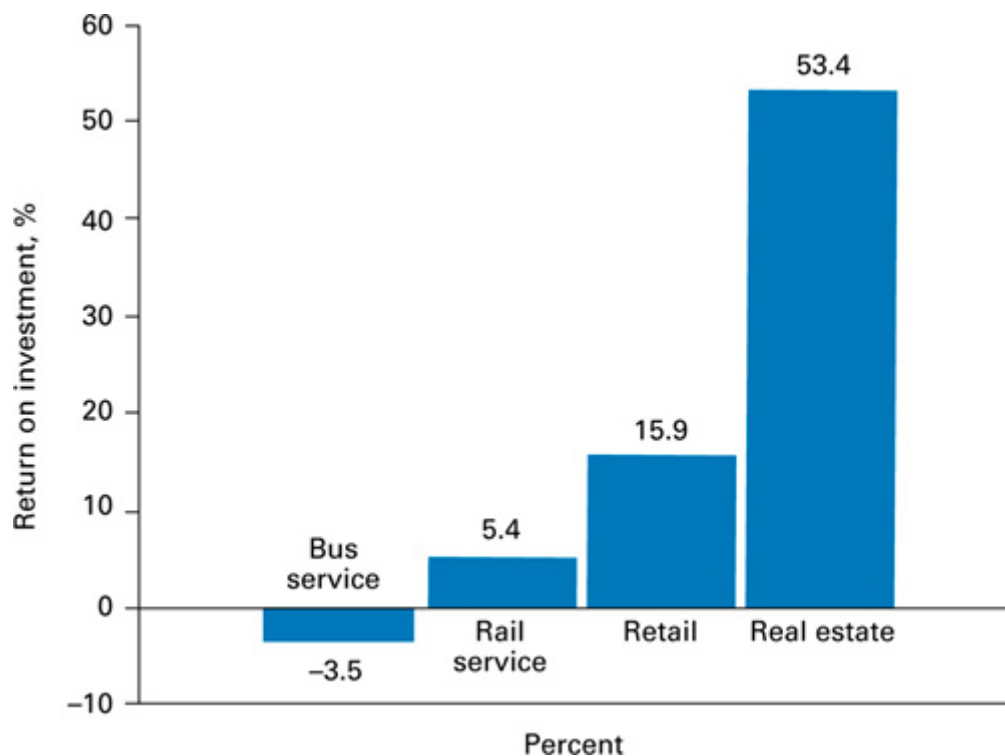
The role of rail transit in capturing larger shares of motorized trips has increased, and it is slated to continue to do so in coming years. The length of Singapore’s world-class rapid transit system more than doubled, to 138 kilometers in 2011 from 67 kilometers in 1990, leading to a doubling of ridership, to nearly 2 million riders a day, from a little under 1 million in 1998. In 1999, Singapore added automated light rail services to the mix, with trackage increasing from 8 kilometers in 1999 to 29 kilometers in 2010. Bus ridership in 2011 was only 1.5 times higher than total rail ridership, down from 3 times higher than rail in 1998. Singapore’s latest land transport master plan, released in 2008, embraces “making public transport a choice mode” and “managing road usage” as strategic thrusts toward retaining its status as a world-class transit metropolis.

Tokyo: Public-Private Urban Regeneration

Tokyo's position as a global transit metropolis mimics that of Hong Kong SAR, China, and Stockholm, albeit in different ways. As in Hong Kong SAR, China, the private sector has historically linked transit investments and urban development through value capture mechanisms; as in Stockholm, the focus of the transit/land-use nexus has been on urban regeneration rather than new-town development in recent years, albeit more for market than urban policy reasons.

Tokyo's railway network—owned and operated by a mix of public, private, and quasi-private entities—is by far the world's largest. Most of the region's extensive network of suburban railway lines was built by private companies that received government concessions and exclusive rights to design, build, and operate rail services. Tokyo's railway companies have historically leveraged real estate development to both pay for infrastructure and produce profits for shareholders. They have opened convenience stores and shopping malls within and adjacent to stations.

What most distinguishes Tokyo's railway companies is their construction of large-scale new towns on once virgin lands (Cervero 1998). West of central Tokyo, where many of the region's most up-market suburbs are located, entire communities are the domains of powerful conglomerates that are best known for their department store chains—Keio, Odakyu, Seibu, and Tokyu—but which first and foremost are in the business of railway and real estate development. All started as private railway companies that over time branched into businesses closely related to the railway industry, including real estate, retailing, bus operations, and electric power generation. Such business expansion made good economic sense; placing shopping malls, apartments, and entertainment complexes near stations generated rail traffic, and railways brought customers to these establishments. During the 1980s, at the height of railway/new-town co-development and a surge in Japanese real estate prices, railway companies were earning investment returns on ancillary real estate projects in the range of 50–70 percent, with profit margins from real estate far outstripping profits from transit services ([figure 2.10](#)).

Figure 2.10 Rates of return of private railway corporations in metropolitan Tokyo, 1980–96

Source: Cervero 1998.

Note: Data cover 1980–96.

Tokyu Corporation is Greater Tokyo's largest private railway enterprise. It was among the first companies to advance the business model of railway/new-town co-development. From 1960 to 1984, Tokyu Corporation's 23-kilometer rail line transformed a vast, hilly, scarcely inhabited area into a planned community, Tama Den-en Toshi (Tama Garden City), of a half million residents. Tokyu used land consolidation techniques to assemble farmland at cheap prices in advance of rail construction and to finance neighborhood infrastructure. Under this approach, landowners formed a cooperative that consolidated (often irregularly shaped) properties and returned them as smaller but fully serviced (usually rectangular) parcels. Roads, drainage, sewerage, parks, and other infrastructure were funded through the sale of the "extra" reserved land contributed by cooperative members. Land consolidation relieved railway companies such as Tokyu from the up-front burden and risks of acquiring land and financing infrastructure.

A new era for Tokyo's private railway companies began in the 1990s, when the bursting of Japan's real estate price bubble saw the market valuations of rail companies' landholdings fall. Powerful demographic trends, such as declining birth rates and an aging population, combined with the slowing

of the economy, reduced the demand for new-town construction. To spread the risks of a shakier real estate market, private railway companies partnered with third parties to pursue large-scale development projects. Recent real estate projects of Tokyu Corporation, for example, have relied on Real Estate Investment Trust (REIT) funding.

Changing traffic conditions have also had a hand in changing the portfolios of Tokyo's private railways. Greater Tokyo's rail-served new towns and subcenters featured housing and retail services, but most white-collar jobs remained in the urban core (Cervero 1998; Sorensen 2001). The result was tidal radial patterns of commuting, which exacerbated traffic congestion in the urban core. Long commutes combined with crowded trains and roadways helped trigger a return-to-the-city movement. Several large-scale redevelopment projects are now underway as joint ventures between private railways and real estate companies, targeted at young professionals, empty-nesters, and other less traditional niche markets drawn to central-city living. In a break from tradition, when buildings above subway stations were exclusively office or commercial projects, major subway stations now feature high-end housing and consumer services. Residential and commercial districts around several central-city stations, notably Akihabara, Shibuya, Shinagawa, and Shinjuku, are now abuzz with activity 24/7 (24 hours a day, 7 days a week) (figure 2.11).

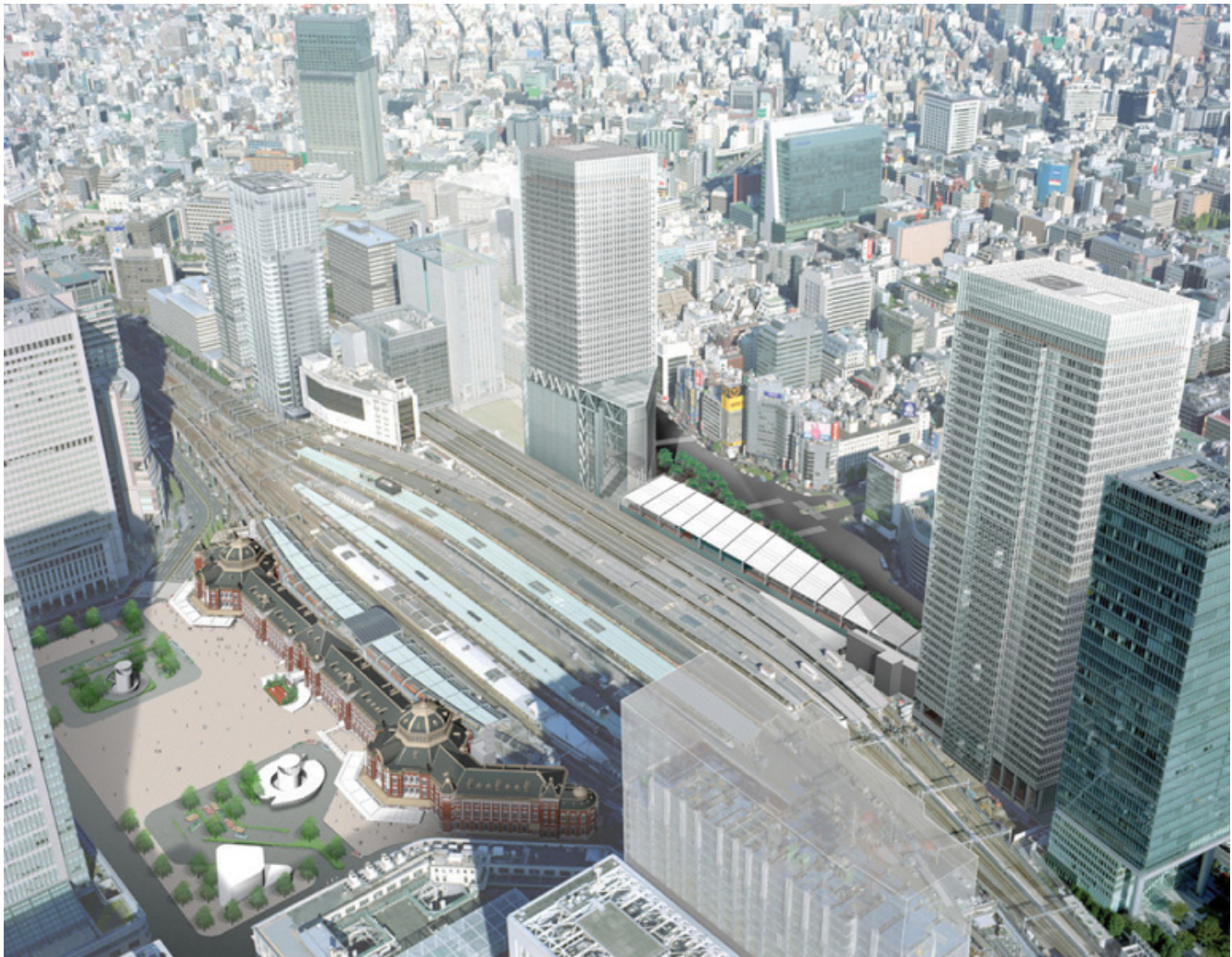
Figure 2.11 Street life near Shinjuku metrorail station, Tokyo



Photo by Robert Cervero.

Tokyo's two former public railways, JR East and Tokyo Metro, are also pursuing the redevelopment and infilling of strategic central-city land parcels. In the case of JR East, mounting fiscal losses incurred by the former Japan National Railway (with an accumulated debt of \$300 billion) led to privatization in 1987. At the time, the national government gave JR East large developable land parcels around terminal stations, prime for commercial redevelopment. Borrowing from the practices of the Tokyu Corporation and other private railway consortia, JR East and Tokyo Metro aggressively transformed these properties into high-rise commercial ventures. In 2006, real estate yielded returns on investment of more than 40 percent for both former public railways.

JR East's showcase real estate project is Tokyo Station City, developed jointly with other private interests. Tokyo Station City features high-rise, class-A office buildings; retail centers; and hotels ([figure 2.12](#)). It is well suited for large-scale redevelopment, thanks to the large amount of buildable space above depots as well as high pedestrian traffic volumes. On a typical weekday in 2005, about half a million passengers passed through Tokyo station (JR East 2005).

Figure 2.12 Tokyo Station City

Source: Japan Railway Corporation East (<http://www.jreast.co.jp>; reproduced with permission).

Note: This high-rise commercial project mixes the old and new, with the restored historical station (lower left) flanked by towering office, retail, and hotel structures.

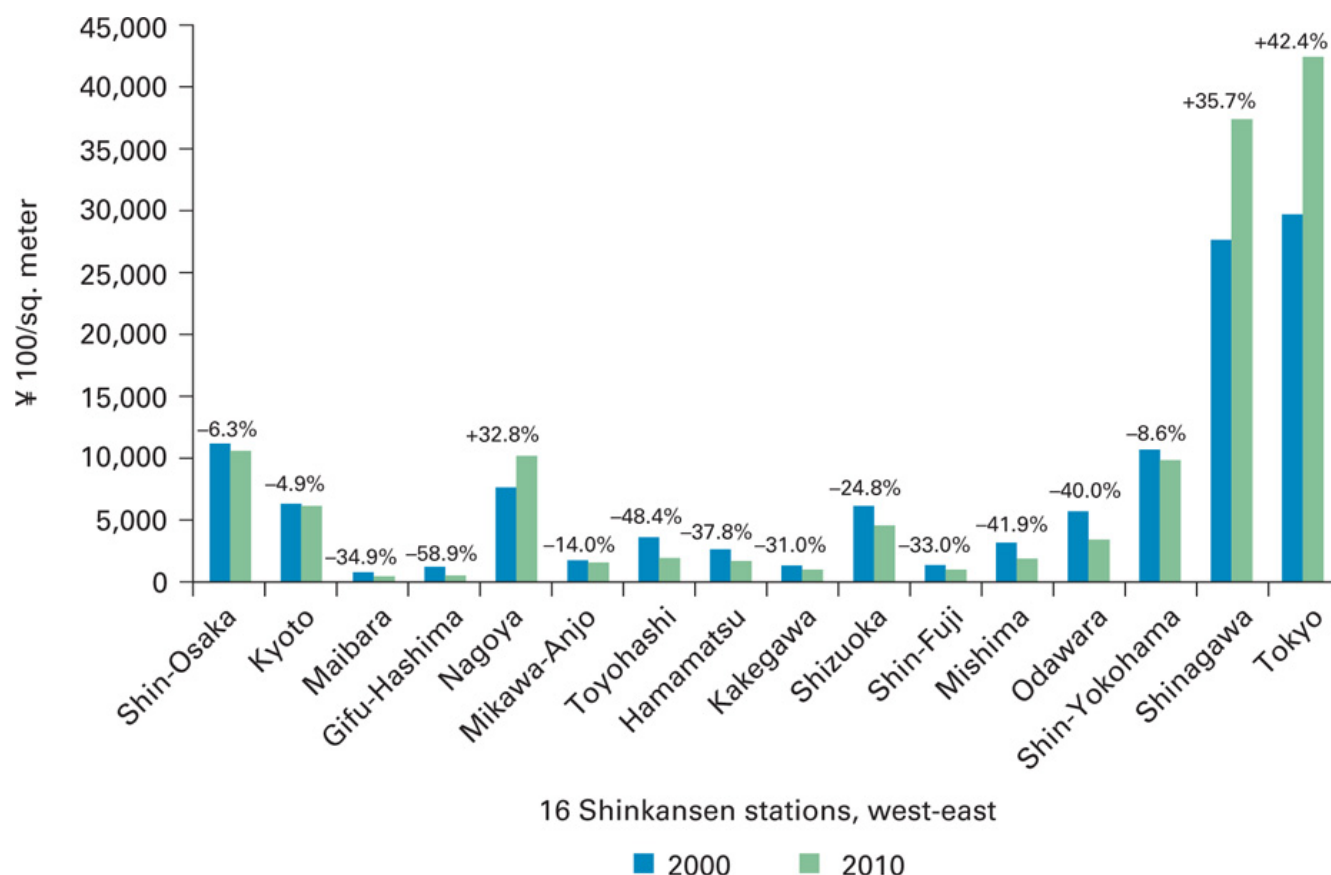
Like MTRC in Hong Kong SAR, China, Tokyo's private railways are clearly responding to market price signals, as indicated by an analysis of 2005 residential land prices along 16 mostly private railway corridors as a function of distance to central Tokyo. Within and along the Yamanote loop, where most large-scale redevelopment projects have been built on land owned by private railway companies, residential prices are generally twice what they are 15–20 kilometers from the center. Since 2000, the only area where residential land has gained value has been around terminal stations on the Yamanote loop.

It is at intermodal rail stops served by high-speed rail—notably, the Tokaido Shinkansen line—where urban development activities in Greater Tokyo have been most prominent. Around the newly opened Shinagawa Shinkansen station in central Tokyo, for example, the metropolitan government,

the national government, the privatized Central Japan Railway Company, and private real estate developers joined forces to co-develop prestigious office towers and shopping malls. The project featured high-quality public green plazas and well-designed pedestrian circulation systems as a lure to firms and workers that place a premium on livability and are drawn to urban amenities when deciding where to open a business or to work.

These joint transit–commercial redevelopment efforts aim not only to increase the number of business passengers on the Tokaido Shinkansen line, but also to raise the potential of land value capture around the terminal stations. Nagoya, Shinagawa, and Tokyo and have considerably increased commercial land values within 5 kilometers of Shinkansen stations where large-scale redevelopment projects were delivered through public-private partnerships ([figure 2.13](#)). Compared with many private intercity railway corporations in Osaka and Tokyo, the former Japanese National Railways was historically reticent to capture land development benefits around the Nagoya Shinkansen stations. However, in response to Japan’s urban regeneration boom in the past decade, real estate revenue streams of the privatized Central Japan Railway Company (JR Central) soared—from ¥24.3 billion (\$0.21 billion) in fiscal 1999 to ¥66.7 billion (\$0.74 billion) in fiscal 2009—largely from the new commercial property packages redeveloped around the Nagoya Shinkansen station (JR Central 2011).¹

Figure 2.13 Average commercial land values within five kilometers of Shinkansen line stations in Tokyo, 2000 and 2010



Source: Authors, based on data from Japan's Ministry of Land, Infrastructure, Transport and Tourism 2011.

Note: Percentages at top of bars represent the average percentage change in land values between 2000 and 2010.

Washington, DC, and Arlington County, Virginia: A Transit-Development Success Story

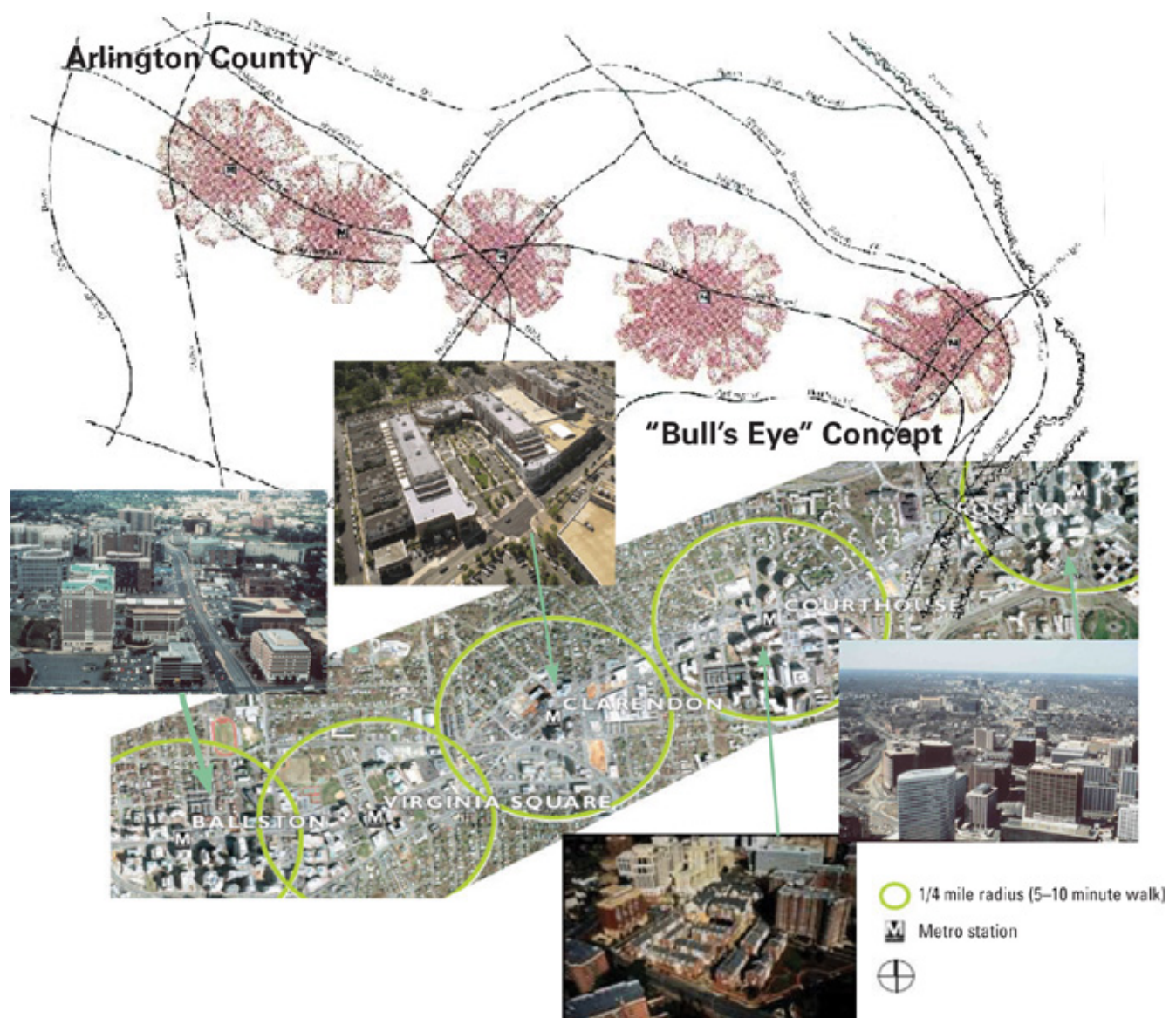
More growth has occurred near metropolitan Washington, DC's heavy rail system in the past quarter century than anywhere in the United States. From 1980 to 1990, 40 percent of the region's office and retail space was built within walking distance of a Metrorail station (Cervero and others 2004). The fact that the timing of the railway investment (late 1970s through 1980s) coincided with a rapid period of growth (more jobs were added in metropolitan Washington, DC than anywhere in the United States) helped steer new development to rail-served corridors. Combined with height limit restrictions within the District of Columbia and federal policy that mandates that government offices be located near rail stations, these factors encouraged TOD.

The recipient of most spillover growth from Washington, DC has been Arlington County, Virginia, just across the Potomac River (Cervero and others 2004). This 26-square-mile county just south of the nation's capital has experienced a tremendous increase in building activity since the opening of Washington Metrorail in 1978. More than 25 million square feet of office space, 4 million square feet

of retail space, 25,000 mixed-income dwelling units, and 6,500 hotel rooms were built over the past three decades.

Much of this growth has been wrapped around Metrorail stations, guided by a “necklace-of-pearls” vision articulated in the “bull’s eye” concept plan adopted in the late 1960s ([figure 2.14](#)). Of the nearly 190,000 people living in Arlington County, 26 percent reside within a Metrorail-served corridor (roughly a quarter mile walk shed from the station), even though these corridors represent only 8 percent of county land area. If the development added to these two corridors had been built at suburban density standards, as in neighboring Fairfax County, Virginia, seven times as much land area would have been required.

Figure 2.14 “Bull’s eye” concept of rail development in Arlington County, Virginia



Source: Cervero and others 2004.

Note: The bull's eye concept targets urban growth along the Rosslyn-Ballston corridor. Over the entire corridor, most land uses and urban activities found throughout the greater metropolitan area are clustered within a five-minute walk of one of the five stations. Rosslyn is predominantly an office center; Courthouse functions as a government center; Clarendon features retail and commercial uses; Virginia Square has museums and performing arts centers; and Ballston has offices, hotels, stores, condominiums, and government buildings. Collectively, the mix of land uses along the Rosslyn-Ballston corridor ensures that during peak hours, Metrorail trains are full in both directions, as passengers travel up and down the corridor.

The transformation of once rural Arlington County into a showcase of compact, mixed-use TOD has been the product of ambitious, laser-focused station-area planning and investment. Before Metrorail,

planners in Arlington County understood that high-performance transit provided an unprecedented opportunity to shape growth. They introduced various strategies—targeted infrastructure improvements, incentive zoning, development proffers, and permissive and as-of-right zoning—to entice private investment around stations. After preparing countywide and station-area plans on desired land-use outcomes, density and set-back configurations, and circulation systems, they changed zoning classifications, allowing developments that complied with these classifications to proceed unencumbered. The ability of complying developers to create TODs “as-of-right” (which ensures the issuance of development permits as long as projects adhere to the requirements of local plans) was particularly important. It allowed developers to line up capital, secure loans, incur up-front costs, and phase in construction without fear that the local government might change its mind regarding permitted uses, building densities, and the like.

Another key factor that gave rise to TODs was the decision not to set Arlington County’s Metrorail rail corridor in the median of Interstate 66. Instead, county officials persuaded the region’s transit authority to align the corridor in traditional urban centers to help jump-start the process of urban regeneration, even though doing so substantially increased construction costs.

Arlington County’s transit ridership statistics reveal the payoff of concentrated growth along rail corridors. The county boasts one of the highest percentages of transit use in the Washington region, with 39.3 percent of Metrorail corridor residents commuting to work by public transit—twice the share among county residents who live outside the Metrorail corridors. At several apartments and condominium buildings near the Rosslyn and Ballston stations, about two-thirds of employed residents take public transit to work.

An important outcome of promoting mixed-use development along rail corridors has been balanced jobs and housing growth, which in turn has produced balanced two-way travel flows. Counts of station entries and exits in Arlington County are nearly equal during both peak and off-peak hours. As a result, trains and buses are largely full in both directions. The presence of so much retail, entertainment, and hotel activity along the county’s Metrorail corridors has filled trains and buses during midday and on weekends. Arlington County averages more passengers at its stations in off-peak hours than other jurisdictions in the region except downtown Washington, DC. Balanced mixed-use development has translated into as close to a 24/7 ridership profile as any U.S. setting outside a central business district.

Part of the credit for the transit-oriented growth of Arlington County is the entrepreneurial leanings of the region’s rail authority. Over the past two decades, the Washington Metropolitan Area Transit Authority (WMATA)—an independent regional transportation authority responsible for designing,

building, and operating the region's rail transit and public bus services—has aggressively sought to recapture value through joint development activities. Metrorail's joint development program of air-rights leases and station connection fees generate about 2 percent of the system's annual revenues; increased ridership—and thus fare-box intake—at least doubles this percentage (Cervero and others 2004).

Proactivism accounts for much of WMATA's joint development success. A vital step in WMATA pursuing value capture was the creation of a real estate development department within the agency at the very beginning, before the construction and opening of railway services. By hiring seasoned real estate professionals to staff and manage this department, WMATA positioned itself to seek out remunerative joint development possibilities. Private-sector experiences helped create a more entrepreneurial approach to land development than is found at most U.S. transit agencies. Staff members were given the financial resources to purchase land around planned rail stations on the open market, often before formal plans were announced and thus at fairly reasonable prices. Rather than waiting and reacting to developer proposals, WMATA's real estate office actively sought out mutually advantageous joint development opportunities. With financial and institutional support from board members, WMATA's real estate office has over time amassed an impressive portfolio of landholdings, much of it purchased on the open market. To date, WMATA had undertaken more than 30 development projects at a value of more than \$2 billion on land the agency owns.

Over time, WMATA has refined its joint development activities. Station sites are carefully screened according to a set of criteria that gauge development potential. For sites selected, a request for proposals is issued to solicit developer interests. Through negotiations, a developer team is chosen and contracts specifying the financial terms of the deal are drawn up. With the help of a private real estate firm, WMATA now rates potential sites according to the likely degree of private sector interests and development constraints.

Because joint development is controlled by a transit agency, it is the form of value capture that has the greatest potential for financing rail investments in much of the world. It requires, however, an institutional capacity to expand beyond the traditional mission of transit agencies (that is, building and operating transit services) to venture into other entrepreneurial realms, such as real estate development and co-development. As suggested by experiences in metropolitan Washington, institutional reforms are often a necessary part of any successful transit value capture scheme.

Integrating Transit and Land-Use Planning through Adaptive Transit and Bus Rapid Transit

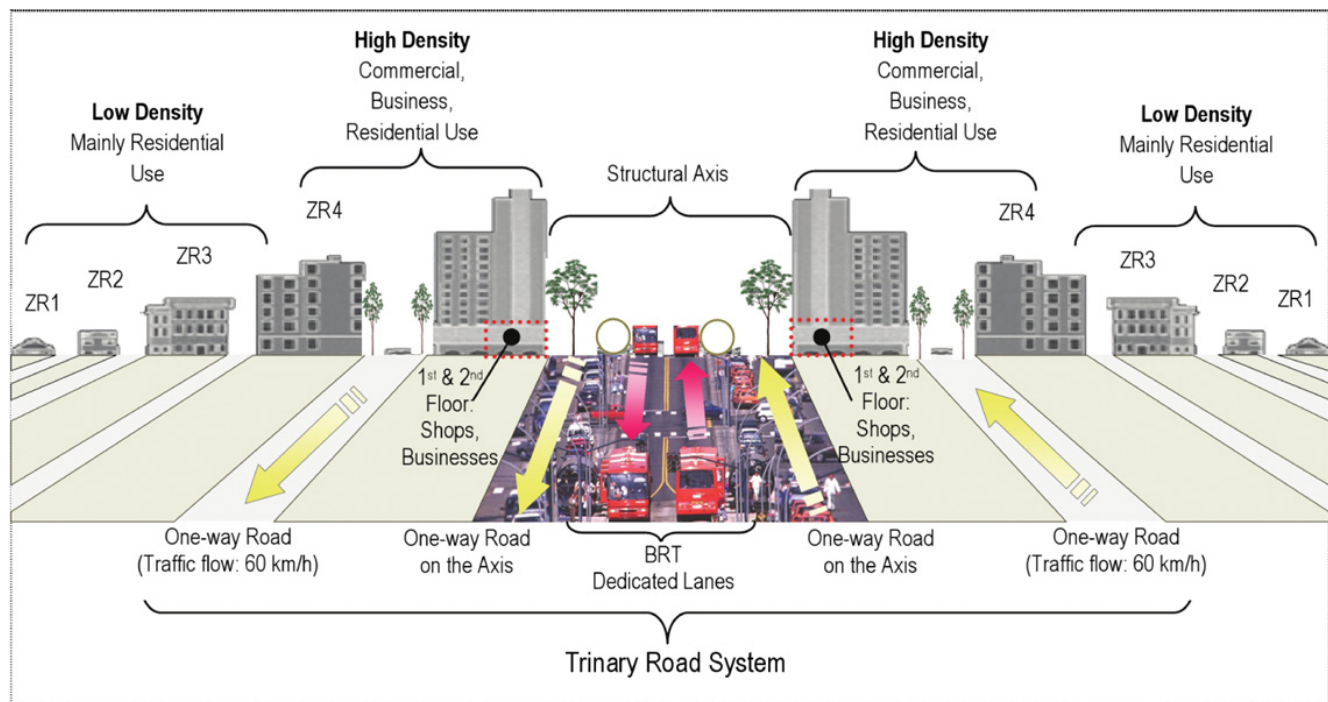
The two cases reviewed in this section—Curitiba and Ottawa—are given the “adaptive transit” moniker mainly because they turned to an inherently flexible technology—the rubber-tire bus—as high-capacity transit carriers. Both cities feature high-rise, mixed-use development around a number of BRT stations but also retain many lower-density neighborhoods that are served by feeder lines. Thus, they rely on adaptive bus-based transit to provide a versatile mix of transit services while also supporting rail-like TOD.

Curitiba, Brazil: The Lineal Bus Rapid Transit Metropolis

Curitiba is internationally renowned as one of the world’s most sustainable, well-planned cities, in large part because of its success at integrating BRT investments and urban development. The city’s experiences underscore the environmental benefits of balancing urban growth along bus-served linear axes and aggressively pursuing a “transit first” policy. By emphasizing planning for people rather than cars, Curitiba has evolved along well-defined radial axes that are intensively served by dedicated busways. Along some corridors, streams of double-articulated (that is, two accordion section) buses haul 16,000 passengers an hour, comparable to the number much pricier metrorail systems carry. The city’s system of 390 routes served by 2,000 vehicles carries 2.1 million passengers a day, twice as many as in 1990. To ensure a transit-oriented built form, Curitiba’s government mandates that all medium- and large-scale urban development be sited along a BRT corridor. Orchestrating regional growth is the Institute for Research and Urban Planning (IPPUC), an independent entity charged with ensuring integration of all elements of urban growth.

A design element used to enhance accessibility in Curitiba is the “trinary”—three parallel roadways with compatible land uses and building heights that taper with distance from the BRT corridor ([figure 2.15](#)). Zoning ordinance and urban design standards promote ridership productivity and environmental quality. The first two floors of the busway that do not count against permissible plot ratios (building height/land area) are devoted to retail uses. Above the second floor, buildings must be set back at least five meters from the property line, to allow sun to cast on the transitway. The inclusion of upper-level housing entitles property owners to density bonuses, which has led to vertical mixing of uses within buildings. An important benefit of mixed land uses and transit service levels along these corridors, in addition to extraordinarily high ridership rates, has been balanced bidirectional flows, ensuring

efficient use of bus capacity, just as in Stockholm and Arlington County, Virginia. The influences of the trinary structure on channeling trips are revealed by 2009 origin-destination statistics, which indicate that 78 percent of passengers boarding at the terminus of the north-south corridor are destined to a bus stop on the same corridor (Duarte and Ultramari 2012).

Figure 2.15 Curitiba's trinary road system

Source: Suzuki and others 2010; reproduced with permission of the Transportation Research Board.

Note: Bi-articulated buses operate alongside sidewalks and slow-speed frontage roads. Ground-floor retail on the lower floor of buildings and upper-level office and residential space—set back several meters to allow sunlight to reach the street level—creates a rich mix of land uses along trinary corridors. The inclusion of mixed land uses and affordable housing allows developers to increase building heights, adding density to the corridor. Streetscaping enhancements have created a pleasant pedestrian milieu. Frequent-stop, bi-articulated buses operate on the mainline of the trinary corridor; limited-stop “speedy” buses flank parallel one-way roads. People traveling longer distances often opt for the limited-stop buses, whereas people traveling shorter distances or going to destinations bypassed by the speedy buses usually patronize the mainline frequent-stop services. The market has responded to this rich, differentiated mix of bus-based services along the corridor with high-rise development, supported by zoning and public improvements like sidewalks and landscaping.

Curitiba is one of Brazil’s wealthiest cities, yet it averages considerably more transit trips per capita than Rio de Janeiro and São Paulo, which are much larger. Its share of motorized trips by transit (45 percent) is the highest in Latin America (Santos 2011). High transit use has appreciably shrunk the city’s environmental footprint. Curitiba’s annual congestion cost per capita of \$0.67 is a fraction of São Paulo’s (\$7.34) (Suzuki and others 2010). The city also boasts the cleanest air of any Brazilian

city with more than 1 million inhabitants, despite being a provincial capital with a sizable industrial sector. The strong, workable nexus that exists between Curitiba's bus-based transit system and its mixed-use linear settlement pattern deserves most of the credit.

Sustained political commitment has been an important part of Curitiba's success. The harmonization of transit and land use took place over 40 years of political continuity, marked by a progression of forward-looking, like-minded mayors who built on the work of their predecessors. A cogent long-term vision and the presence of a politically insulated regional planning organization, the IPPUC, to implement the vision have been crucial in allowing the city to chart a sustainable urban pathway.

In recent years, Curitiba has begun to experience the limits of rubber-tire technologies. With buses operating on 30-second intervals on main routes during peak hours, bunching problems have disrupted and slowed services. Veritable "elephant trains" of buses have increased operating costs and precluded the kinds of economies of scale enjoyed by trains operated by a single driver. Extreme overcrowding has prompted many middle-class car-owning travelers to switch to driving. In the words of one urban planner, "Many Curitibaños view [BRT] as noisy, crowded and unsafe. Undermining the thinking behind the master plan, even those who live alongside the high-density rapid-bus corridors are buying cars" (Lubow 2007, p. M8). A long-discussed light rail line, to replace overcrowded buses, has yet to gain momentum, mainly because of cost concerns.

The Green Line is the city's first new BRT corridor in years, an 18-kilometer corridor that was converted from a federal highway. Like Bogota's celebrated BRT, the Green Line has passing lanes, which greatly increase capacity by supporting express services.

Meanwhile, recent legislation has taken bold steps to alter zoning and land use along the Green Line to promote TOD. Formerly a national highway dotted with truck stops and lumberyards, this hodgepodge of industrial uses is slated to become a pedestrian-friendly, mixed-use corridor that can accommodate up to half a million new residents.

As important to the enrichment of services in Curitiba is an evolved view of BRT corridors as rights of way that also accommodate linear parks (strips of long park areas located to next to railways, bus lines, roads, canals, or rivers) and bike paths. A law passed in 2010 promotes the preservation of green space along the corridor by giving developers increased building rights if they purchase or preserve land along the corridor as parks.

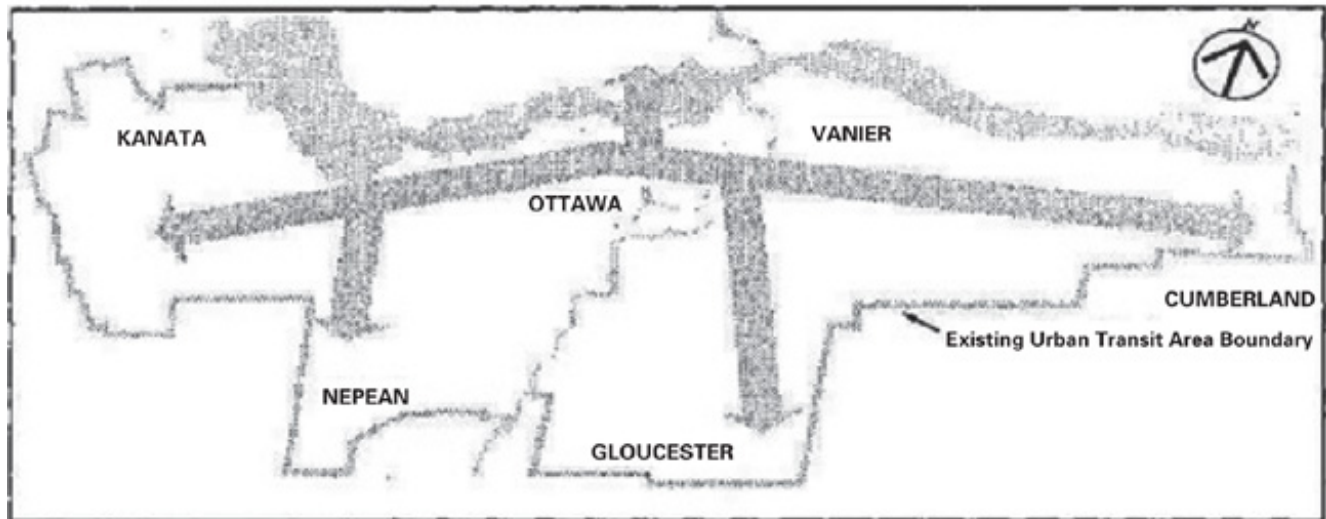
Ottawa: North America's First High-End Bus Rapid Transit

North America's best example of BRT-based TOD comes from Canada's capital, Ottawa. Borrowing a chapter from Scandinavian metropolises like Copenhagen and Stockholm, Ottawa's leaders began

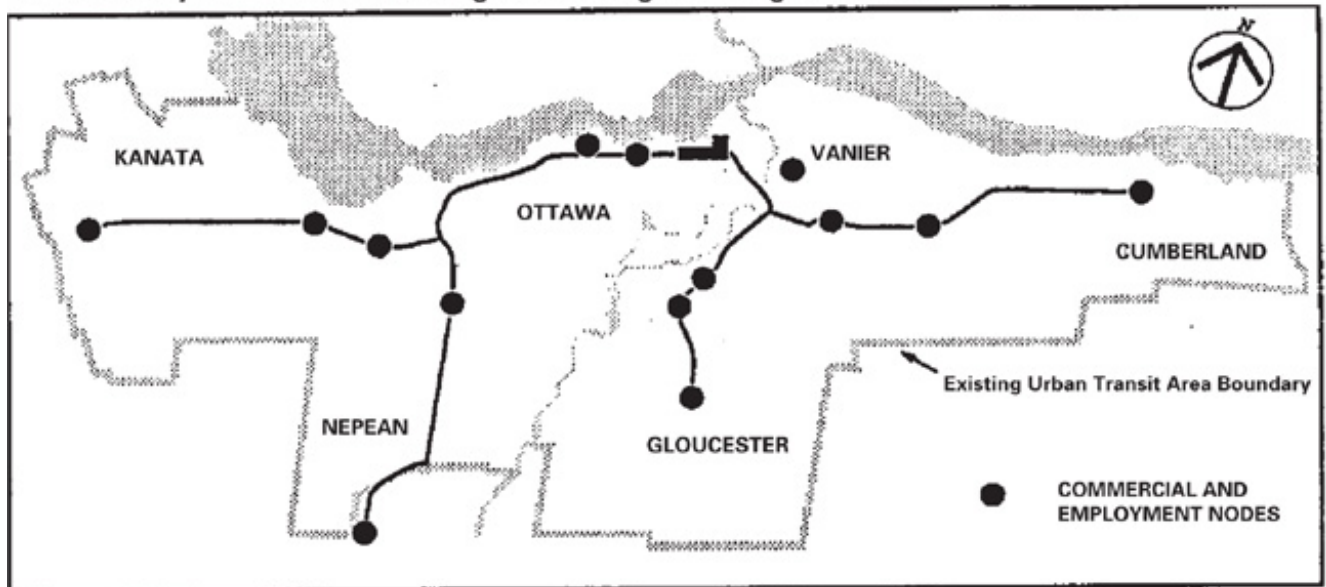
with a concept plan that defined desired growth axes and then strategically invested in a high-quality, high-capacity transitway to drive growth along these corridors (figure 2.16). As a result of supportive zoning and world-class bus services, growth gravitated to bus corridors between 1985 and 2000.

Figure 2.16 Transitway development in Ottawa, Canada

a. Urban form vision of desired growth axes



b. Transitway investment to drive growth along desired growth axes



Source: Cervero 1998; reproduced with permission from Island Press, Washington, DC.

Institutional factors partly account for Ottawa's success in implementing a long-range vision. A regional planning body, the Regional Council, was formed in 1969 to carry out comprehensive planning, invest in major infrastructure, and provide regional services such as air quality management. A checks-and-balances system is in place in which localities oversee land-use decisions, which the regional authority can override if it considers them incompatible with the regional plan. In practice, the regional authority rarely overrules the wishes of municipalities.

Based on broad-based citizen input, in 1974, the Regional Council endorsed a multicentered urban structure: downtown Ottawa would retain its position as the dominant commercial, employment, and cultural center of the region, and it would be surrounded by a hierarchy of primary and secondary urban centers, interconnected by high-quality transit. Market-driven (predominantly low-density) patterns of development would be permitted outside these centers.

The chief instrument for achieving this desired physical form was the busway. With a vision in place and the agreement to build a busway to make the vision a reality, the Regional Council turned its attention to land-use management. TOD policies were introduced that called for substantial increases in the share of regional jobs located near Transitway stations. The long-term goal called for 40 percent of the region's jobs to be within walking distance (400 meters) of the Transitway. The two principal suburban catchments for job growth, the Orléans and Kanata urban centers, were slated to host more than 10,000 new jobs. The official plan also required that regional shopping centers with more than 375,000 square feet (34,840 square meters) of gross leasable space be sited near the Transitway or future extensions.

Ottawa officials also adopted a “transit first” policy early in their BRT planning: improvements to the existing transit system and the development of rapid transit were to take precedence over all forms of road construction and widening. The regional plan specifically called for creating rapid transit services. No commitments were made on preferred routing or transit technologies.

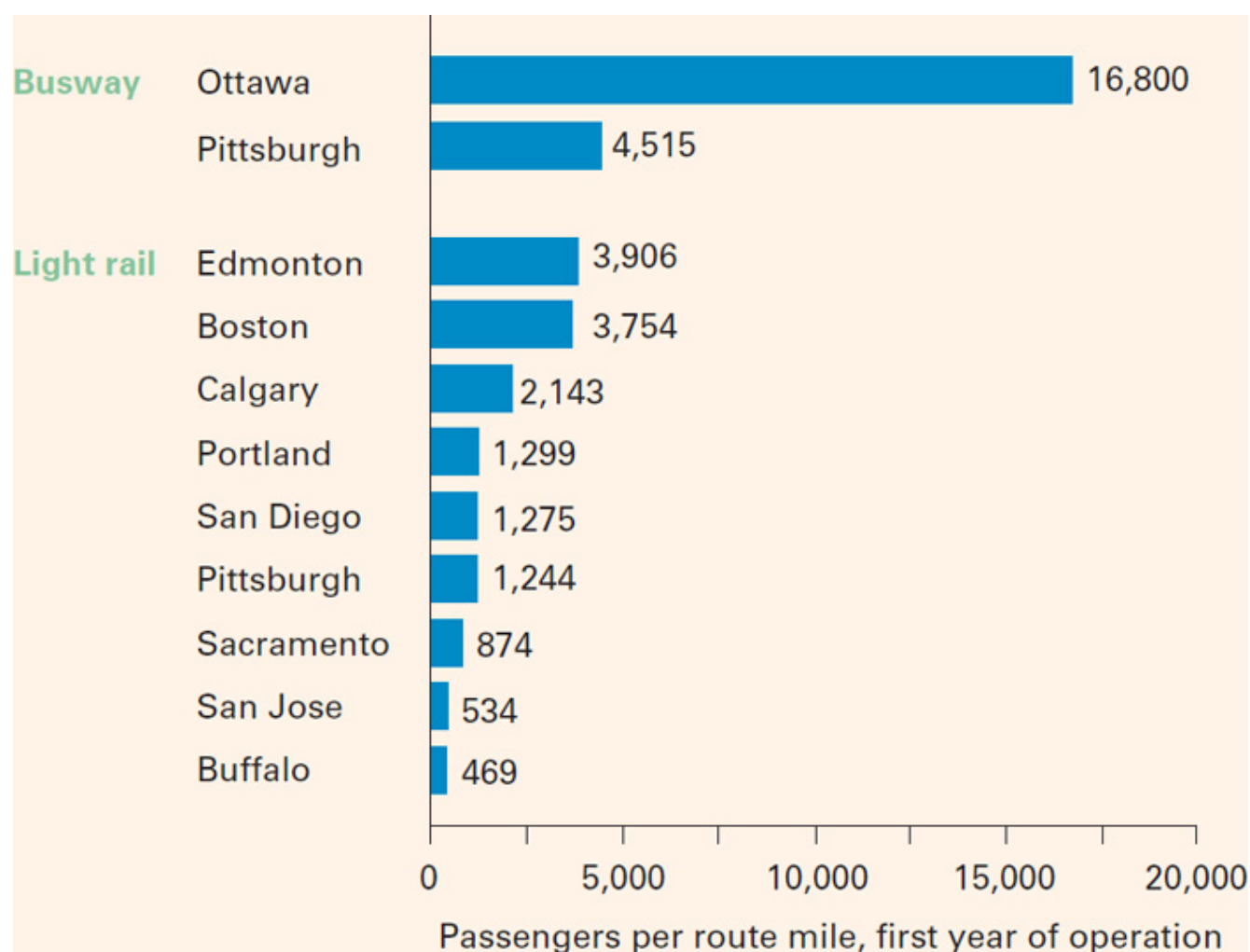
Ottawa opted for a busway at a time when every other medium-size North American metropolis investing in new transit systems selected the eminently more popular light rail transit technology. Similar in size to Ottawa, Calgary and Edmonton both constructed regional light rail systems in the 1970s and 1980s; Vancouver, Canada's third-largest metropolis, built an elevated “advanced” light rail system, called SkyTrain. The decision to go with a busway made the Ottawa-Carleton region a maverick of sorts, but in dollars and cents it made good sense, as busways were shown to be 30 percent less expensive to build and 20 percent less expensive to operate than a light rail. Because of the busway's relatively high operating speeds, the region has been able to get by with 150 fewer buses than it would have needed to carry the same number of passengers on surface streets. These savings exceeded the \$275 million capital outlay for the first 20 kilometers of the busway.

Perhaps most important, a busway was better suited to the region's future land-use vision of concentrated workplaces and retail destinations encircled by largely low-density, single-family detached housing. From a travel standpoint, such a settlement pattern translates into “many-to-few” trip origins and destinations. A point-to-point rail system, planners reasoned, was incompatible with this spatial pattern of trips. Rather, they reasoned, the geometry of a flexible busway system that

allowed trunkline buses to morph into neighborhood feeders was much more compatible with the geography of future travel. As in all good transit metropolises, a close correspondence between the physical design of transit and spatial pattern of trip making was achieved.

Within the first year of operation, Ottawa's busway outperformed all other North American busway and light rail systems built in the early 1990s by nearly four to one on a passenger per guideway-mile basis (figure 2.17). With more than 140 public transit trips per person per year, Ottawa has one of the highest transit utilization rates in North America, even when compared with much larger rail-served cities like Chicago and Philadelphia.

Figure 2.17 Passengers per guideway mile in selected North American busway and light rail systems



Source: Cervero 1998; adapted with permission from Island Press, Washington, DC.

Note: Data are for 1991–93.

As with successful transit metropolises in Scandinavia and elsewhere, Ottawa's TOD vision was buttressed by a series of TDM measures that aimed to level the playing field by either regulating car use or passing on price signals to motorists that began to reflect broader societal costs. Most notable

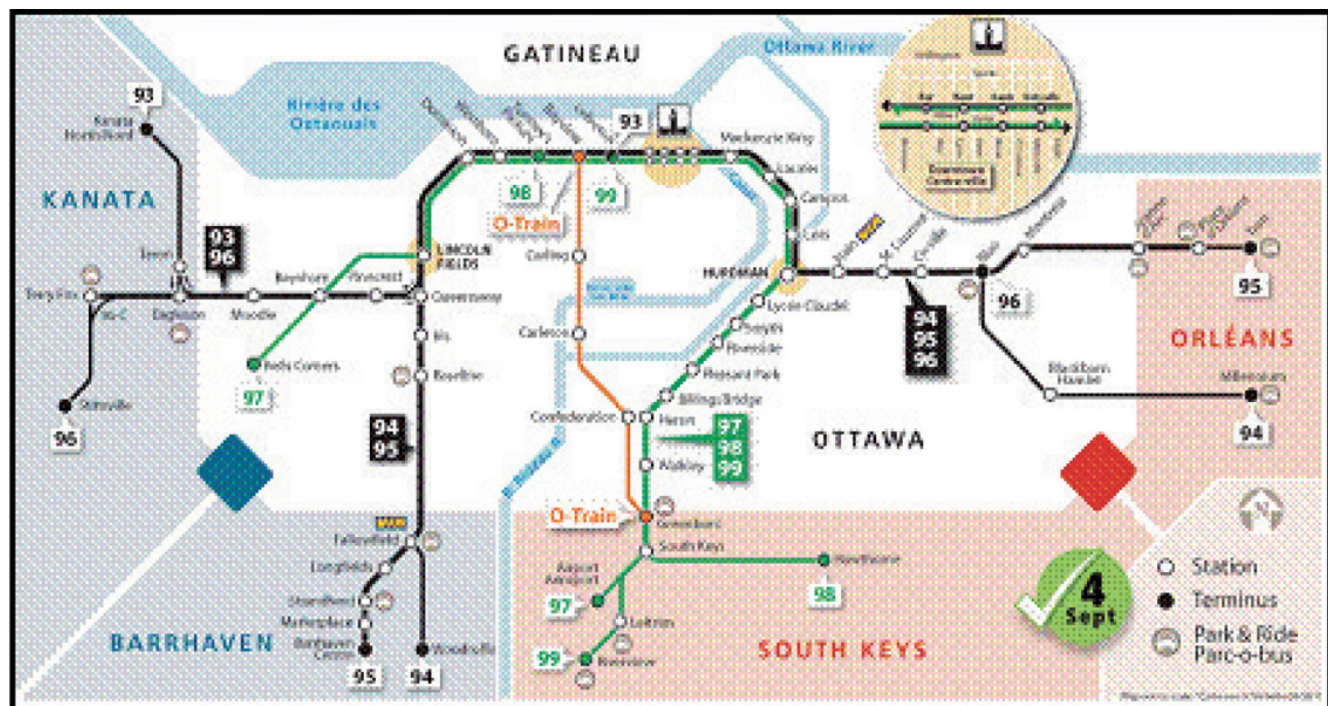
were parking policies. When the Transitway opened in 1983, the federal government began eliminating free parking for its employees and reducing the number of downtown parking spaces. By 1984, downtown Ottawa had 15 percent fewer parking spaces than in 1975, despite a near doubling of office space. The federal government also introduced flexible working schedules for its employees, producing a more even distribution of transit usage over the course of the day. Supportive parking policies were also introduced at busway stations. OC Transpo (the Ottawa-Carleton Regional Transit Commission) restricted park-and-ride facilities to the busway's terminuses to encourage the use of feeder and express services as well as to increase the development potential of selected stations. Other requirements included zoning targets of placing 40 percent of future job growth and all regional trip generators of more than 100,000 square meters within 400 meters of a busway stop.

In addition to "sticks," various pro-transit "carrots" were concurrently introduced, including one of the first bus-based, real-time passenger information systems anywhere; targets for transit mode splits for cordon lines throughout the region (which govern where service improvements are directed); and eco-passes, which provide regular transit users with deep fare discounts. In 1982–85, the city adopted a set of TOD design guidelines that call for, among other things, building setbacks to create human-scale development; public art to enliven station areas; and short street blocks, cut-through walkways, and easy way-finding to enhance the quality of pedestrian environments linked to busway stations (Municipality of Ottawa 2007).

Like Curitiba, Ottawa has experienced some of the growing pains, and limitations, of a highly successful BRT system. Historically, more than 60 percent of regional buses on the Transitway passed through a downtown one-way, surface-level couplet (two one-way streets that run parallel to each other) causing peak-period traffic snarls and the bunching of buses. With passenger throughput having reached the capacity of a surface transit system, a decision was made to convert the downtown busway into a light-rail subway alignment.

The Downtown Tunnel, expected to cost about \$1.4 billion, is the single costliest transit investment under consideration in Ottawa. It would extend the O-Train light rail, which began operation in 2001 as a pilot project to evaluate the benefits of converting Ottawa's Transitways to light rails, north through downtown (figure 2.18). Local officials are quick to point out that BRT is not being abandoned but simply converted to light rail transit along the highest-capacity downtown Transitway corridor. Indeed, new or expanded BRT Transitways are planned in the east, west, and south portions of metropolitan Ottawa.

Figure 2.18 Ottawa's Transitway network, including O-Train light rail line linked to busway



Source: Municipality of Ottawa 2008.

Note: Current plans call for replacing the downtown surface-street busway segment, shown in the circular insert, with an underground light rail system tied to the O-Train light rail line to the south.

Ottawa's past three decades of "transit first" policies continue to pay off in terms of high ridership levels. Transit use has increased steadily since 1998, by about 3.5 percent a year. From 70 million passengers in 1998, annual ridership reached about 100 million in 2010. Ottawa's transit mode share has remained steady, at nearly 15 percent of daily trips, despite declining mode shares in almost all other Canadian cities (Municipality of Ottawa 2008).

Conclusion

A number of lessons can be drawn from international best-case experiences that have direct relevance to cities in developing countries and elsewhere that are investing in BRT and other high-capacity transit systems. [Chapter 4](#) highlights these lessons.

One overarching principle followed by all cities is that successful transit and land-use integration requires a cogent vision of the future city. Visions of how the city will ideally grow and the role of transportation investments and policy in achieving this urban-form vision were well articulated in all

cases. Moreover, land-use visions shaped transportation decisions far more than vice versa, reflecting the core notion that transportation is a means, not an end unto itself. For adaptive cities, the vision of compact, mixed-use, often lineal corridors produced “necklace-of-pearls”-style built forms and induced sustainable mobility choices. For adaptive transit, policy focused on investing in flexible, lower-cost transit systems, like BRT, which can better serve market-driven patterns of development. It is this unwavering commitment to linking transit investments and urban development in mutually beneficial and reinforcing ways that distinguishes these successful global cases.

Applying such principles and practices to rapidly growing and motorizing cities in developing countries is a challenge. [Chapter 3](#) describes how four cities—Ahmedabad, India; Bogota, Colombia; Guangzhou, China; and Ho Chi Minh City, Vietnam—are meeting this challenge.

Annex Table 2A.1 Modes of transportation in selected cities

City (Region)	Population		Population density (residents per square kilometer)		GDP per capita (US dollars)	Area (square kilometers)		Car ownership (cars owned per 1,000 residents)	Transit ridership (annual riders per capita)	Mode of transportation in city (percent)			
	City	Metro- politan area	City	Metro- politan area		City	Metro- politan area			Public transit	Private auto- mobile	Bicycle	Walking
Copenhagen, Denmark	662,600 (2011)	1,199,300 (2011)	5,407	2,452	68,000 (2010)	123	456	184	228	27	40	33	—
Curitiba, Brazil	1,764,500 (2010)	3,210,000 (2010)	4,062	211	8,000 (2007)	430	15,417	400	355	45	—	—	—
Hong Kong SAR, China	7,061,200 (2010)	7,061,200 (2010)	6,480	6,480	45,736 (2008)	1,104	1,104	82	574	88	11	1	—
Ottawa, Canada	883,400 (2011)	1,236,300 (2011)	317	197	73,500 (2008)	2,778	5,716	557	142	21	68	2	9
Singapore	5,076,700 (2010)	5,076,700 (2010)	7,315	7,315	63,867 (2010)	694	—	100	263	63	27	0	10
Seoul, Republic of Korea	10,646,000 (2010)	23,616,000 (2010)	17,000	12,446	27,809 (2008)	605	1,897	214	1,028	69	26	5	—
Stockholm, Sweden	1,372,565 (2010)	2,064,000 (2010)	3,597	317	70,950 (2008)	382	6,519	373	213	45	52	3	—
Tokyo, Japan	13,185,500 (2011)	36,682,500 (2011)	6,027	2,629	40,337 (2008)	2,188	13,752	308	1,107	51	12	14	22
Washington, DC Metro Area	618,000 (2011)	5,580,000 (2011)	3,886	372	76,200 (2008)	177	14,412	680	63	37	48	3	12

Source: Authors' compilation based on Cervero 1998; Pucher and Buehler 2005; PricewaterhouseCoopers 2009; Singapore Land Transport Authority 2010; Santos 2011; <http://en.wikipedia.org>; <http://international.stockholm.se/Future-Stockholm/Urban-development/>; <http://www.kk.dk/sitecore/content/Subsites/CityOfCopenhagen/SubsiteFrontpage/Press/FactsOnCopenhagen/Statistics/Income/HouseholdsFamilyType.aspx>; <http://www.kk.dk/sitecore/content/Subsites/CityOfCopenhagen/SubsiteFrontpage/Press/FactsOnCopenhagen/Statistics/Carsforprivateuse/CarsForPrivateUse.aspx>; <http://ltaacademy.gov.sg/doc/J11Nov-p60PassengerTransportModeShares.pdf>; http://www.ottawa.ca/visitors/about/economy_en.html; http://en.wikipedia.org/wiki/List_of_cities_by_GDP; <http://www.mwcog.org/uploads/committee-documents/Z11fWFIW20111110130359.pdf>; and <http://ltaacademy.gov.sg/doc/J11Nov-p60PassengerTransportModeShares.pdf>.

Note: Figures in parentheses show the year of the data. — = Not available.

Note

1. Conversion calculated based on exchange rate of \$1 = ¥ 114.37 in 1999 and ¥ 93.52 in 2009.

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